

Final project: Emotional Intelligence as A Measure of Potential Academic Success

Clark University School of Business

STAT 4600

Intermediary Statistical Modeling for Analytics

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(Logo created with Canva.com)

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I. Abbreviations

<u>Abbreviation</u>	<u>Description</u>
AI.....	Artificial Intelligence
ANOVA.....	Analysis of Variance
CP.....	Complexity Parameter
CSV.....	Comma-Separated Values
EI	Emotional Intelligence
EQI	Emotional Quotient Inventory
ERIC	Education Report Information Center
ETS	Educational Testing Service
F-statistic	Statistical Test to Compare Variance
GMAT.....	Graduate Management Admission Test
GPA	Grade point Average
GRE.....	Graduate Record Examination
IQ	Intelligence Quotient
ISI	Institute of Scientific Information
MSCEIT	Mayer-Salovey-Caruso Emotional Intelligence Test
MSE	Mean-squared Error
n	Sample Size
NaN.....	Not a Number
PDP.....	Partial Dependence Plot
perc_white.....	Percentage of White Participants in the Sample
pub_type	Publication Type
p-value	Probability Value
KB, JK, KS, and AZ	

Abbreviation**Description**

Yi	Effect Size
Q-Q plot	Quantile-Quantile Plot
R-squared.....	Co-efficient of Determination
SAT	Scholastic Aptitude Test
SD	Standard Deviation
STAT4600	Course code: Intermediary Statical Model Analytics
vi	Variance of Independent Variable
xerror	Cross-Validation Error

II. Introduction

A. *The Research Topic*

The focus of this research is to test if measures of Emotional Intelligence (EI) are accurate predictors of academic success in graduate school, by comparing measures of student success (standardized test scores, Grade Point Average (GPA)) to measures of Emotional Intelligence. In the last ten years, EI has become known as a potent predictor of potential workplace and academic achievement. Unlike conventional measures of aptitude including the GPA and scores on achievement tests, EI focuses on personality characteristics that involve coping strategies, social and emotional interactions and resilience. These attributes are especially important in academically competitive graduate programs, as cognitive ability alone is not sufficient.

B. *The Research Problem.*

The primary issue with traditional predictors of academic performance is that they are limited in assessing a student's true potential in a rigorous academic setting. Traditional measures of academic performance include standardized criteria such as test scores, IQ, grade point average (GPA), SATs, and GREs, which measure cognitive ability. Recent studies have shown that these measures are not accurate when evaluating academic performance (Haynes, 2023; MacCann, 2019). A 2008 study evaluating the SAT scores for 50,000 students found a correlation of 0.41, accounting for 16% of the variance reported in college GPAs (Haynes, 2023; Kobrin et al., 2008). Such scores provide a limited scope of a student's aptitude, excluding a student's socio-economic circumstances, inequalities in access to educational opportunities, or the behavioral/emotional aspects of the student's personality (Haynes, 2023; Sepehrian, 2012). This limitation distorts a very crucial phase of admissions where skills like emotional stability, flexibility and interpersonal skills are left out. New research indicates that these socio-emotional attributes are important and can be assessed using Emotional Intelligence (EI) as far as the effects on long term academic achievement are concerned. Therefore, this research aims to assess if EI measures can fill the above-mentioned gap, as an effective way to add up or replace the regular measures to consider students' potential into a more suitable approach.

C. Importance of the Research Problem.

This research is important because it will address the limitations of traditional measures of academic potential, while highlighting the utility of EI measures in admissions criteria. GRE and GMAT tests traditionally evaluate general cognitive ability, including verbal, quantitative, and writing skills (Moneta-Koehler et.al., 2017). These exams are admission requirements for many graduate programs. The Education Testing Service (ETS), which creates and proctors the GRE examination, cautions against using the test as a valid predictor of academic success, as it overlooks other skills vital to succeeding in a rigorous academic program (Haynes, 2023; Moneta-Koehler et al., 2017). EI encompasses constructs such as perseverance, emotional resilience, and adaptability, which are essential to navigating the challenges of an academic environment (Haynes, 2023; Moneta-Koehler et al., 2017; Romanelli et al., 2006). The importance placed on standardized testing scores and GPA has been questioned in many studies and institutions, resulting in approximately 50% of science-based graduate programs dropping their GRE admissions requirement by 2020 (Langin, 2019). Considering this statistic, EI measurements could be a more accurate replacement for the GRE. In graduate-level education, skills such as coping with stress, people management skills, and time management are vital tools (MacCann et.al., 2019; Moneta-Koehler et.al., 2017; Romanelli et. al., 2006). Based on this preliminary evidence, can emotional intelligence be an accurate metric for academic performance compared to standardized tests?

B. Study Question and Study Aim

This aim of this study is to address the following research question: can measures of Emotional Intelligence be accurate predictors of academic success in graduate school, by comparing measures of student success (standardized test scores, GPA) to measures of Emotional Intelligence? By asking this question, this study hopes to inform future graduate enrollment policies, paving the way for the use of more comprehensive measures of potential student success in higher education, rather than a focus on mere measures of cognitive ability.

D. How does this project relate to our work in STAT4600?

This project utilizes advanced statistical modeling and analytical techniques taught in the STAT 4600 course to address a real-world problem. Specifically:

1. **Regression Analysis:** Linear regression is used to evaluate the relationship between Emotional Intelligence (EI) and academic success metrics (e.g., GPA, standardized test scores). This aligns with the course's emphasis on understanding and applying regression models to interpret and predict outcomes.
2. **Decision Tree Models:** Decision tree algorithms are implemented to classify and identify patterns within the dataset, providing insights into which variables (e.g., ability EI or mixed EI) significantly impact academic success. These models reinforce the course's focus on decision-making tools in statistical analytics.
3. **Data Cleaning and Descriptive Statistics:** The initial stages of the project involve cleaning the dataset and calculating basic descriptive statistics, crucial components in statistical modeling for ensuring data quality and interpretability.
4. **Application of Machine Learning:** The inclusion of machine learning algorithms, such as decision trees, reflects the course's interdisciplinary approach, bridging traditional statistical methods with modern data science practices.
5. **Research Context:** By exploring EI as a predictor of academic success, the project demonstrates the applicability of statistical models to address significant questions in education and admissions policies, a relevant and practical extension of STAT 4600 concepts.

III. Methods

A. Type/Source/Content of the Data

In 2019, MacCann et al. of the University of New South Wales wrote the first comprehensive meta-analysis addressing the correlation between EI and academic performance

(MacCann, 2019). The original study included the results from 1,276 studies exploring the connection between EI and academic performance (MacCann, 2019). The collected studies represent nine databases: ERIC, Google Scholar, ISI Web of Science, Medline, ProQuest, PsycINFO, PubMed, Science Direct, and Scopus (MacCann, 2019). Papers were included based on the following criteria: presented in English, the measure of EI used were published in a test manual or journal, measures of academic performance were included (GPA, SAT, GRE), presented original data, and reported the sample size (MacCann, 2019). The included studies were published between 1997 and 2019 and represented 27 different countries (MacCann, 2019).

The study analyzes a dataset of 553 graduate-level students collected between 2000 and 2019. The dataset includes global participants, with sources like journal articles (52.04%), dissertations (40.77%), unpublished data (6.95%), and other sources (0.24%). Two EI streams are used as the focus of the trimmed data: ability EI (48.68%), measured by MSCEIT, which evaluates cognitive abilities related to emotion-based information and mixed EI (51.32%), measured by EQI, assessing emotional and social functioning in workplace-related outcomes. MSCEIT measures four branches of emotional abilities: observing, producing, recognizing, and controlling emotions. EQI assesses personality and behavioral aspects, such as teamwork, stress management, and conflict resolution. EI measures are weakly correlated with each other. Cognitive success metrics (GPA: 83.21%, standardized tests: 16.79%) were compared with EI measures. Regression and machine learning models are used to calculate the effect sizes and evaluate EI's predictive validity for academic performance.

B. Analysis and Modeling Methods

1. **Holistic Approach:** The approach discussed herein proceeds systematically from elementary data to useful information, leaving out nothing that may be of vital importance.
2. **Interdisciplinary applicability:** Combining sound statistics with convenient aid from decision trees, this will find common application in psychology, education, and business analysis.
3. **Iterative Improvement:** This approach helps make the hypotheses and methods better over time through checking results and looking at models.

Step1: Cleaning Data and Basic Statistics

Data cleaning can be defined as the process of preparing the raw data for analysis and this includes error correction, treatment of missing information as well as consistency checking. Descriptive statistics presents the key features of a dataset. Descriptive statistics can also be used to calculate each part to analyse and find trends and differences in the dataset. The summary includes measures of centre, such as mean and median, and measures of variability, such as standard deviation and range.

Why It Is Important: Ensures that the data is clean and ready for analysis by reducing the presence of errors due to mistakes or unusual values. Descriptive statistics give a full picture of the dataset, thereby giving a better understanding of the data before using more complex statistical models. Untrimmed data is looked at first to find basic characteristics. Important factors drop down data like education and measures of Emotional Intelligence to ensure that parts of the data are relevant to the goals of the study. Basic statistics are then calculated for each component to determine the trends and differences within our data set.

Step 2: Preliminary Results

This can also mean finding the dispersion measures, such as variance and interquartile range, so that we can analyse how the data spreads across different variables.

Why It Is Important: Knowing about variability helps in the detection of patterns or unusual cases in the data that may affect future analysis and that makes way for the selection of the right models or methods in performing predictive analysis. It helps to find out how alike or different the specific variables are by calculating the spread.

Step 3: Regression Analysis

Regression analysis is the study of the relation of one variable to another variable, or to more than one other variable.

Why It Is Important: It helps to see how strong and what kind of relationships exist between the variables. This is important for testing ideas and making predictions. It shows how each variable affects the outcome variable. Dependent variables can also be referred to as a measure of outcome such as performance or ability. Independent variables: This covers factors about people or education, group size or sample size, and emotional intelligence ability measures EI, or mixed EI measures. Regression helps to find out how important EI is compared to other factors on the dependent variable.

Step 4: Decision Tree

A decision tree predicts the outcomes by flowing the data through branches depending on certain conditions. Each of the branches stands for a decision rule.

Why It Is Important: Shows a complex connections and interactions between the variables that are not easy to see with simpler methods. It is not ambiguous, hence easy to understand by non-experts. The goal was to create a decision tree with two main branches, one for ability EI and one for mixed EI. Based on research into implementing the tree model, a different approach was taken. In the Branched Tree Model used for this study the effect size was compared to one of the following variables: variance, population size, age, ei_construct, ei_measure, and achievement_type. The variables used in the model were expanded, based on which variables were most influential in the linear regression models.

Step 5: Understanding

Interpretation of the results involves drawing conclusions from the results of analyses, finding their implications, and consequently recommending future actions.

Why It Is Important: It helps connect statistical results to the research questions or ideas. Helps decide what to do for policy, intervention, or more research. Results are interpreted in terms of Importance of findings: Are the results significant in the data and real life? Do the results provide enough evidence to accept or reject the hypotheses that were proposed? What should be done

because of these findings, such as gathering additional data, conducting more in-depth analysis, or implementing ideas?

IV. Results (Add page numbers to tables and figure references once complete)

The focus of this study was to investigate whether EI can accurately predict academic outcomes for graduate students, compared to standardized tests. The dataset that was used included pre-calculated values for the correlation coefficients comparing EI to the academic measures of success (MacCann, 2020). These values were transformed into Pearson's R Correlations in the form of an effect size, represented by the variable y_i (MacCann, 2020, Using Effect Size, 2024). To test the proposed hypothesis a regression analysis was performed.

A regression analysis is an algorithm that calculated the linear relationship between an independent variable and one or more independent variables (GeeksforGeeks3, 2024). For this study, the effect size (y_i) was used as the dependent variable and compared to several independent variables including the following: v_i (variance), n (population size), age, $ei_measure$, $ei_construct$, and ei_stream . Seven models were performed with one variable alone or in different combinations. The algorithm calculates the difference between the observed and predicted values, also known as the residuals (GeeksforGeeks3, 2024; Outlier, 2022).

The residuals help measure how well each model fits the data (GeeksforGeeks3, 2024; Outlier, 2022). Residuals help determine the Goodness of Fit or how well each model depicts the relationship between the variables (GeeksforGeeks3, 2024; Outlier, 2022). The patterns of the residuals help identify linearity, variability, and outliers (Outlier, 2022). The analyses were performed using R software and the output includes four plots: residuals vs. fitted, a Q-Q plot, a scale-location, and residuals vs leverage. The first plot, residuals vs fitted, checks for non-linearity and how well the model fits the data (Gelman, 2006; Outlier, 2022). The Q-Q plot measures if the residuals are normally distributed (Gelman, 2006; Outlier, 2022). Scale-location depicts variance within the residuals, also known as heteroscedasticity (Gelman, 2006; Outlier, 2022). Finally, the residuals vs leverage demonstrates if certain points are more influential on the models than others (Gelman, 2006; Outlier, 2022).

A. Regression Analysis

Seven models were performed to test the hypothesis: can measures of emotional intelligence be accurate predictive markers for academic performance in graduate students, compared to standardized test scores?

Model 1 compares the dependent variable y_i (effect size) to independent variables (vi : variance, n : population, ei_stream : ability or mixed EI, and $ei_measure$ (1.1: MSCEIT, 3.1: EQI)). The residuals suggest a non-linear pattern that is mostly normally distributed with some deviation at the tails (Figure 1, p. 19). The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent (Figure 1, p. 19). Clustering of the data points in this plot shows that some data points are influencing the outcome of the model more than others and warrant further investigation (Figure 1, p. 19). The residuals have a range of -0.75165 to -0.11863 (Table 1, p. 19 - 20). The coefficients were all significant and are as follows: intercept (0.217), vi (-6.614), n (-0.00015), and EI-stream (0.0139) (Table 1 p. 19 - 20). EI_measure was excluded due to redundancy. The F-statistics were large (F-statistic = 6.766), with a significant p-value (Table 1 p. 19 - 20). However, the adjusted R-squared value was equal to 0.03992, only explaining ~4.0% of the variance in the model (Table 1 p. 19 - 20). Overall, these findings show that this model was not a good fit for the data (Figure 1, Table 1 p. 19 - 20).

Model 5 compares the dependent variable y_i (effect size) and $ei_measure$: 1.1 (MSCEIT) or 3.1 (EQI)). The residuals have a narrow range, are tightly clustered, and are possibly overfitted (Figure 5, p. 25). The pattern of the residuals in the Q-Q plot, are mostly normally distributed, but show some deviation at the tails (Figure 5, p. 25). The variance of residuals seems to be consistent, with no clear pattern (Figure 5, p. 25). There are two clusters within the last plot that suggest that these two sets of points may be important to the outcome of the regression analysis, however, the low adjusted R-squared value suggests that this is not the case (Figure 5, p. 25, Table 5, p. 25 - 26). The residuals have a range of -0.73102 to 0.56953 (Table 5 p. 25 - 26). The coefficients were all significant and are as follows: intercept (0.143746) and $ei_measure$ (0.004552). The F-statistic is very small and is equal to 0.2249, with an insignificant p-value (Table 5 p. 25 - 26). The adjusted R-squared value was equal to -0.001867, meaning there is little to no explanatory power behind this model (Table 5). Overall, these findings show that this model was not a good fit for the data (Figure 5, p. 25, Table 5 p. 25 - 26).

Model 6 compares the dependent variable y_i (effect size) achievement_type. The residuals have a narrow range, are tightly clustered, and are possibly overfitted (Figure 6, p. 26 - 28). The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails (Figure 6 p. 26 - 28). The variance of the residuals seems to be consistent, with no clear pattern present (Figure 6 p. 26 - 28). There are two clusters within the plot that may suggest that these two sets of points may be important to the outcome of the regression analysis, however, the low adjusted r-squared value suggests that this is not the case. The residuals have a range of -0.73667 to 0.56967 (Table 6, p. 27 - 28). The coefficients were all significant and are as follows: intercept (0.154395) and achievement_type (-0.005785) (Table 6 p. 27 - 28). The F-statistics were very small (F-statistic = 0.05075), with an insignificant p-value. The adjusted R-squared value was equal to -0.002287, demonstrating that this model has little to no explanatory power. Overall, these findings show that this model was not a good fit for the data (Figure 6, p. 26 - 28, Table 6 p. 27 - 28).

Model 2 compares the dependent variable y_i (effect size) to independent variables (v_i : variance, n : age, ei_stream : ability or mixed EI, $ei_construct$: subcategories for EI stream, and $ei_measure$ (1.1: MSCEIT, 3.1: EQI). The residual pattern suggests that the data is non-linear, meaning the model is not a good fit (Figure 2, p. 20 - 21). The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails (Figure 2 p. 20 - 21). The variance of the residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, another indicator that the model is not a good fit (Figure 2 p. 20 - 21). Clustering of the data points in the last plot shows that some data points are influencing the outcome of the model more than others and warrant further investigation. The residuals have a range of -0.75763 to 0.53177 (Table 2, p. 21 - 22). The coefficients were all significant and are as follows: intercept (0.2078), v_i (-6.296), n (-0.0001456), EI-stream (0.007576), and EI-construct (0.004743) (Table 2, p. 21 - 22). EI_measure was excluded due to redundancy. The F-statistics were large (F-statistic = 5.297), with a significant p-value (Table 2, p. 21 - 22). However, the adjusted R-squared value was equal to 0.04071, only explaining ~4.1% of the variance in the model (Table 2, p. 21 - 22). Variance (v_i) and population size (n) seem to have the most influence on the model (Table 2, p. 21 - 22). A seventh model was generated to explore if the adjusted R-squared value would change significantly if these two variables were isolated.

Model 7 compares the dependent variable y_i (effect size) to independent variables (v_i : variance and n : population). The pattern of the residuals is more uniform compared to Model 2 (Figure 7, p. 28 - 29). The residuals start off having an even distribution around zero but then create a funnel pattern towards the right of the plot (Figure 7, p. 28 - 29). This pattern of the residuals is still non-linear, and mostly normally distributed, with some deviation from a normally distributed pattern in the left tail (Figure 7, p. 28 - 29). The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, another indicator that the model is not a good fit. Clustering of the data points in this plot shows that some data points are influencing the outcome of the model more than others and warrant further investigation. The residuals have a range of -0.76164 to -0.12101 (Table 7, p. 29). The coefficients were all significant and are as follows: intercept (0.2413), v_i (-6.776), and n (-0.0001286) (Table 7). The F-statistics were large (F-statistic = 9.293), approximately twice the value of Model 2, with a significant p-value (Table 7, p. 29). However, the adjusted R-squared value was equal to 0.03834, only explaining ~3.8% of the variance in the model (Table 7, p. 29). This value was approximately 0.3% lower than the adjusted R-squared value in Model 2. To investigate if the variables variance or population size have improved adjusted R-squared values when compared to effect size individually, a simple linear regression was performed for each variable.

Model 3 compares the dependent variable y_i (effect size) to independent variable (v_i : variance) and Model 4 compares the dependent variable y_i to the independent variable (n : population). The plots for both Model 3 and Model 4 suggest a non-linear pattern within the data, that is mostly normally distributed, with some deviation at the tails (Figures 3 and 4, p. 22 - 25). The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, indicating that the models are not a good fit. The clustered pattern of the data points in this plot shows that some data points are influencing the outcome of the model more than others. The residuals in Model 3 have a range of -0.7339 to 0.5644 (Table 3, p. 23). The coefficients were all significant and are as follows: intercept (0.17602) and v_i (-3.01211) (Table 3, p. 23). The F-statistics were large (4.294), with a significant p-value (Table 3, p. 23). However, the adjusted R-squared value was equal to 0.007855, only explaining ~ 0.79% of the variance in the model (Table 3, p. 23). In Model 4, the residuals had a range of -0.74794 to 0.55261 (Table 4, p. 24 - 25). The coefficients were all significant and are as follows: intercept (0.01686) and n (-5.259e-05) (Table 4, p. 24 - 25). The F-statistics were large (F-statistic =

3.411), with a significant p-value (Table 4, p. 24 - 25). However, the adjusted R-squared value was equal to 0.005762, only explaining ~ 0.58% of the variance in the model (Table 4, p. 24 - 25). Neither model was a good fit for the data, but based on the adjusted R-squared values, variance is more influential on the outcome for the algorithm, than population size.

The results of the simple and multiple linear regression models demonstrate that none of our models fit well within the given dataset. Despite large and significant F statistics present in several of the models, the adjusted R-squared values demonstrate little to no explanatory power. This suggests that further testing needs to be completed to better understand any confounding variables that could be influencing the data. To further investigate the relationship between the variables of interest, a drill down study was performed using a Branched Decision Tree Model.

B. Branched Decision Tree Model, Cross-validation Model, and Pruned Tree Summary

A Branched Decision Tree model is a type of machine learning algorithm that looks and acts like a decision tree (Almog, 2024; GeeksforGeeks2,2020). Data is sub-setted based on the specifications of the algorithm, and branches are formed based on the “decisions” of those specifications (Almog, 2024; GeeksforGeeks2,2020). Each node on the tree represents a subset of data, the splits or branches are formed based on the conditions of the model, and each terminal node where the data no longer branches is known as a leaf. The complexity parameter (CP) controls how each branch is formed (Almog, 2024; GeeksforGeeks2,2020). The lower the CP value, the more branches are formed, and the higher the CP values, the less branches are formed (Almog, 2024; GeeksforGeeks2,2020). The less branches a tree has, the less complex and more accurate it becomes, reducing the cross-validation error of the tree (Almog, 2024). Cross-validation error or xerror measures the error between the observed and predicted values within the tree (Almog, 2024; GeeksforGeeks1,2020; GeeksforGeeks2,2020). In the Branched Tree Model used for this study the effect size was compared to one of the following variables: variance, population size, age, ei_construct, ei_measure, and achievement_type (Figure 8, p. 30-31, Tables 8 and 9, p. 32-33).

The summary for the branched Decision Tree is as follows. Variables ei_construct and achievement_type were excluded due to low representation. The optimal tree was selected with a complexity parameter = 0.01214359 (Table 8, p. 32). The tree had an n = 417, as 136

observations were deleted due to NaN values (Table 8, p. 32). The mean squared error (root node error) = 0.03823. The cross-validation error (xerror) = 0.6697 (Table 8, p. 32). Variance (vi) was the top split, indicating that it is the most important variable ($vi \geq 2.6089$) (Table 8, p. 32). The cross-validation error (xerror) = 0.6697 (Table 8, p. 32). The optimal tree was constructed using a cross-validation model to decrease cross-validation error. In Table 9 the variables included in the model are ranked by importance, most important at the top to least important at the bottom (Table 9, p. 33). Variance (vi) and population (n) were calculated as being the most important variables in the construction of the tree, with importance values of 5.73 and 4.36, respectively (Table 9, p. 33).

V. Conclusions

A. Data Summary/ Key Findings

Emotional intelligence (EI) is identified as a potential predictor of academic success. However, the results from the study did not provide us any sufficient statistical or practical evidence to confirm our hypothesis. The results of the simple and multiple linear regression models demonstrate that none of our models fit well within the given dataset. Despite large and significant F statistics present in several of the models, the adjusted R-squared values show that there is little to no explanatory power for the variance in the data. This suggests that further testing needs to be completed to better understand any confounding variables that could be influencing the data. The drill down study, using a Branched Decision Tree, variance (vi) and population (n) were the most important variables in the construction of the tree, with importance values of 5.73 and 4.36, respectively. This is contrary to the assumptions at the beginning of this study, that the EI-stream and EI_measures would be the most significant variables. Due to certain limitations, such as samples with high variance or differing population sizes, the conclusions of this study illustrate a very weak or even non-significant associations between EI and academic success. These conclusions do not support the proposed hypothesis.

B. Limitations

There were several limitations within the study that may have affected the outcome of the results. The pre-calculated correlation values restrict flexibility in analysing relationships within

the raw data, reducing potential insights into unexplored associations. Varying population sizes within the dataset constrained statistical power, and contributed to the variance of the residuals, skewing the outcome of the calculated models. High variance means that there is low heterogeneity in the data, making generalization difficult. This can also be seen by wide confidence intervals in the regression outputs or the inconsistent importance of variables in decision trees. The missing values, known as NaN values, in important variables were excluded, therefore possibly introducing bias and minimizing representation of key data points. Our study analysed data that was from a single point in time, and as such, one cannot strongly establish causality in comparison to a longitudinal study, which are carried out over long periods of time.

C. Business Recommendations/ Future Directions

Based on the results of this study; it is recommended that the next steps include the following:

1. ***Perform a longitudinal study:*** A longitudinal design could be used to monitor changes in EI and academic success over time, which would have much stronger implications for causality. The study design would include a sample group of students entering graduate school. Before entering the program, each student would be given a GRE exam, along with at least two tests measuring each of the following EI streams: ability EI, self-rate EI, and mixed EI. Over the course of the student's programs a GPA score will be calculated for each semester. Retention rates will also be included at the end of each semester and scored based on the number of semesters completed. After the completion of this study, the linear regression analysis and branched decision tree model will be repeated.
2. ***Future Directions:*** While the future is uncertain for many students, the threat of technology advancing, such as with AI systems, may add to that uncertainty. Exploring the role of EI as a uniquely human skill could mitigate these concerns. Future applications and projects involving EI could investigate how high levels of EI further benefit human-AI collaboration using advanced machine learning models. Emotional intelligence in the sense of empathy, interpersonal skills, and self-regulation, can be highlighted as a human strength, furthering the success of students in both academically and professionally.

Appendix A

Tables/Figures

Regression Analysis	p. 19 - 29
Model 1: Comparing yi (effect size) with vi (variance), n (population), ei_stream (ability or mixed) and ei_measure (1.1: MSCEIT or 3.1: EQI) (Figure 1/ Table 1)	p. 19 - 20
Model 2: Comparing yi (effect size) with vi (variance), n (age), ei_stream (ability or mixed), ei_construct (stream sub-categories), and ei_measure (1.1: MSCEIT or 3.1: EQI) (Figure 2/ Table 2)	p. 20 - 22
Model 3: Comparing yi (effect size) with vi (variance) (Figure 3/ Table 3)	p. 22 - 23
Model 4: Comparing yi (effect size) with n (population) (Figure 4/ Table 4)	p. 24 - 25
Model 5: Comparing yi (effect size) with ei_measure (1.1: MSCEIT or 3.3: EQI) (Figure 5/ Table 5)	p. 25 - 26
Model 6: Comparing yi (effect size) with achievement_type (Figure 6/ Table 6)	p. 26 - 28
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Model 1: Comparing y_i (effect size) with v_i (variance), n (population), ei_stream (ability or mixed) and $ei_measure$ (1.1: MSCEIT or 3.1: EQI)

Figure 1: Output of Regression Analysis, Model # 1

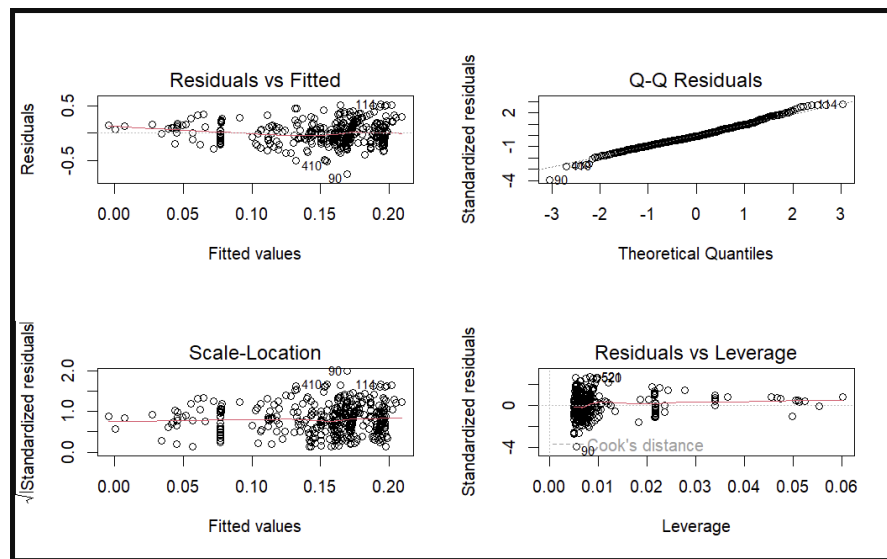


Figure 1 Note: Output of Regression Analysis, Model # 1, comparing the dependent variable y_i (effect size) to independent variables (v_i : variance, n : population, ei_stream : ability or mixed EI, and $ei_measure$ (1.1: MSCEIT, 3.1: EQI). Plot 1 (Residuals v. Fitted): The residual pattern suggests a non-linear pattern, meaning the model is not a good fit. Plot 2 (Q-Q Residuals): The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails. Plot 3 (Scale- Location): The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, another indicator that the model is not a good fit. Plot 4 (Residuals vs. Leverage): Clustering of the data points in this plot show that some data points are influencing the outcome of the model more than others and warrant further investigation.

Table 1: Output of Regression Analysis, Model # 1

	Minimum	1 st Quarter	Median	3 rd Quarter	Maximum
Residuals	-0.75165	-0.11863	-0.01518	0.12247	0.52410
Coefficient	Estimate	Standard Error	T value	Pr (> t)	Significance

Intercept	2.17e-01	2.920e-02	7.434	6.15e-13	***
vi	-6.614	1.749	-3.781	0.000179	***
n	-1.465e-04	3.684e-05	-3.977	8.24e-05	***
EI_stream	1.39e-02	1.072e-02	1.297	0.195440	Not Significant
EI_measure	NA	NA	NA	NA	
Residual Standard Error	Degrees pf Freedom	Multiple R- squared	Adjusted R- squared	F-statistic	P-value
0.1918	413	0.04685	0.03992	6.766	0.0001833

Table 1 Note: Output of Regression Analysis, Model # 1, comparing the dependent variable *yi* (effect size) to independent variables (*vi*: variance, *n*: population, *ei_stream*: ability or mixed EI, and *ei_measure* (1.1: MSCEIT, 3.1: EQI). The residuals had a range of -0.75165 to -0.11863. The coefficients were all significant and are as follows: intercept = 0.217, *vi* = -6.614, *n* = -0.00015, and *EI-stream* = 0.0139. *EI_measure* was excluded due to redundancy. The *F*-statistics were large = 6.766, with a significant *p*-value. However, the adjusted *R*-squared value was = 0.03992, only explaining ~4.0% of the variance in the model.

Model 2: Comparing *yi* (effect size) with *vi* (variance), *n* (age), *ei_stream* (ability or mixed), *ei_construct* (stream sub-categories), and *ei_measure* (1.1: MSCEIT or 3.1: EQI).

Figure 2: Output of Regression Analysis, Model # 2

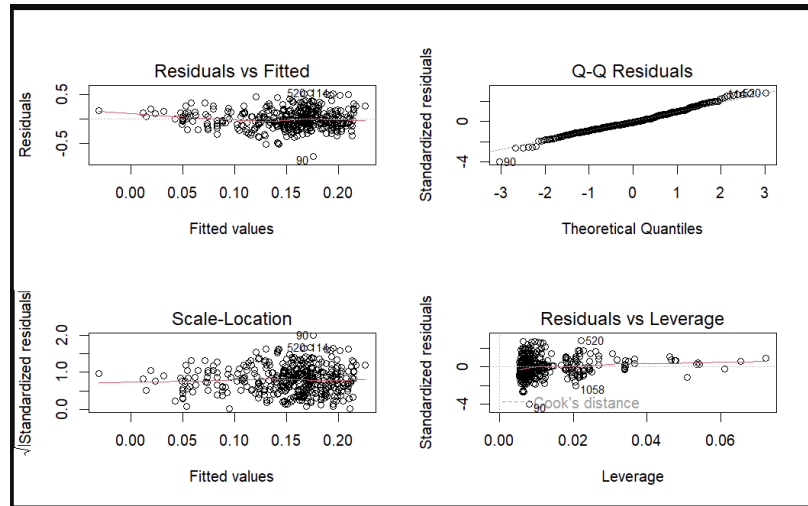


Figure 2 Note: Output of Regression Analysis, Model # 2, comparing the dependent variable *yi* (effect size) to independent variables (*vi*: variance, *n*: age, *ei_stream*: ability or mixed EI, *ei_construct*: subcategories for EI stream, and *ei_measure* (1.1: MSCEIT, 3.1: EQI). Plot 1 (Residuals v. Fitted): The residual pattern suggests a non-linear pattern, meaning the model is not a good fit. Plot 2 (Q-Q Residuals): The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails. Plot 3 (Scale-Location): The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, another indicator that the model is not a good fit. Plot 4 (Residuals vs. Leverage): Clustering of the data points in this plot shows that some data points are influencing the outcome of the model more than others and warrant further investigation.

Table 2: Output of Regression Analysis, Model # 2

	Minimum	1 st Quarter	Median	3 rd Quarter	Maximum
Residuals	-0.75763	-0.11608	-0.01895	0.12192	0.53177
Coefficient	Estimate	Standard Error	T value	Pr (> t)	Significance
Intercept	2.078e-01	3.009e-02	6.905	1.98e-11	***
vi	-6.296	1.760	-3.577	0.00039	***
n	-1.456e-04	3.673e-05	-3.965	8.68e-05	***

EI_stream	7.576e-03	1.321e-02	0.574	0.56652	Not Significant
EI_construct	4.743e-03	3.869e-03	1.226	0.22096	Not Significant
EI_measure	NA	NA	NA	NA	
Residual Standard Error	Degrees of Freedom	Multiple R-squared	Adjusted R-squared	F-statistic	P-value
0.1911	401	0.05018	0.04071	5.297	0.0003636

Table 2 Note: Output of Regression Analysis, Model # 2, comparing the dependent variable y_i (effect size) to independent variables (v_i : variance, n : population, ei_stream : ability or mixed EI, $ei_construct$ (stream sub-categories), and $ei_measure$ (1.1: MSCEIT, 3.1: EQI). The residuals had a range of -0.75763 to 0.53177. The coefficients were all significant and are as follows: intercept = 0.2078, v_i = -6.296, n = -0.0001456, EI_stream = 0.007576, and $EI_construct$ = 0.004743. $EI_measure$ was excluded due to redundancy. The F-statistics were large = 5.297, with a significant p-value. However, the adjusted R-squared value was = 0.04071, only explaining ~4.1% of the variance in the model.

Model 3: Comparing y_i (effect size) with v_i (variance)

Figure 3: Output of Regression Analysis, Model # 3

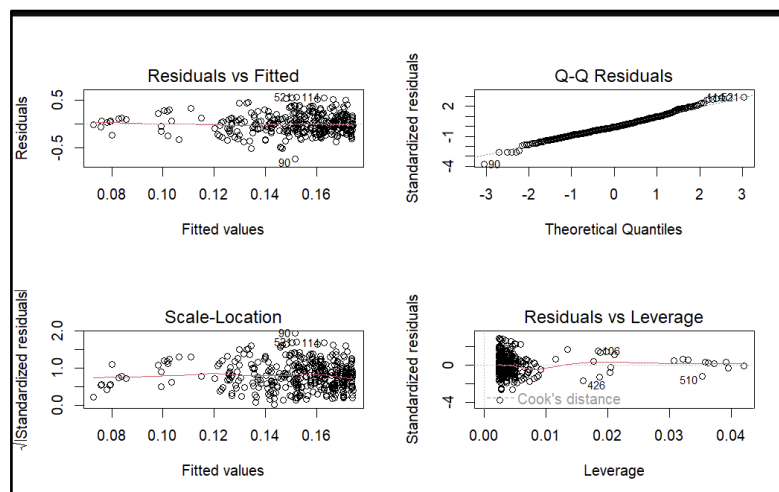


Figure 3 Note: Output of Regression Analysis, Model # 3, comparing the dependent variable y_i (effect size) to independent variables (v_i : variance). Plot 1 (Residuals v. Fitted): The residual pattern suggests a non-linear pattern, meaning the model is not a good fit. Plot 2 (Q-Q Residuals): The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails. Plot 3 (Scale- Location): The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, another indicator that the model is not a good fit. Plot 4 (Residuals vs. Leverage): Clustering of the data points in this plot show that some data points are influencing the outcome of the model more than others and warrant further investigation.

Table 3: Output of Regression Analysis, Model # 3

	Minimum	1 st Quarter	Median	3 rd Quarter	Maximum
Residuals	-0.7339	-0.1305	-0.0156	0.1195	0.5644
Coefficient	Estimate	Standard Error	T value	Pr (> t)	Significance
Intercept	0.17602	0.01449	12.145	<2e-16	***
v_i	-3.01211	1.45363	-2.072	0.0389	*
Residual Standard Error	Degrees of Freedom	Multiple R-squared	Adjusted R-squared	F-statistic	P-value
0.195	415	0.01024	0.007855	4.294	0.03887

Table 3 Note: Output of Regression Analysis, Model # 3, comparing the dependent variable y_i (effect size) to independent variables (v_i : variance). The residuals had a range of -0.7339 to 0.5644. The coefficients were all significant and are as follows: intercept = 0.17602, v_i = -3.01211. The F-statistics were large = 4.294, with a significant p-value. However, the adjusted R-squared value was = 0.007855, only explaining ~ 0.79% of the variance in the model.

Model 4: Comparing y_i (effect size) with n (population)

Figure 4: Output of Regression Analysis, Model # 4

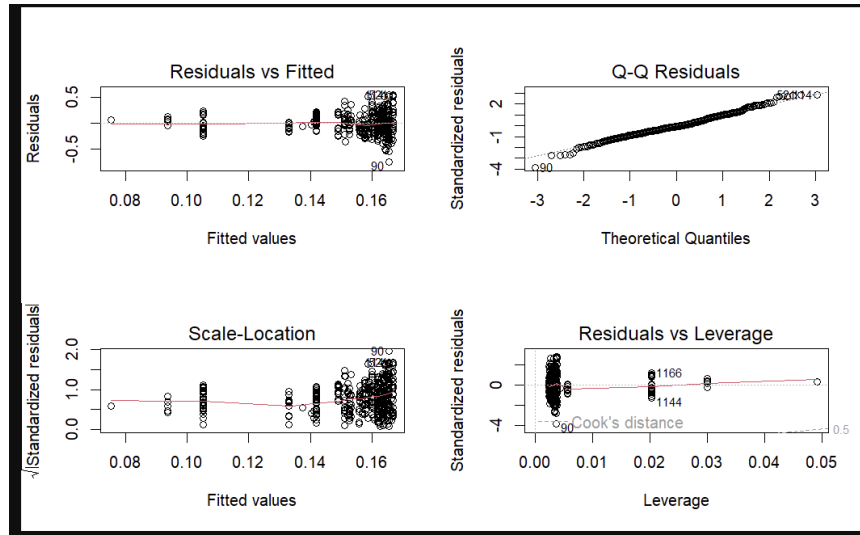


Figure 4 Note: Output of Regression Analysis, Model # 4, comparing the dependent variable y_i (effect size) to independent variables (n : population). Plot 1 (Residuals v. Fitted): The residual pattern suggests a non-linear pattern, meaning the model is not a good fit. Plot 2 (Q-Q Residuals): The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails. Plot 3 (Scale- Location): The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, another indicator that the model is not a good fit. Plot 4 (Residuals vs. Leverage): Clustering of the data points in this plot show that some data points are influencing the outcome of the model more than others and warrant further investigation.

Table 4: Output of Regression Analysis, Model # 4

	Minimum	1 st Quarter	Median	3 rd Quarter	Maximum
Residuals	-0.74794	-0.11608	-0.01759	0.13014	0.55261
Coefficient	Estimate	Standard Error	T value	Pr (> t)	Significance
Intercept	0.1686e-01	1.259e-02	13.386	<2e-16	***
n	-5.259e-05	2.848e-05	-1.847	0.0655	Not Significant

Residual Standard Error	Degrees of Freedom	Multiple R-squared	Adjusted R-squared	F-statistic	P-value
0.1952	415	0.008152	0.005762	3.411	0.06549

Table 4 Note: *Output of Regression Analysis, Model # 4, comparing the dependent variable n (population) to independent variables (vi : variance). The residuals had a range of -0.74794 to 0.55261. The coefficients were all significant and are as follows: intercept = 0.01686, $n = 5.259e-05$. The F-statistics were large = 3.411, with a significant p-value. However, the adjusted R-squared value was = 0.005762, only explaining ~ 6.6% of the variance in the model.*

Model 5: Comparing y_i (effect size) with $ei_measure$ (1.1: MSCEIT or 3.3: EQI)

Figure 5: Output of Regression Analysis, Model # 5

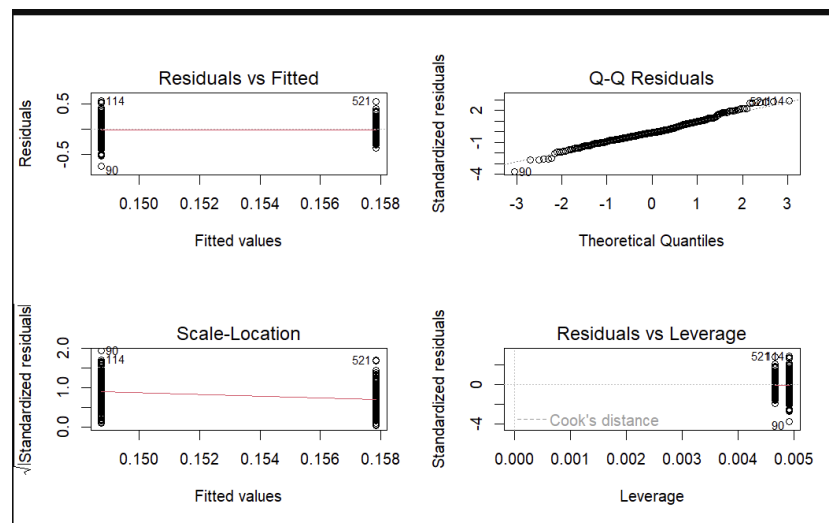


Figure 5 Note: *Output of Regression Analysis, Model # 5, comparing the dependent variable y_i (effect size) and $ei_measure$: 1.1 (MSCEIT) or 3.1 (EQI)). Plot 1 (Residuals v. Fitted): The residuals have a narrow range, tightly clustered, and are possibly overfitted. Plot 2 (Q-Q Residuals): The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails. Plot 3 (Scale- Location): The variance of*

residuals seems to have a consistent pattern at the tails. Plot 4 (Residuals vs. Leverage): There are two clusters within the plot that may suggest that these two sets of points may be important to the outcome of the regression analysis, however, the low adjusted r-squared value suggests that this is not the case.

Table 5: Output of Regression Analysis, Model # 5

	Minimum	1 st Quarter	Median	3 rd Quarter	Maximum
Residuals	-0.73102	-0.12193	-0.01583	0.12596	0.56953
Coefficient	Estimate	Standard Error	T value	Pr (> t)	Significance
Intercept	0.143746	0.022553	6.374	4.92e-10	***
ei_measure	0.004552	0.009599	0.474	0.636	Not Significant
Residual Standard Error	Degrees of Freedom	Multiple R-squared	Adjusted R-squared	F-statistic	P-value
0.1959	415	0.0005416	-0.001867	0.2249	0.6356

Table 5 Note: Output of Regression Analysis, Model # 5, comparing the dependent variable yi (effect size) and ei_measure: 1.1 (MSCEIT) or 3.1 (EQI)). The residuals had a range of -0.73102 to 0.56953. The coefficients were all significant and are as follows: intercept = 0.143746, ei_measure = 0.004552. The F-statistics were very small = 0.2249, with an insignificant p-value. The adjusted R-squared value was = -0.001867, demonstrating that there is little to no explanatory power.

Model 6: Comparing yi (effect size) with achievement_type

Figure 6: Output of Regression Analysis, Model # 6

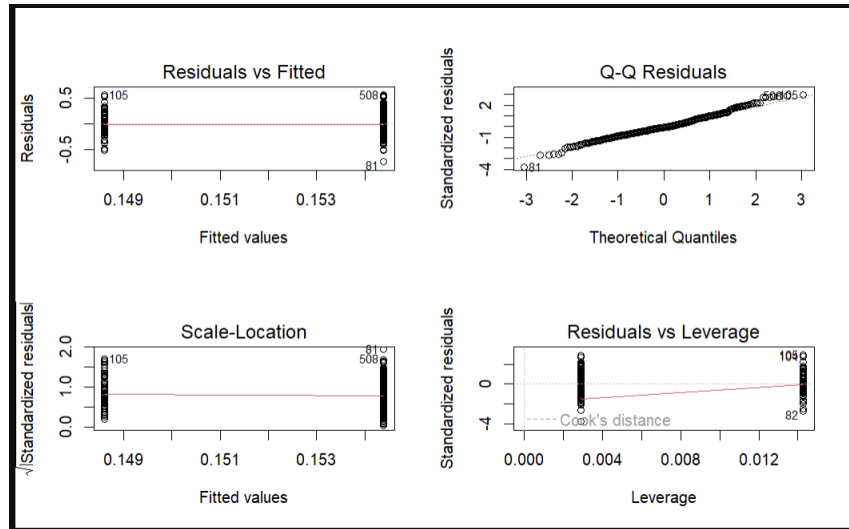


Figure 6 Note: Output of Regression Analysis, Model # 6, comparing the dependent variable *yi* (effect size) *achievement_type*. Plot 1 (Residuals v. Fitted): The residuals have a narrow range, tightly clustered, and are possibly overfitted. Plot 2 (Q-Q Residuals): The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern at the tails. Plot 3 (Scale- Location): The variance of residuals seems to have a consistent pattern at the tails. Plot 4 (Residuals vs. Leverage): There are two clusters within the plot that may suggest that these two sets of points may be important to the outcome of the regression analysis, however, the low adjusted *r*-squared value suggests that this is not the case.

Table 6: Output of Regression Analysis, Model # 6

	Minimum	1 st Quarter	Median	3 rd Quarter	Maximum
Residuals	-0.73667	-0.12387	-0.01659	0.12114	0.56967
Coefficient	Estimate	Standard Error	T value	Pr (> t)	Significance
Intercept	0.154395	0.010521	14.675	<2e-16	***
Achievement_type	-0.005785	0.025679	-0.225	0.822	Not Significant

Residual Standard Error	Degrees pf Freedom	Multiple R- squared	Adjusted R-squared	F-statistic	P-value
0.196	415	0.0001223	-0.002287	0.05075	0.8219

Table 6 Note: *Output of Regression Analysis, Model # 6, comparing the dependent variable yi (effect size) and ei_measure: 1.1 (MSCEIT) or 3.1 (EQI)). The residuals had a range of -0.73667 to 0.56967. The coefficients were all significant and are as follows: intercept = 0.154395 and achievement_type = -0.005785. The F-statistics were very small = 0.05075, with an insignificant p-value. The adjusted R-squared value was = -0.002287, demonstrating little to no explanatory power.*

Model 7: Comparing yi (effect size) with vi (variance) an n (age)

Figure 7: Output of Regression Analysis, Model # 7

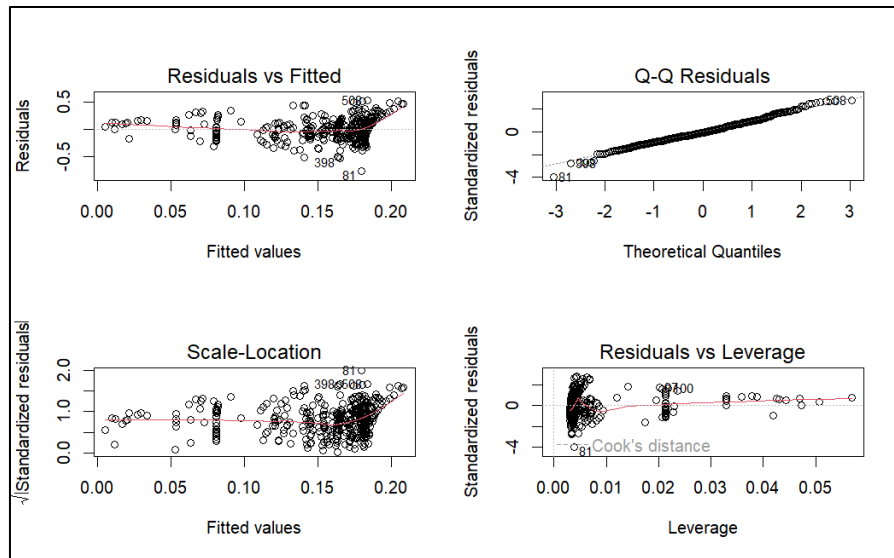


Figure 7 Note: *Output of Regression Analysis, Model # 7, comparing the dependent variable yi (effect size) to independent variables (vi: variance and n: population). Plot 1 (Residuals v. Fitted): The residuals start off having an even distribution around 0, but then create a funnel pattern towards the right of the plot. This residual pattern suggests the data is non-linear,*

meaning the model is not a good fit. Plot 2 (Q-Q Residuals): The pattern of the residuals is mostly normally distributed but shows some deviation from a normally distributed pattern in the left tail. Plot 3 (Scale- Location): The variance of residuals shows a funnel shaped pattern suggesting that the variance isn't consistent, another indicator that the model is not a good fit. Plot 4 (Residuals vs. Leverage): Clustering of the data points in this plot show that some data points are influencing the outcome of the model more than others and warrant further investigation.

Table 7: Output of Regression Analysis, Model # 7

	Minimum	1 st Quarter	Median	3 rd Quarter	Maximum
Residuals	-0.76164	-0.12101	-0.01275	0.12642	0.5328
Coefficient	Estimate	Standard Error	T value	Pr (> t)	Significance
Intercept	2.413e-01	2.246e-02	10.743	<2e16	***
vi	-6.776	1.746	-3.881	0.000121	***
n	-1.286e-04	3.417e-05	-3.763	0.000192	***
Residual Standard Error	Degrees of Freedom	Multiple R-squared	Adjusted R-squared	F-statistic	P-value
0.192	414	0.04297	0.03834	9.293	0.0001127

Table 7 Note: Output of Regression Analysis, Model # 7, comparing the dependent variable y_i (effect size) to independent variables (vi : variance and n : population). The residuals had a range of -0.76164 to -0.12101. The coefficients were all significant and are as follows: intercept = 0.2413, vi = -6.776, and n = -0.0001286. The F-statistics were large = 9.293, with a significant p-value. However, the adjusted R-squared value was = 0.03834, only explaining ~3.8% of the variance in the model.

B. Branched Decision Tree Model, Cross-validation Model, and Pruned Tree Summary

Figure 8: Output of the Branched Decision Tree Model

Figure 8 Note: Output of the Branched Decision Tree Model, located on the following page, comparing the dependent variable y_i (effect size) to independent variables (v_i : variance, n : population, age, and $ei_measure$). Variables $ei_construct$ and $achievement_type$ were excluded due to low representation. The optimal tree was selected with a complexity parameter = 0.01214359. The tree had an $n = 417$, as 136 observations were deleted due to NaN values. The mean squared error (root node error) = 0.03823. The cross-validation error ($xerror$) = 0.6697. Variance (v_i) was the top split, indicating that it is the most important variable ($v_i \geq 2.6089$).

Figure 8: Branched Decision Tree

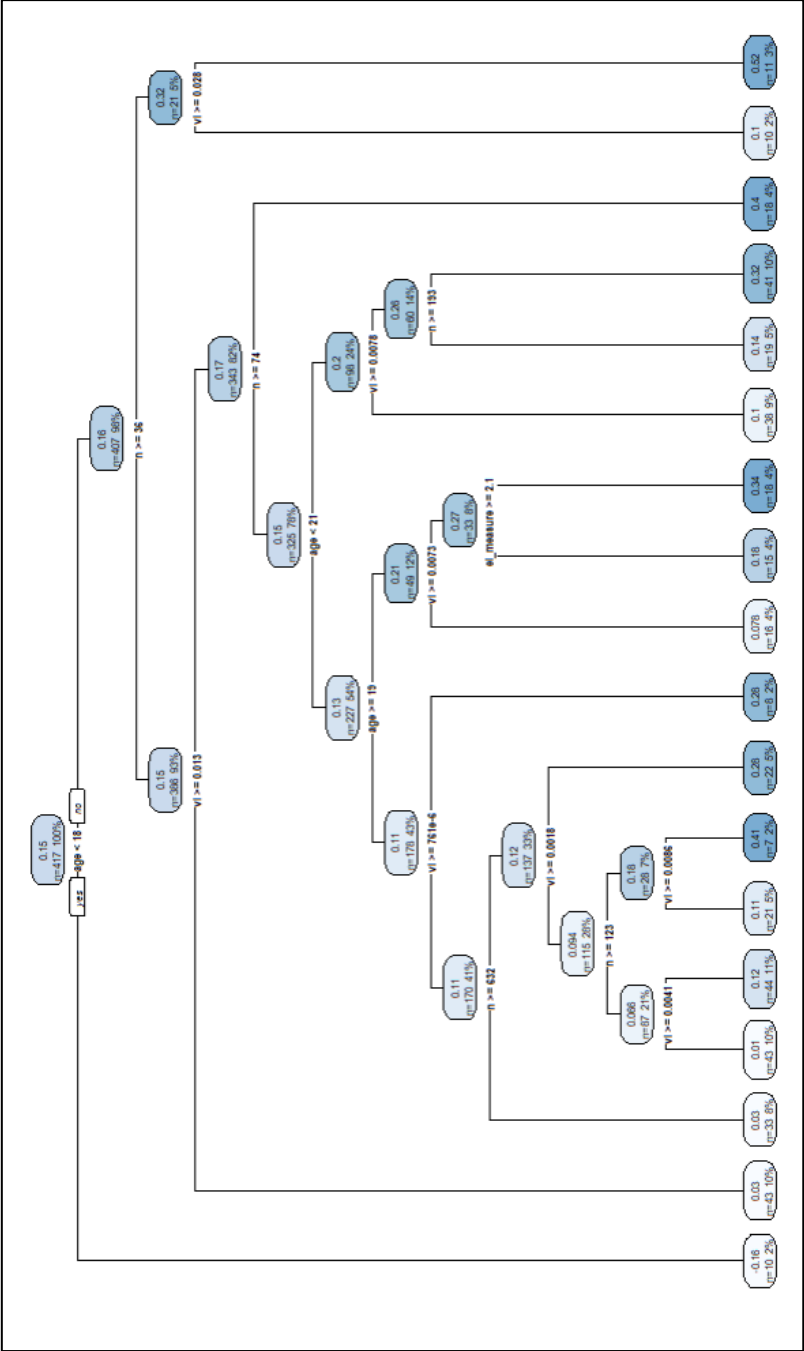


Table 8: Summary of the Branched Decision Tree Model

Branched Regression Tree Results				
Minimum Cross-Validation Error		0.6697429		
Optimal complexity Parameter (CP)		0.01214359		
Variables	vi (variance)	n (population)	age	ei_measure
Root Node Error (15.942/417)		0.03823	n = 417	136 NaN
CP	Nsplit	Relative Error	Xerror	Xstd
0.064967	0	1.00000	1.00250	0.079790
0.051687	1	0.93503	0.96971	0.075752
0.029078	5	0.72828	0.85731	0.092885
0.026732	7	0.67013	0.80278	0.092601
0.023390	8	0.64340	0.76646	0.092016
0.016779	15	0.47813	0.71044	0.089926
0.013258	16	0.46135	0.68670	0.088910
0.012144	17	0.44809	0.66974	0.086760

Table 8 Note: Summary of the Branched Decision Tree Model, located on the previousg page, comparing the dependent variable y_i (effect size) to independent variables (vi : variance, n : population, age, and $ei_measure$). Variables $ei_construct$ and $achievement_type$ were excluded due to low representation. The optimal tree was selected with a complexity parameter = 0.01214359. The tree had an $n = 417$, as 136 observations were deleted due to NaN values. The mean squared error (root node error) = 0.03823. The cross-validation error ($xerror$) = 0.6697. Variance (vi) was tne top split, indicating that it is the most important variable ($vi \geq 2.6089$).

Table 9: Summary of the Cross-validation Model

Variable	Variable Importance
vi (variance)	5.7290335
n (population)	4.3578561
age	2.6134019
ei_measure	0.724815
ei_construct	0.4275492
achievement_type	0.2232582

Table 9: Summary of the Cross-validation Model. Summary of the variables of importance included in the branched decision tree model comparing the dependent variable y_i (effect size) to independent variables (v_i : variance, n : population, age, and $ei_measure$). Variables $ei_construct$ and $achievement_type$ were excluded due to low representation. The optimal tree was constructed using a cross-validation model to decrease cross-validation error. The variables are ranked by importance (most important at the top). Variance (v_i) and population (n) were calculated as being the most important variables in the tree construction.

Appendix B

Full Code

A. Fully Embedded PDF Link



20241212_STAT4600_Final_project_Code_EI_Measures_KB_JK_KS_AZ.html

B. Final Project: Can Emotional Intelligence Predict Academic Performance

Title: Can EI be used as an alternative measure of potential student success

when evaluating admissions for graduate students?

Date: 12/1/2024

Group Members: K. Banu, J. Kawal, K. Shanmugasundaram, and A. Zuger

Course: STAT 4600

Professor: Dr. Jin Fang

Attestation Statement

The code provided was taken from the following sources:

Grolemond, Garrett, and Hadley Wickham. R for Data Science. O'Reilly Media, 2017.

Long, JD, and Paul Teetor. R Cookbook. 2nd ed., O'Reilly Media, 2019. ISBN: 978-1-492-04068-2.

Almog, Uri. "Decision Trees, Explained." Medium, 2 June 2022,
<https://towardsdatascience.com/decision-trees-explained-d7678c43a59e>.

"Decision Tree for Regression in R Programming." GeeksforGeeks, 26 July 2020,
<https://www.geeksforgeeks.org/decision-tree-for-regression-in-r-programming/>.

KB, JK, KS, and AZ

```
# https://topepo.github.io/caret/variable-importance.html

# https://www.datacamp.com/tutorial/decision-trees-R

# https://christophm.github.io/interpretable-ml-book/pdp.html

# https://journal.r-project.org/archive/2017/RJ-2017-016/RJ-2017-016.pdf

# https://www.science.smith.edu/~jcrouser/SDS293/labs/lab14-r.html

# https://stackoverflow.com/questions/39886562/how-to-check-interaction-effects-for-a-lot-of-predictors-in-r

# https://stats.stackexchange.com/questions/278800/predict-using-trained-model-on-dataset

# https://www.geeksforgeeks.org/cross-validation-in-r-programming/

# OpenAI. (2024). ChatGPT (November 2024 version) [Large language model]. Retrieved from
https://chat.openai.com/

# AI was used solely for debugging.
# Part 1: View Data, Trim Data, Calculate Descriptive Statistics.

# Step 1: Downloading Data Package.

# Data upload and visualization

#install.packages("psymetadata")
#library(psymetadata)
```

```

#data("maccann2020")

#setwd("C:/Users/angel/OneDrive/Desktop/Final_code_files")

#write.csv(maccann2020, "maccann2020.csv", row.names = FALSE)

# Note: excluded code after running and downloading the csv file. Could not stitch to html with
code present

# Start here: MccCan 2020 CSV is in the accompanying folder
maccann2020 <-
read.csv("C:\\Users\\angel\\OneDrive\\Desktop\\Final_code_files\\maccann2020.csv")

summary(maccann2020)
##   study_id   sample_id   author      pub_year
## Min.   : 1.0   Min.   : 1.0   Length:1246   Min.   :1997
## 1st Qu.: 61.0  1st Qu.: 64.0   Class :character 1st Qu.:2007
## Median :130.0  Median :143.0   Mode  :character Median :2010
## Mean   :127.3  Mean   :139.2           Mean   :2010
## 3rd Qu.:195.0  3rd Qu.:214.0           3rd Qu.:2013
## Max.   :247.0  Max.   :278.0           Max.   :2019
##
##      yi          vi      pub_type      n
## Min.   :-0.69615   Min.   :0.0003938   Min.   :0.0000   Min.   : 18.0
## 1st Qu.: 0.04664   1st Qu.:0.0026962   1st Qu.:0.0000   1st Qu.: 99.0
## Median : 0.15498   Median :0.0051739   Median :0.0000   Median :180.0
## Mean   : 0.17593   Mean   :0.0072094   Mean   :0.6268   Mean   :256.8
## 3rd Qu.: 0.30657   3rd Qu.:0.0091286   3rd Qu.:1.0000   3rd Qu.:292.8
## Max.   : 0.72471   Max.   :0.0551461   Max.   :4.0000   Max.   :2195.0
##

```

```

## ed_level1 ed_level2 country perc_white
## Min. :0.000 Min. :0.000 Length:1246 Min. : 0.00
## 1st Qu.:1.000 1st Qu.:1.000 Class :character 1st Qu.: 33.90
## Median :2.000 Median :2.000 Mode :character Median : 52.00
## Mean :1.535 Mean :1.451 Mean : 54.62
## 3rd Qu.:2.000 3rd Qu.:2.000 3rd Qu.: 79.00
## Max. :3.000 Max. :3.000 Max. :100.00
## NA's :735
## age perc_female ei_construct ei_stream ei_measure
## Min. : 7.68 Min. : 0.00 Min. :0.00 Min. :1.000 Min. :1.10
## 1st Qu.:15.40 1st Qu.: 51.04 1st Qu.:0.00 1st Qu.:1.000 1st Qu.:1.10
## Median :19.52 Median : 62.00 Median :3.00 Median :3.000 Median :3.10
## Mean :18.76 Mean : 61.31 Mean :3.15 Mean :2.257 Mean :2.35
## 3rd Qu.:21.33 3rd Qu.: 73.03 3rd Qu.:6.00 3rd Qu.:3.000 3rd Qu.:3.10
## Max. :40.00 Max. :100.00 Max. :9.00 Max. :3.000 Max. :3.30
## NA's :1 NA's :173 NA's :229 NA's :224
## subject humanities achievement_type
## Min. : 0.000 Min. :0.0000 Min. :0.0000
## 1st Qu.: 0.000 1st Qu.:0.0000 1st Qu.:0.0000
## Median : 0.000 Median :0.0000 Median :0.0000
## Mean : 1.161 Mean :0.1573 Mean :0.1573
## 3rd Qu.: 2.000 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :10.000 Max. :1.0000 Max. :1.0000
## NA's :11

```

```
# Step 2: Install packages.
```

```
# Install Packages
```

```
#Manual CRAN mirror
```

```
options(repos = c(CRAN = "https://cloud.r-project.org/"))
```

```
# Fix for html
```

```

unlink("C:/Users/angel/AppData/Local/R/win-library/4.4/00LOCK", recursive = TRUE)
install.packages("rpart")
## Installing package into 'C:/Users/angel/AppData/Local/R/win-library/4.4'
## (as 'lib' is unspecified)
## package 'rpart' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\angel\AppData\Local\Temp\RtmpEjErRu\downloaded_packages
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(ggplot2)
library(corrplot)
## corrplot 0.95 loaded
library(tidyr)
library(kableExtra)
##
## Attaching package: 'kableExtra'
## The following object is masked from 'package:dplyr':
##
##   group_rows
library(formattable)
library(rlang)

# Important Data Before Trim

```

```
# Publication dates:1997-2019
# Age of respondents: Min 7,Max 40, mean = 18.76
# countries represented: Kuala Lumpur, Malaysia, USA, Nigeria, and Italy
# respondents: 77% white
# respondents: 62% female
```

```
# Step 3: Read McCann Data into a data frame.
```

```
EI_AP_Data <- data.frame(maccann2020)
```

```
# Descriptive Statistics
```

```
head(EI_AP_Data)
```

```
## study_id sample_id author
## 1 1 1 Abdo
## 2 2 2 Abdullah, Habibah, Mahyuddin, & Uli
## 3 3 3 Abel
## 4 5 5 Adeyemo
## 5 6 6 Afolabi, Okediji, Ogunmwonji
## 6 7 7 Agnoli, Mancini, Pozzoli, Baldaro, Russo & Surcinelli
## pub_year yi vi pub_type n ed_level1 ed_level2
## 1 2012 0.2183201 0.003422432 1 266 1 1
## 2 2004 0.3003224 0.004057587 0 205 1 1
## 3 2013 0.1376696 0.012499394 1 78 1 1
## 4 2007 0.4101488 0.002313895 0 300 2 2
## 5 2009 0.2933921 0.007662860 0 110 2 2
## 6 2012 0.1771894 0.002672916 0 352 0 0
## country perc_white age perc_female ei_construct ei_stream
## 1 USA 77 17.28 62.00 0 3
## 2 Kuala Lumpur & Malaysia NA 16.50 NA 0 1
## 3 USA 0 17.47 63.30 0 3
```

```

## 4      Nigeria      NA 19.40   53.33      0      2
## 5      Nigeria      NA 21.55   41.80      0      3
## 6      Italy       NA  9.35   53.41      0      3
## ei_measure subject humanities achievement_type
## 1      3.1      0      0      0
## 2      1.2      0      0      0
## 3      3.2      0      0      0
## 4      2.2      6      0      0
## 5      NA       0      0      0
## 6      3.2      1      0      0
str(EI_AP_Data)
## 'data.frame':  1246 obs. of  20 variables:
## $ study_id      : int  1 2 3 5 6 7 7 8 8 8 ...
## $ sample_id     : int  1 2 3 5 6 7 7 8 8 8 ...
## $ author        : chr  "Abdo" "Abdullah, Habibah, Mahyuddin, & Uli" "Abel" "Adeyemo" ...
## $ pub_year      : int  2012 2004 2013 2007 2009 2012 2012 2011 2011 2011 ...
## $ yi           : num  0.218 0.3 0.138 0.41 0.293 ...
## $ vi           : num  0.00342 0.00406 0.0125 0.00231 0.00766 ...
## $ pub_type      : int  1 0 1 0 0 0 0 0 0 0 ...
## $ n            : int  266 205 78 300 110 352 352 204 204 204 ...
## $ ed_level1     : int  1 1 1 2 2 0 0 2 2 2 ...
## $ ed_level2     : int  1 1 1 2 2 0 0 2 2 2 ...
## $ country       : chr  "USA" "Kuala Lumpur & Malaysia" "USA" "Nigeria" ...
## $ perc_white    : num  77 NA 0 NA NA NA NA NA NA NA ...
## $ age          : num  17.3 16.5 17.5 19.4 21.6 ...
## $ perc_female   : num  62 NA 63.3 53.3 41.8 ...
## $ ei_construct  : int  0 0 0 0 0 0 0 2 3 0 ...
## $ ei_stream     : int  3 1 3 2 3 3 3 1 1 1 ...
## $ ei_measure    : num  3.1 1.2 3.2 2.2 NA 3.2 3.2 1.1 1.1 1.1 ...
## $ subject       : int  0 0 0 6 0 1 2 0 0 0 ...
## $ humanities    : int  0 0 0 0 0 0 0 0 0 0 ...

```



```

## $ achievement_type: int 0 0 0 0 0 0 0 0 0 ...
summary(EI_AP_Data)
##   study_id   sample_id   author       pub_year
## Min.   : 1.0   Min.   : 1.0   Length:1246   Min.   :1997
## 1st Qu.: 61.0   1st Qu.: 64.0   Class :character   1st Qu.:2007
## Median :130.0   Median :143.0   Mode  :character   Median :2010
## Mean   :127.3   Mean   :139.2           Mean   :2010
## 3rd Qu.:195.0   3rd Qu.:214.0           3rd Qu.:2013
## Max.   :247.0   Max.   :278.0           Max.   :2019
##
##   yi          vi          pub_type      n
## Min.   :-0.69615   Min.   :0.0003938   Min.   :0.0000   Min.   : 18.0
## 1st Qu.: 0.04664   1st Qu.:0.0026962   1st Qu.:0.0000   1st Qu.: 99.0
## Median : 0.15498   Median :0.0051739   Median :0.0000   Median :180.0
## Mean   : 0.17593   Mean   :0.0072094   Mean   :0.6268   Mean   :256.8
## 3rd Qu.: 0.30657   3rd Qu.:0.0091286   3rd Qu.:1.0000   3rd Qu.:292.8
## Max.   : 0.72471   Max.   :0.0551461   Max.   :4.0000   Max.   :2195.0
##
##   ed_level1   ed_level2   country       perc_white
## Min.   :0.000   Min.   :0.000   Length:1246   Min.   : 0.00
## 1st Qu.:1.000   1st Qu.:1.000   Class :character   1st Qu.: 33.90
## Median :2.000   Median :2.000   Mode  :character   Median : 52.00
## Mean   :1.535   Mean   :1.451           Mean   : 54.62
## 3rd Qu.:2.000   3rd Qu.:2.000           3rd Qu.: 79.00
## Max.   :3.000   Max.   :3.000           Max.   :100.00
##
##                                     NA's :735
##   age   perc_female   ei_construct   ei_stream   ei_measure
## Min.   : 7.68   Min.   : 0.00   Min.   :0.00   Min.   :1.000   Min.   :1.10
## 1st Qu.:15.40   1st Qu.: 51.04   1st Qu.:0.00   1st Qu.:1.000   1st Qu.:1.10
## Median :19.52   Median : 62.00   Median :3.00   Median :3.000   Median :3.10
## Mean   :18.76   Mean   : 61.31   Mean   :3.15   Mean   :2.257   Mean   :2.35

```

```
## 3rd Qu.:21.33 3rd Qu.: 73.03 3rd Qu.:6.00 3rd Qu.:3.000 3rd Qu.:3.10
## Max. :40.00 Max. :100.00 Max. :9.00 Max. :3.000 Max. :3.30
## NA's :1 NA's :173 NA's :229 NA's :224
## subject humanities achievement_type
## Min. :0.000 Min. :0.0000 Min. :0.0000
## 1st Qu.: 0.000 1st Qu.:0.0000 1st Qu.:0.0000
## Median : 0.000 Median :0.0000 Median :0.0000
## Mean : 1.161 Mean :0.1573 Mean :0.1573
## 3rd Qu.: 2.000 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :10.000 Max. :1.0000 Max. :1.0000
## NA's :11
# Variance and standard error calculated for quantitative variables
```

```
var(EI_AP_Data$pub_year, na.rm = TRUE)
## [1] 19.28302
sd(EI_AP_Data$pub_year, na.rm = TRUE)
## [1] 4.391243
var(EI_AP_Data$yi, na.rm = TRUE)
## [1] 0.04440424
sd(EI_AP_Data$yi, na.rm = TRUE)
## [1] 0.2107231
var(EI_AP_Data$vi, na.rm = TRUE)
## [1] 4.59168e-05
sd(EI_AP_Data$vi, na.rm = TRUE)
## [1] 0.006776193
var(EI_AP_Data$n, na.rm = TRUE)
## [1] 95577.25
sd(EI_AP_Data$n, na.rm = TRUE)
## [1] 309.1557
var(EI_AP_Data$perc_white, na.rm = TRUE)
## [1] 763.3892
```

```

sd(EI_AP_Data$perc_white, na.rm = TRUE)
## [1] 27.6295
var(EI_AP_Data$age, na.rm = TRUE)
## [1] 25.75317
sd(EI_AP_Data$age, na.rm = TRUE)
## [1] 5.074759
var(EI_AP_Data$perc_female, na.rm = TRUE)
## [1] 476.402
sd(EI_AP_Data$perc_female, na.rm = TRUE)
## [1] 21.82663
# Visualizations of Descriptive Statistics

# Visualization of outliers for all variables in the `vars` list using boxplots
# Assuming you have a list of variable names in the `vars` list
vars <- c("pub_year", "yi", "vi", "n", "perc_white", "age", "perc_female")

# Check for non-finite values in each variable
summary(EI_AP_Data[, vars])
##   pub_year      yi          vi          n
## Min.   :1997  Min.   :-0.69615  Min.   :0.0003938  Min.   : 18.0
## 1st Qu.:2007  1st Qu.: 0.04664  1st Qu.:0.0026962  1st Qu.: 99.0
## Median :2010  Median : 0.15498  Median :0.0051739  Median : 180.0
## Mean   :2010  Mean   : 0.17593  Mean   :0.0072094  Mean   : 256.8
## 3rd Qu.:2013  3rd Qu.: 0.30657  3rd Qu.:0.0091286  3rd Qu.: 292.8
## Max.   :2019  Max.   : 0.72471  Max.   :0.0551461  Max.   :2195.0
##
##   perc_white      age      perc_female
## Min.   : 0.00  Min.   : 7.68  Min.   : 0.00
## 1st Qu.: 33.90  1st Qu.:15.40  1st Qu.: 51.04
## Median : 52.00  Median :19.52  Median : 62.00

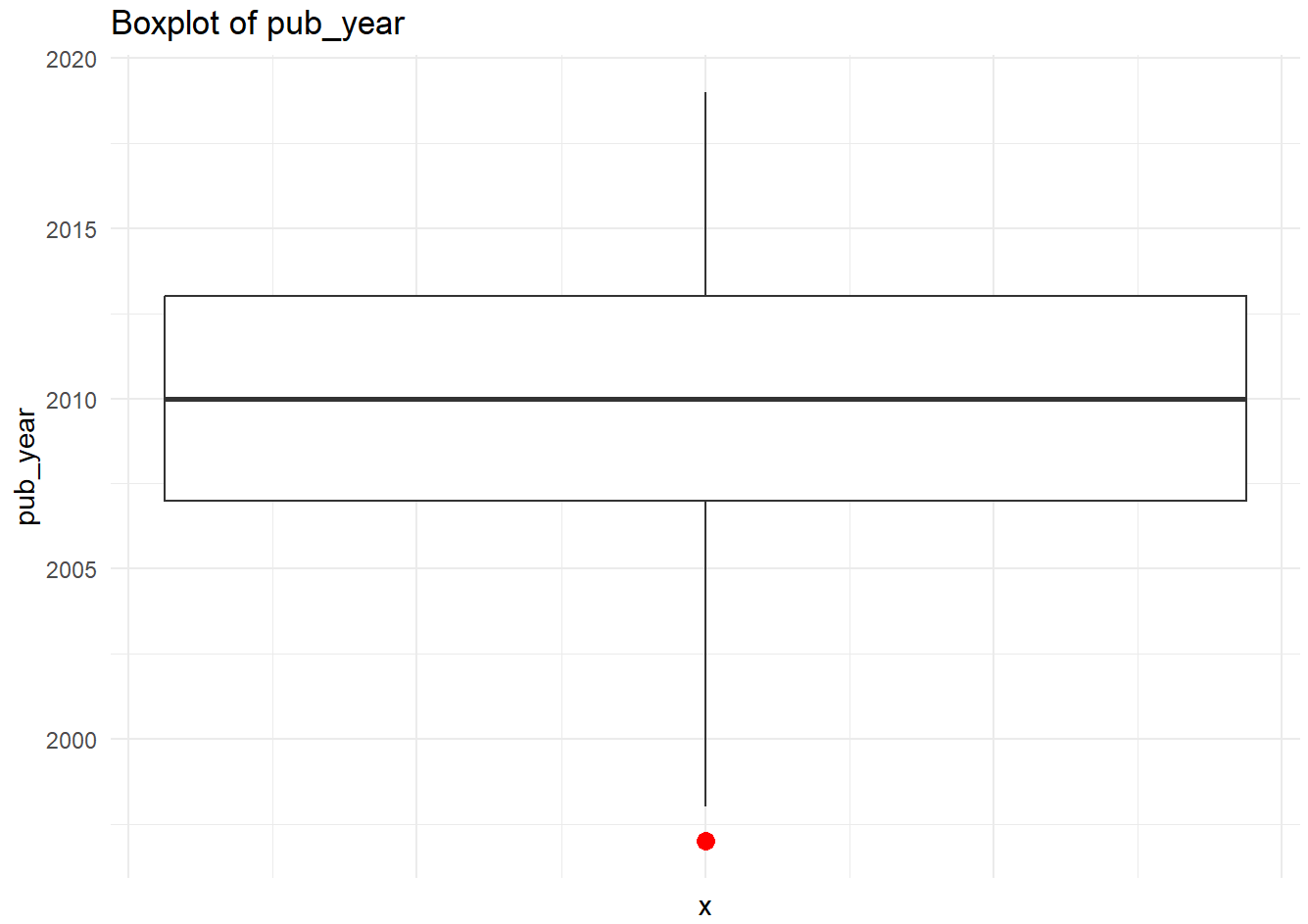
```

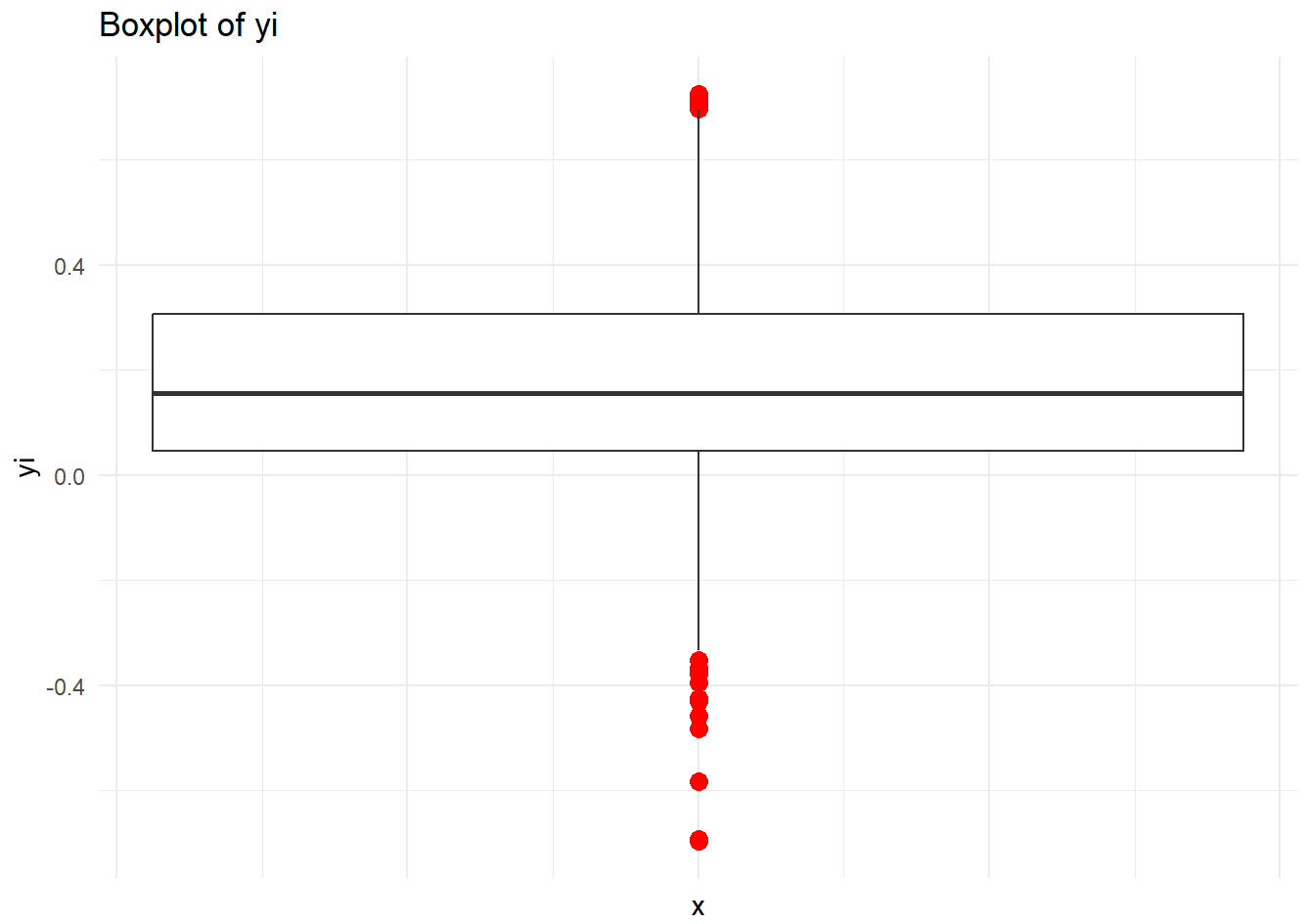
```

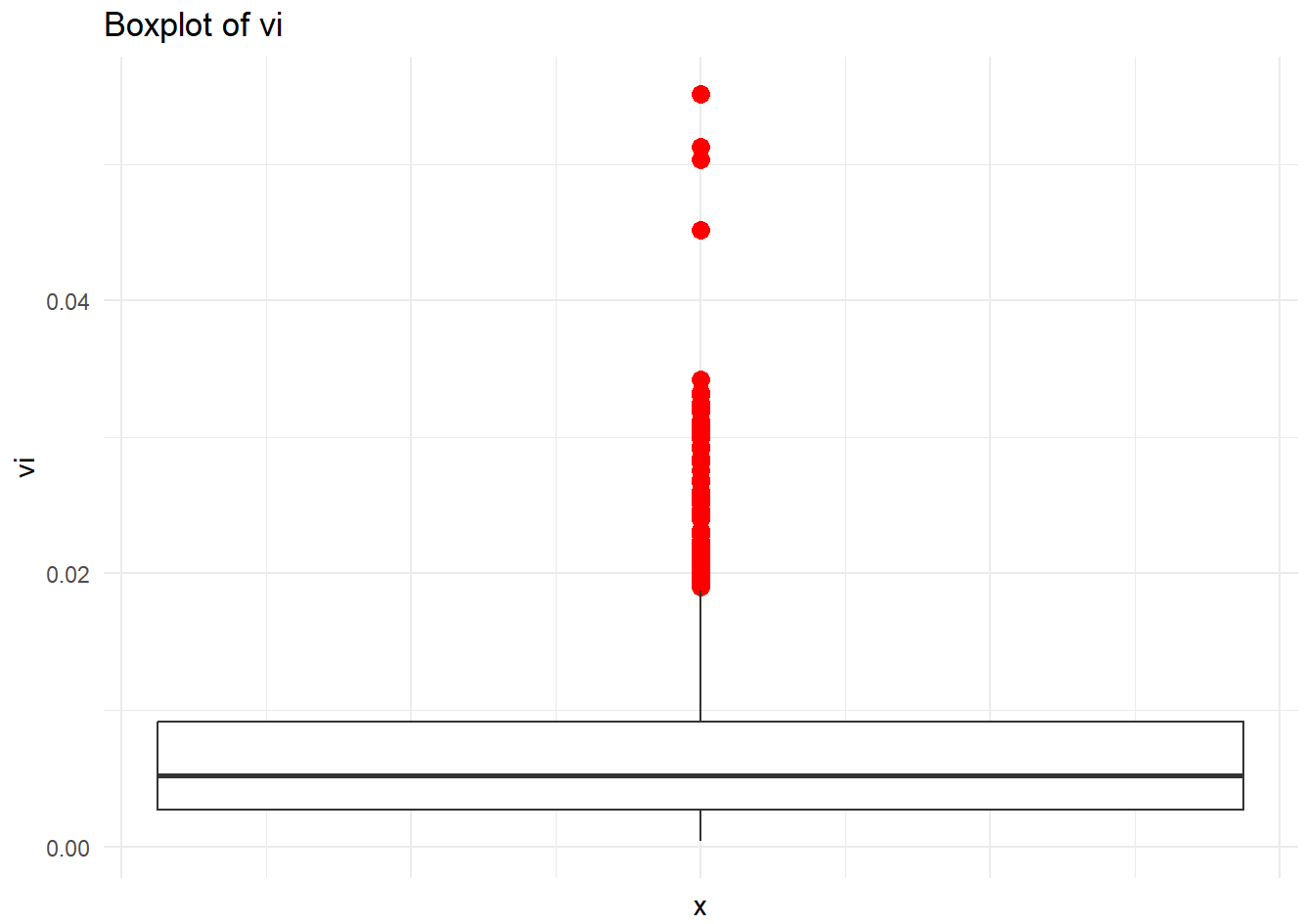
## Mean : 54.62 Mean :18.76 Mean : 61.31
## 3rd Qu.: 79.00 3rd Qu.:21.33 3rd Qu.: 73.03
## Max. :100.00 Max. :40.00 Max. :100.00
## NA's :735 NA's :1 NA's :173
# Alternatively, check the specific variables for NAs
sapply(EI_AP_Data[, vars], function(x) sum(is.na(x)))
## pub_year yi vi n perc_white age
## 0 0 0 0 735 1
## perc_female
## 173
for (var in vars) {
  p <- ggplot(EI_AP_Data, aes(x = 1, y = !!sym(var))) +
    geom_boxplot(outlier.colour = "red", outlier.size = 3) +
    labs(title = paste("Boxplot of", var), y = var) +
    theme_minimal() +
    theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())

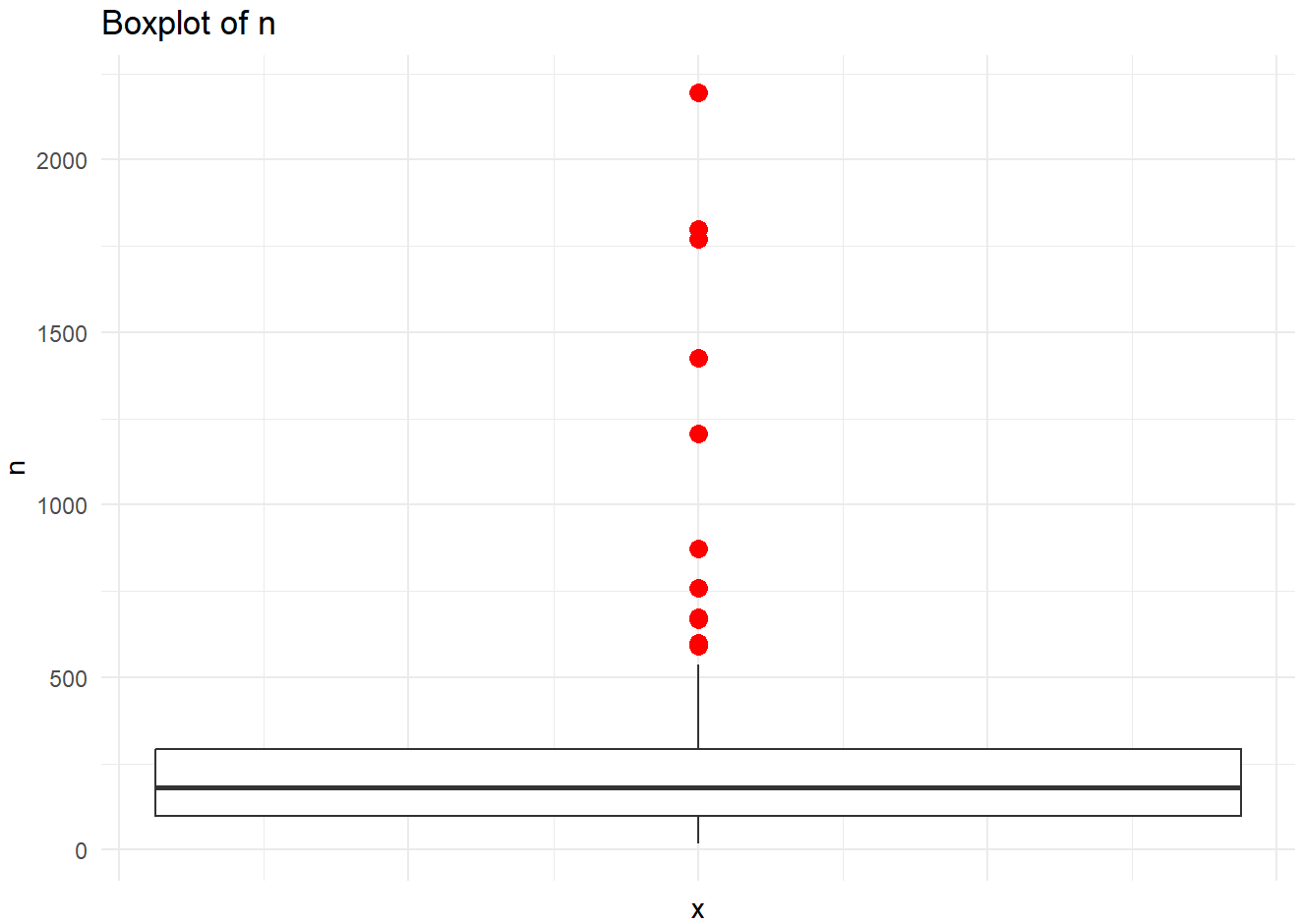
  # Print the plot
  print(p)
}

```

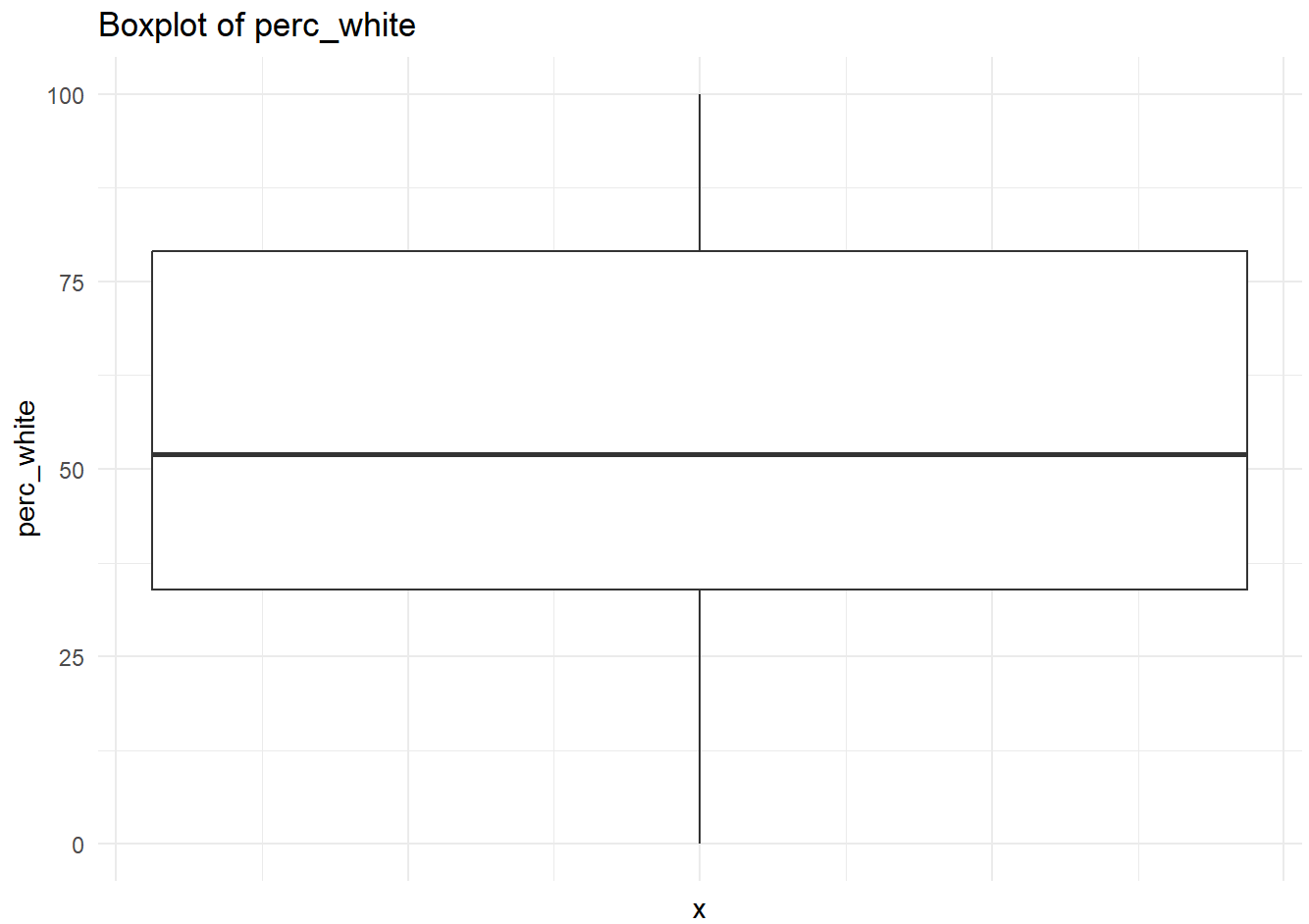




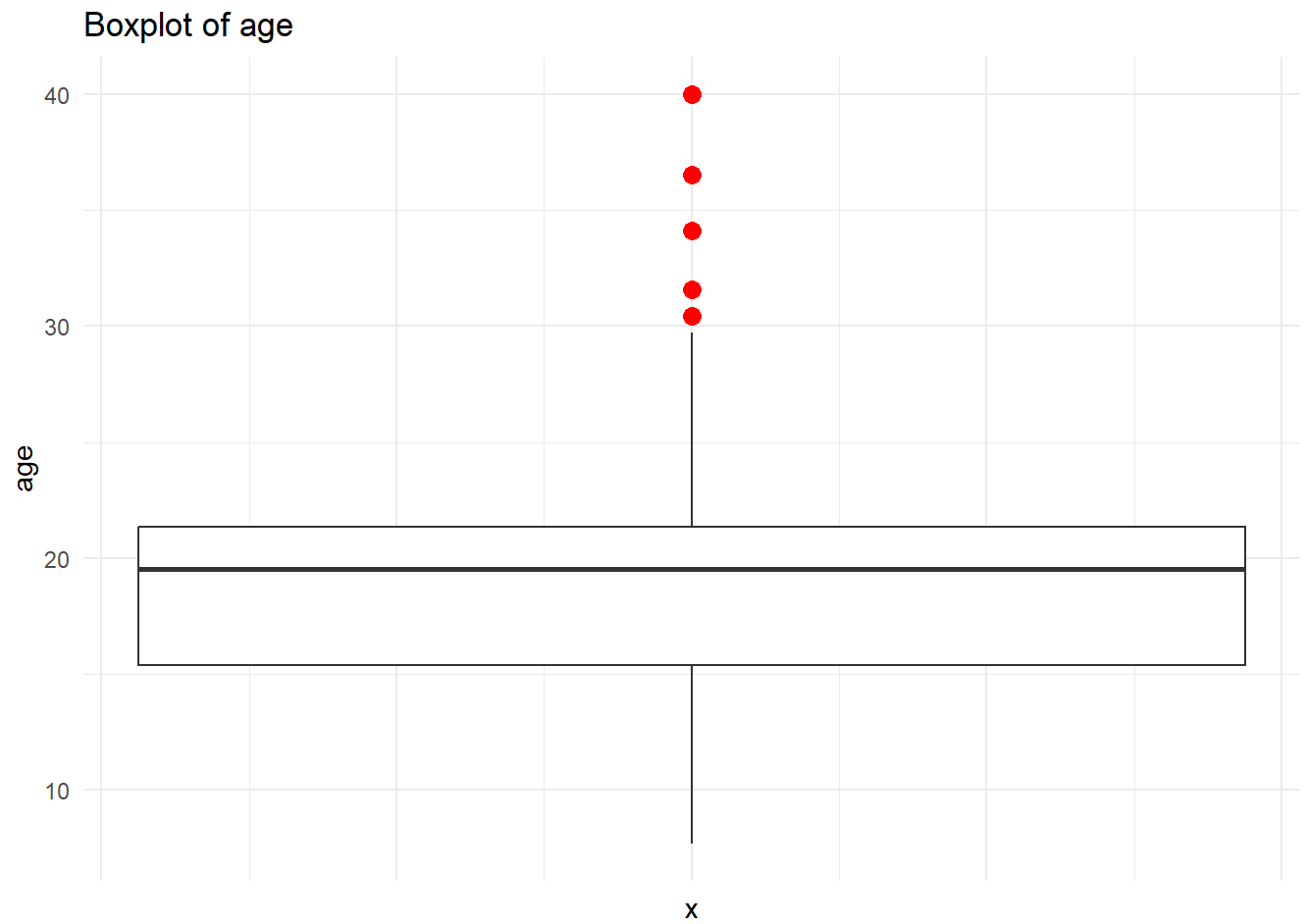




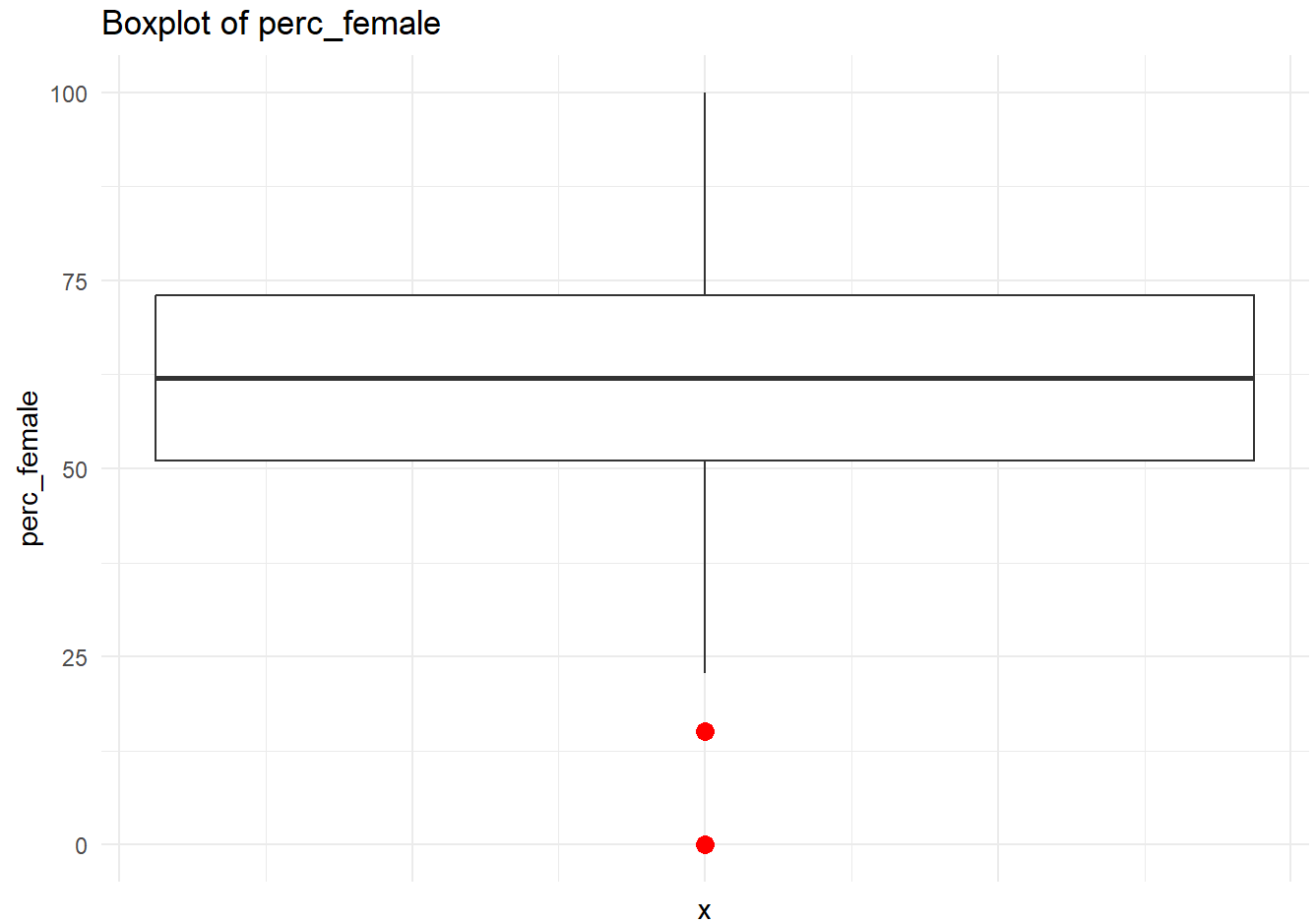
```
## Warning: Removed 735 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 1 row containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



```
## Warning: Removed 173 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Proportion tables Created for Categorical Variables

Country

```
length(EI_AP_Data$country)
```

```
## [1] 1246
```

```
sum(is.na(EI_AP_Data$country))
```

```
## [1] 0
```

```
country_data_o <- EI_AP_Data$country[!is.na(EI_AP_Data$country)]
```

```
prop_table_o1 <- prop.table(table(country_data_o)) * 100
```

```
prop_table_o1 <- round(prop_table_o1, 2)
```

```
prop_df1 <- data.frame(Category = names(prop_table_o1), Percentage = prop_table_o1)
```

```
print(prop_df1)
```

KB, JK, KS, and AZ

##	Category	Percentage.country_data_o	Percentage.Freq
## 1	Australia	Australia	7.95
## 2	Bahrain	Bahrain	0.48
## 3	Barbados	Barbados	0.48
## 4	Belgium	Belgium	0.08
## 5	Canada	Canada	4.17
## 6	China	China	0.24
## 7	Croatia	Croatia	0.24
## 8	Cyprus	Cyprus	0.08
## 9	Egypt	Egypt	1.61
## 10	Europe	Europe	0.48
## 11	India	India	0.24
## 12	Iran	Iran	5.30
## 13	Ireland	Ireland	1.61
## 14	Israel	Israel	0.16
## 15	Italy	Italy	1.28
## 16	Kuala Lumpur & Malaysia	Kuala Lumpur & Malaysia	0.08
## 17	Malaysia	Malaysia	1.44
## 18	Nigeria	Nigeria	0.40
## 19	Pakistan	Pakistan	0.96
## 20	Portugual	Portugual	4.33
## 21	Romania	Romania	0.16
## 22	Saudi Arabia	Saudi Arabia	0.16
## 23	Slovak Republic	Slovak Republic	0.80
## 24	Spain	Spain	2.73
## 25	Turkey	Turkey	0.64
## 26	UAE	UAE	0.56
## 27	UK	UK	8.27
## 28	UK or Canada	UK or Canada	7.22
## 29	USA	USA	43.90
## 30	USA & Canada	USA & Canada	3.85

```

## 31      USA, Canada      USA, Canada      0.08
# Publication Type
length(EI_AP_Data$pub_type)
## [1] 1246
sum(is.na(EI_AP_Data$pub_type))
## [1] 0
pub_type_data_o <- EI_AP_Data$pub_type[!is.na(EI_AP_Data$pub_type)]
prop_table_o2 <- prop.table(table(pub_type_data_o)) * 100
prop_table_o2 <- round(prop_table_o2, 2)
prop_df2 <- data.frame(Category = names(prop_table_o2), Percentage = prop_table_o2)
print(prop_df2)
##  Category Percentage.pub_type_data_o Percentage.Freq
## 1      0              0          54.90
## 2      1              1          38.20
## 3      3              3           3.13
## 4      4              4           3.77
# Publication Year
pub_year_data_o <- EI_AP_Data$pub_year[!is.na(EI_AP_Data$pub_year)]
prop_table_o3 <- prop.table(table(pub_year_data_o)) * 100
prop_table_o3 <- round(prop_table_o3, 2)
prop_df3 <- data.frame(Category = names(prop_table_o3), Percentage = prop_table_o3)
print(prop_df3)
##  Category Percentage.pub_year_data_o Percentage.Freq
## 1    1997              1997          0.08
## 2    1998              1998          0.08
## 3    2000              2000          1.12
## 4    2002              2002          3.05
## 5    2003              2003          8.43
## 6    2004              2004          4.09
## 7    2005              2005          4.09
## 8    2006              2006          2.89

```

```
## 9    2007          2007      7.54
## 10   2008          2008      8.03
## 11   2009          2009      8.03
## 12   2010          2010      5.38
## 13   2011          2011      5.38
## 14   2012          2012     14.53
## 15   2013          2013      6.82
## 16   2014          2014      6.02
## 17   2015          2015      7.54
## 18   2016          2016      1.61
## 19   2017          2017      1.52
## 20   2018          2018      3.53
## 21   2019          2019      0.24
```

```
#Education Level 1
```

```
length(EI_AP_Data$ed_level1)
```

```
## [1] 1246
```

```
sum(is.na(EI_AP_Data$ed_level1))
```

```
## [1] 0
```

```
ed_level1_data_o <- EI_AP_Data$ed_level1[!is.na(EI_AP_Data$ed_level1)]
```

```
prop_table_o4 <- prop.table(table(ed_level1_data_o)) * 100
```

```
prop_table_o4 <- round(prop_table_o4, 2)
```

```
prop_df4 <- data.frame(Category = names(prop_table_o4), Percentage = prop_table_o4)
```

```
print(prop_df4)
```

```
##   Category Percentage.ed_level1_data_o Percentage.Freq
```

```
## 1      0              0          7.30
```

```
## 2      1              1         32.34
```

```
## 3      2              2         59.87
```

```
## 4      3              3          0.48
```

```
# EI Stream
```

```
length(EI_AP_Data$ei_stream)
```

```
## [1] 1246
```

```

sum(is.na(EI_AP_Data$ei_stream))
## [1] 0
ei_stream_data_o <- EI_AP_Data$ei_stream[!is.na(EI_AP_Data$ei_stream)]
prop_table_o5 <- prop.table(table(ei_stream_data_o)) * 100
prop_table_o5 <- round(prop_table_o5, 2)
prop_df5 <- data.frame(Category = names(prop_table_o5), Percentage = prop_table_o5)
print(prop_df5)
##  Category Percentage.ei_stream_data_o Percentage.Freq
## 1      1              1      30.74
## 2      2              2      12.84
## 3      3              3      56.42
# EI Measure
length(EI_AP_Data$ei_measure)
## [1] 1246
sum(is.na(EI_AP_Data$ei_measure))
## [1] 224
ei_measure_data_o <- EI_AP_Data$ei_measure[!is.na(EI_AP_Data$ei_measure)]
prop_table_o6 <- prop.table(table(ei_measure_data_o)) * 100
prop_table_o6 <- round(prop_table_o6, 2)
prop_df6 <- data.frame(Category = names(prop_table_o6), Percentage = prop_table_o6)
print(prop_df6)
##  Category Percentage.ei_measure_data_o Percentage.Freq
## 1      1.1              1.1      28.57
## 2      1.2              1.2       1.57
## 3      1.3              1.3       1.08
## 4      1.4              1.4       0.68
## 5      1.5              1.5       1.08
## 6      2.1              2.1      6.56
## 7      2.2              2.2       2.74
## 8      2.3              2.3       1.76
## 9      2.4              2.4       1.27

```

```

## 10    2.5            2.5      1.66
## 11    3.1            3.1     39.33
## 12    3.2            3.2      5.28
## 13    3.3            3.3      8.41

# Achievement Type
length(EI_AP_Data$achievement_type)
## [1] 1246
sum(is.na(EI_AP_Data$achievement_type))
## [1] 0
achieve_type_data_o <-
EI_AP_Data$achievement_type[!is.na(EI_AP_Data$achievement_type)]
prop_table_o7 <- prop.table(table(achieve_type_data_o)) * 100
prop_table_o7 <- round(prop_table_o7, 2)
prop_df7 <- data.frame(Category = names(prop_table_o7), Percentage = prop_table_o7)
print(prop_df7)
##   Category Percentage.achieve_type_data_o Percentage.Freq
## 1      0            0      84.27
## 2      1            1     15.73

# Part 2: Trim Data Part 1-> By education level
# filter df by column value, education level 1 = 2 (graduate)

library(dplyr)
summary(EI_AP_Data)
##   study_id   sample_id   author      pub_year
## Min.   : 1.0   Min.   : 1.0   Length:1246   Min.   :1997
## 1st Qu.: 61.0  1st Qu.: 64.0   Class :character 1st Qu.:2007
## Median :130.0  Median :143.0   Mode  :character  Median :2010
## Mean   :127.3  Mean   :139.2           Mean   :2010
## 3rd Qu.:195.0  3rd Qu.:214.0           3rd Qu.:2013
## Max.   :247.0  Max.   :278.0           Max.   :2019
##

```



```

##      yi          vi          pub_type          n
## Min. :-0.69615 Min. :0.0003938 Min. :0.0000 Min. : 18.0
## 1st Qu.: 0.04664 1st Qu.:0.0026962 1st Qu.:0.0000 1st Qu.: 99.0
## Median : 0.15498 Median :0.0051739 Median :0.0000 Median :180.0
## Mean : 0.17593 Mean :0.0072094 Mean :0.6268 Mean :256.8
## 3rd Qu.: 0.30657 3rd Qu.:0.0091286 3rd Qu.:1.0000 3rd Qu.:292.8
## Max. : 0.72471 Max. :0.0551461 Max. :4.0000 Max. :2195.0
##
##      ed_level1      ed_level2      country      perc_white
## Min. :0.000 Min. :0.000 Length:1246 Min. : 0.00
## 1st Qu.:1.000 1st Qu.:1.000 Class :character 1st Qu.: 33.90
## Median :2.000 Median :2.000 Mode :character Median : 52.00
## Mean :1.535 Mean :1.451 Mean : 54.62
## 3rd Qu.:2.000 3rd Qu.:2.000 3rd Qu.: 79.00
## Max. :3.000 Max. :3.000 Max. :100.00
##
##                                     NA's :735
##      age      perc_female      ei_construct      ei_stream      ei_measure
## Min. : 7.68 Min. : 0.00 Min. :0.00 Min. :1.000 Min. :1.10
## 1st Qu.:15.40 1st Qu.: 51.04 1st Qu.:0.00 1st Qu.:1.000 1st Qu.:1.10
## Median :19.52 Median : 62.00 Median :3.00 Median :3.000 Median :3.10
## Mean :18.76 Mean : 61.31 Mean :3.15 Mean :2.257 Mean :2.35
## 3rd Qu.:21.33 3rd Qu.: 73.03 3rd Qu.:6.00 3rd Qu.:3.000 3rd Qu.:3.10
## Max. :40.00 Max. :100.00 Max. :9.00 Max. :3.000 Max. :3.30
## NA's :1 NA's :173 NA's :229 NA's :224
##      subject      humanities      achievement_type
## Min. : 0.000 Min. :0.0000 Min. :0.0000
## 1st Qu.: 0.000 1st Qu.:0.0000 1st Qu.:0.0000
## Median : 0.000 Median :0.0000 Median :0.0000
## Mean : 1.161 Mean :0.1573 Mean :0.1573
## 3rd Qu.: 2.000 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :10.000 Max. :1.0000 Max. :1.0000

```

```

## NA's :11
filtered_mccann1 <- EI_AP_Data[EI_AP_Data$ed_level1 == 2, ]
summary(filtered_mccann1)
##  study_id    sample_id    author      pub_year
## Min.   : 5.0   Min.   : 5.00   Length:746    Min.   :1997
## 1st Qu.: 67.0   1st Qu.: 69.25   Class :character 1st Qu.:2005
## Median :153.0   Median :167.00   Mode  :character Median :2010
## Mean   :135.6   Mean   :148.19           Mean   :2009
## 3rd Qu.:195.0   3rd Qu.:220.00           3rd Qu.:2013
## Max.   :247.0   Max.   :278.00           Max.   :2019
##
##    yi          vi          pub_type      n
## Min.   :-0.69319   Min.   :0.0003938   Min.   :0.000   Min.   : 18.0
## 1st Qu.: 0.02799   1st Qu.:0.0026885   1st Qu.:0.000   1st Qu.: 85.0
## Median : 0.13076   Median :0.0054171   Median :0.000   Median :163.0
## Mean   : 0.14799   Mean   :0.0079933   Mean   :0.756   Mean   :287.5
## 3rd Qu.: 0.27544   3rd Qu.:0.0103272   3rd Qu.:1.000   3rd Qu.:306.0
## Max.   : 0.71828   Max.   :0.0551461   Max.   :4.000   Max.   :2195.0
##
##  ed_level1 ed_level2    country      perc_white
## Min.   :2   Min.   :1.000   Length:746    Min.   : 0.00
## 1st Qu.:2   1st Qu.:2.000   Class :character 1st Qu.: 38.40
## Median :2   Median :2.000   Mode  :character Median : 52.00
## Mean   :2   Mean   :1.862           Mean   : 56.73
## 3rd Qu.:2   3rd Qu.:2.000           3rd Qu.: 78.00
## Max.   :2   Max.   :2.000           Max.   :100.00
##
##                                NA's :409
##  age      perc_female   ei_construct   ei_stream
## Min.   : 7.68   Min.   : 0.00   Min.   :0.000   Min.   :1.000
## 1st Qu.:19.90   1st Qu.: 56.97   1st Qu.:0.000   1st Qu.:1.000
## Median :21.33   Median : 66.22   Median :3.000   Median :3.000

```

```
## Mean :22.06 Mean : 63.87 Mean :3.125 Mean :2.221
## 3rd Qu.:23.00 3rd Qu.: 73.62 3rd Qu.:5.000 3rd Qu.:3.000
## Max. :40.00 Max. :100.00 Max. :9.000 Max. :3.000
## NA's :1 NA's :157 NA's :179
## ei_measure subject humanities achievement_type
## Min. :1.100 Min. :0.0000 Min. :0.0000 Min. :0.0000
## 1st Qu.:1.100 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:0.0000
## Median :2.500 Median :0.0000 Median :0.0000 Median :0.0000
## Mean :2.253 Mean :0.8587 Mean :0.1501 Mean :0.1501
## 3rd Qu.:3.100 3rd Qu.:0.0000 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :3.300 Max. :7.0000 Max. :1.0000 Max. :1.0000
## NA's :136 NA's :10
# 746 available samples at the graduate level
```

2. Summary Statistics

```
head(filtered_mccann1)
```

```
## study_id sample_id author pub_year yi
## 4 5 5 Adeyemo 2007 0.4101488
## 5 6 6 Afolabi, Okediji, Ogunmwonji 2009 0.2933921
## 8 8 8 Ahammed, Abdullah, & Hassane 2011 -0.1367060
## 9 8 8 Ahammed, Abdullah, & Hassane 2011 -0.1150284
## 10 8 8 Ahammed, Abdullah, & Hassane 2011 -0.1129918
## 11 8 8 Ahammed, Abdullah, & Hassane 2011 -0.1047417
## vi pub_type n ed_level1 ed_level2 country perc_white age
## 4 0.002313895 0 300 2 2 Nigeria NA 19.40
## 5 0.007662860 0 110 2 2 Nigeria NA 21.55
## 8 0.004743705 0 204 2 2 UAE NA 20.50
## 9 0.004796611 0 204 2 2 UAE NA 20.50
## 10 0.004801127 0 204 2 2 UAE NA 20.50
## 11 0.004818614 0 204 2 2 UAE NA 20.50
## perc_female ei_construct ei_stream ei_measure subject humanities
```

```

## 4    53.33      0    2    2.2    6    0
## 5    41.80      0    3     NA    0    0
## 8    65.20      2    1    1.1    0    0
## 9    65.20      3    1    1.1    0    0
## 10   65.20      0    1    1.1    0    0
## 11   65.20     NA    1    1.1    0    0
## achievement_type
## 4      0
## 5      0
## 8      0
## 9      0
## 10     0
## 11     0
str(filtered_mccann1)
## 'data.frame':  746 obs. of  20 variables:
## $ study_id      : int  5 6 8 8 8 8 8 8 8 9 ...
## $ sample_id     : int  5 6 8 8 8 8 8 8 8 9 ...
## $ author        : chr  "Adeyemo" "Afolabi, Okediji, Ogunmwonji" "Ahammed, Abdullah, &
Hassane" "Ahammed, Abdullah, & Hassane" ...
## $ pub_year      : int  2007 2009 2011 2011 2011 2011 2011 2011 2011 2011 ...
## $ yi           : num  0.41 0.293 -0.137 -0.115 -0.113 ...
## $ vi           : num  0.00231 0.00766 0.00474 0.0048 0.0048 ...
## $ pub_type      : int  0 0 0 0 0 0 0 0 0 0 ...
## $ n            : int  300 110 204 204 204 204 204 204 204 291 ...
## $ ed_level1     : int  2 2 2 2 2 2 2 2 2 2 ...
## $ ed_level2     : int  2 2 2 2 2 2 2 2 2 2 ...
## $ country       : chr  "Nigeria" "Nigeria" "UAE" "UAE" ...
## $ perc_white    : num  NA NA NA NA NA NA NA NA NA NA ...
## $ age          : num  19.4 21.6 20.5 20.5 20.5 ...
## $ perc_female   : num  53.3 41.8 65.2 65.2 65.2 ...
## $ ei_construct  : int  0 0 2 3 0 NA NA 1 4 7 ...

```

```

## $ ei_stream      : int 2 3 1 1 1 1 1 1 1 3 ...
## $ ei_measure     : num 2.2 NA 1.1 1.1 1.1 1.1 1.1 1.1 1.1 3.1 ...
## $ subject        : int 6 0 0 0 0 0 0 0 0 0 ...
## $ humanities     : int 0 0 0 0 0 0 0 0 0 0 ...
## $ achievement_type: int 0 0 0 0 0 0 0 0 0 0 ...
attach(filtered_mccann1)
nrow(filtered_mccann1)
## [1] 746
ncol(filtered_mccann1)
## [1] 20
summary(filtered_mccann1)
##   study_id   sample_id   author      pub_year
## Min.   : 5.0   Min.   : 5.00   Length:746   Min.   :1997
## 1st Qu.: 67.0   1st Qu.: 69.25   Class :character 1st Qu.:2005
## Median :153.0   Median :167.00   Mode  :character Median :2010
## Mean   :135.6   Mean   :148.19           Mean   :2009
## 3rd Qu.:195.0   3rd Qu.:220.00           3rd Qu.:2013
## Max.   :247.0   Max.   :278.00           Max.   :2019
##
##      yi          vi      pub_type      n
## Min.   :-0.69319   Min.   :0.0003938   Min.   :0.000   Min.   : 18.0
## 1st Qu.: 0.02799   1st Qu.:0.0026885   1st Qu.:0.000   1st Qu.: 85.0
## Median : 0.13076   Median :0.0054171   Median :0.000   Median :163.0
## Mean   : 0.14799   Mean   :0.0079933   Mean   :0.756   Mean   :287.5
## 3rd Qu.: 0.27544   3rd Qu.:0.0103272   3rd Qu.:1.000   3rd Qu.:306.0
## Max.   : 0.71828   Max.   :0.0551461   Max.   :4.000   Max.   :2195.0
##
##   ed_level1 ed_level2   country      perc_white
## Min.   :2   Min.   :1.000   Length:746   Min.   : 0.00
## 1st Qu.:2   1st Qu.:2.000   Class :character 1st Qu.: 38.40
## Median :2   Median :2.000   Mode  :character Median : 52.00

```

```

## Mean :2 Mean :1.862 Mean :56.73
## 3rd Qu.:2 3rd Qu.:2.000 3rd Qu.:78.00
## Max. :2 Max. :2.000 Max. :100.00
## NA's :409
## age perc_female ei_construct ei_stream
## Min. :7.68 Min. :0.00 Min. :0.000 Min. :1.000
## 1st Qu.:19.90 1st Qu.:56.97 1st Qu.:0.000 1st Qu.:1.000
## Median :21.33 Median :66.22 Median :3.000 Median :3.000
## Mean :22.06 Mean :63.87 Mean :3.125 Mean :2.221
## 3rd Qu.:23.00 3rd Qu.:73.62 3rd Qu.:5.000 3rd Qu.:3.000
## Max. :40.00 Max. :100.00 Max. :9.000 Max. :3.000
## NA's :1 NA's :157 NA's :179
## ei_measure subject humanities achievement_type
## Min. :1.100 Min. :0.0000 Min. :0.0000 Min. :0.0000
## 1st Qu.:1.100 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:0.0000
## Median :2.500 Median :0.0000 Median :0.0000 Median :0.0000
## Mean :2.253 Mean :0.8587 Mean :0.1501 Mean :0.1501
## 3rd Qu.:3.100 3rd Qu.:0.0000 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :3.300 Max. :7.0000 Max. :1.0000 Max. :1.0000
## NA's :136 NA's :10
# 3. Variables and Summary of Variable Statistics
# Study_id: unique id for each study
# Sample_id: unique id for each sample
# es_id: unique id for each effect size
# effect size measure the strength of the relationship between variables or difference between
groups
# the larger the effect size the more significance the findings have
# ra value: Pearson correlation: -1 (perfect negative relationship - 1 perfect pos. correlation, close
to 0 weak or no relationship)
# primary author
# pub_year: year paper was published

```

yi: effect size converted to an r value
 # vi: sampling variance for the effect size (SE^2)
 # pub_type: type of publication (0- journal article, 1 = dissertation, 2 = conference proceedings
 # 3 = unpublished data)
 # n = number of participants contributing to the effect size
 # ed_level1: level of education at the time of data collection (0 = primary, 1 = secondary, 2 =
 tertiary, 3 = mixed)
 # ed_level2 (academic performance measure): level of education at the time of data collection (0
 = primary, 1 = secondary, 2 = tertiary, 3 = mixed)
 # country: country where the study took place
 # perc_white: % sample categorized as white (USA only)
 # age: mean age of the sample
 # perc_female: % sample that is female
 # ei_construct: (0 = overall ei, 2 = perception, 2 = facilitation, 3 = understanding, 5 =
 intrapersonal
 # 6 = interpersonal, 7 = stress management, 8 = adaptability, 9 = general mood)
 # ei_stream (stream or type of EI instrument used): (1 = maximum-performance ability tests, 2 =
 rating scales based on ability models
 # 3 = other broader models of EI that include non-ability constructs)
 # EI_measure (EI test used): (1.1 = MSCEIT, 1.2 = MEIS, 1.3: DANVA, 1.4 + STEU, 1.5:
 STEM
 # 2.1 = SUEIT, 2.2: SSRI, 2.3: SREIT, 2.4: TMMS, 2.5 = WLEIS, 3.1 = EQI, 3.2 = TEIQue)
 # subject area of academic performance: (0 = general, 1 = verbal/ LA, 2 = math, 3 = science,
 # 4 = social studies, 5 = foreign language, 6 = psychology, 7 = medicine, 8 = engineering
 # 9 = physical education, 10 = art)
 # humanities (science v humanities): (0 = general, 1 = maht and sciences, 2 = humanities and
 verbal abilities)
 # achievement_type: (0 = course gade, 1 = standardized test)

 # Publication Year
 # Min: 1997, max: 2019, median: 2010, mean: 2009, IQR: 8

```

#Effect size (r value)
#yi: min: -0.69319, max: 0.71828, median: 0.13076, mean: 0.14799, IQR: 0.69029

# Sample variance
#vi: min: 0.0003938, max: 0.0551461, median: 0.0054171, mean: 0.0079933, IQR: 0.0099334

#Sample Size

# n: min: 18, max: 2195, median: 163, mean: 287.5, IQR: 2177

#Education level
# ed_level1: 2 = graduate

# Academic performance
#ed_level2: min: 1,max: 2, median: 2

#Demographics (repeat after removing NA's)
# Percent white: min: 0, max: 100, median: 52, mean: 56.73, IQR: 78
# Percent female: min: 0, max: 100, median: 66.22, mean: 73.62, IQR: 100
# age: min: 7.68, max: 40, median: 21.33, mean: 22.096, IQR: 32.32

# Proportion tables for Categorical Variables to inform next steps

#country fixed code
# Check total entries and missing values
print(length(filtered_mccann1$country))    # Total length
## [1] 746
print(sum(is.na(filtered_mccann1$country))) # Number of missing values

```



```

## [1] 0
# Filter non-NA values
country_data_f <- filtered_mccann1$country[!is.na(filtered_mccann1$country)]
print(country_data_f)          # Non-NA values
## [1] "Nigeria"  "Nigeria"  "UAE"      "UAE"      "UAE"
## [6] "UAE"      "UAE"      "UAE"      "UAE"      "Pakistan"
## [11] "Pakistan"  "Pakistan"  "Pakistan"  "Pakistan"  "Pakistan"
## [16] "Pakistan"  "Bahrain"   "USA"      "Iran"      "Iran"
## [21] "Iran"      "Iran"      "USA"      "USA"      "USA"
## [26] "USA"      "USA"      "USA"      "USA"      "USA"
## [31] "USA"      "USA"      "USA"      "USA"      "USA"
## [36] "Australia" "Australia" "Australia" "Australia" "Australia"
## [41] "Australia" "Australia" "Australia" "Australia" "Australia"
## [46] "USA"      "USA"      "USA"      "USA"      "USA"
## [51] "USA"      "USA"      "USA"      "USA"      "USA"
## [56] "USA"      "USA"      "USA"      "USA"      "USA"
## [61] "USA"      "USA"      "USA"      "USA"      "USA"
## [66] "USA"      "USA"      "USA"      "USA"      "USA"
## [71] "USA"      "USA"      "USA"      "USA"      "USA"
## [76] "USA"      "USA"      "USA"      "USA"      "USA"
## [81] "USA"      "USA"      "USA"      "USA"      "USA"
## [86] "USA"      "USA"      "USA"      "USA"      "USA"
## [91] "USA"      "USA"      "USA"      "USA"      "USA"
## [96] "USA"      "USA"      "USA"      "USA"      "USA"
## [101] "USA"      "USA"      "Romania"   "USA"      "USA"
## [106] "USA"      "USA"      "USA"      "USA"      "USA"
## [111] "USA"      "USA"      "USA"      "USA"      "USA"
## [116] "Malaysia"  "Malaysia"  "Malaysia"  "Malaysia"  "Malaysia"
## [121] "Malaysia"  "Malaysia"  "Malaysia"  "Malaysia"  "Malaysia"
## [126] "Malaysia"  "Malaysia"  "Malaysia"  "Malaysia"  "USA"
## [131] "USA"      "USA"      "USA"      "USA"      "USA"

```

## [136]	"USA"	"USA"	"USA"	"USA"	"USA"
## [141]	"USA"	"USA"	"USA"	"USA"	"USA"
## [146]	"USA"	"USA"	"Spain"	"Spain"	"Spain"
## [151]	"Spain"	"Spain"	"Spain"	"Spain"	"Spain"
## [156]	"Spain"	"Spain"	"Spain"	"Spain"	"Spain"
## [161]	"Spain"	"Spain"	"USA"	"USA"	"USA"
## [166]	"USA"	"USA"	"USA"	"USA"	"USA"
## [171]	"USA"	"USA"	"USA"	"USA"	"USA"
## [176]	"USA"	"USA"	"USA"	"USA"	"USA"
## [181]	"USA"	"USA"	"USA"	"USA"	"USA"
## [186]	"USA"	"USA"	"Iran"	"Iran"	"Iran"
## [191]	"Iran"	"Iran"	"Iran"	"Iran"	"Iran"
## [196]	"Iran"	"Iran"	"Iran"	"Iran"	"Barbados"
## [201]	"Barbados"	"Barbados"	"Barbados"	"Barbados"	"Barbados"
## [206]	"USA"	"USA"	"USA"	"USA"	"USA"
## [211]	"Canada"	"Canada"	"Canada"	"Canada"	"Canada"
## [216]	"Canada"	"Canada"	"Canada"	"Canada"	"Canada"
## [221]	"Canada"	"Canada"	"USA"	"USA"	"USA"
## [226]	"USA"	"USA"	"USA"	"UK"	"UK"
## [231]	"UK"	"UK"	"UK"	"UK"	"UK"
## [236]	"UK"	"UK"	"UK"	"UK"	"UK"
## [241]	"USA"	"USA"	"USA"	"USA"	"USA"
## [246]	"USA"	"USA"	"Canada"	"Canada"	"Canada"
## [251]	"Canada"	"Canada"	"Canada"	"Canada"	"Canada"
## [256]	"Canada"	"Canada"	"USA"	"USA"	"USA"
## [261]	"USA"	"USA"	"USA"	"USA"	"USA"
## [266]	"USA"	"USA"	"Canada"	"Canada"	"Canada"
## [271]	"Canada"	"Canada"	"Canada"	"Canada"	"Canada"
## [276]	"Canada"	"Canada"	"Canada"	"Canada"	"USA"
## [281]	"USA"	"USA"	"USA"	"USA"	"USA"
## [286]	"USA"	"USA"	"USA"	"USA"	"USA"

## [291]	"USA"	"USA"	"USA"	"USA"	"USA"
## [296]	"USA"	"USA"	"USA"	"USA"	"USA"
## [301]	"USA"	"USA"	"USA"	"USA"	"USA"
## [306]	"USA"	"USA"	"USA"	"USA"	"USA"
## [311]	"USA"	"USA"	"USA"	"USA"	"India"
## [316]	"Italy"	"Italy"	"Italy"	"USA"	"USA"
## [321]	"USA"	"USA"	"USA"	"Europe"	"Europe"
## [326]	"Europe"	"Europe"	"Europe"	"Europe"	"USA"
## [331]	"USA"	"USA"	"USA"	"USA"	"USA"
## [336]	"USA"	"USA"	"USA"	"USA"	"USA"
## [341]	"USA"	"USA"	"USA"	"USA"	"USA"
## [346]	"USA"	"USA"	"USA"	"USA"	"USA"
## [351]	"USA"	"USA"	"USA"	"Australia"	"Australia"
## [356]	"Australia"	"Australia"	"Australia"	"Australia"	"USA"
## [361]	"USA"	"USA"	"USA"	"USA"	"Saudi Arabia"
## [366]	"Canada"	"Canada"	"Canada"	"Canada"	"Canada"
## [371]	"Canada"	"USA"	"USA"	"USA"	"USA"
## [376]	"USA"	"USA"	"USA"	"USA"	"USA"
## [381]	"USA"	"USA"	"USA"	"USA"	"USA"
## [386]	"USA"	"USA"	"USA"	"USA"	"USA"
## [391]	"USA"	"USA"	"USA"	"Nigeria"	"USA"
## [396]	"USA"	"USA"	"USA"	"USA"	"USA"
## [401]	"Nigeria"	"USA"	"USA"	"USA"	"USA"
## [406]	"USA"	"Canada"	"Canada"	"Canada"	"Canada"
## [411]	"Canada"	"Canada"	"Canada"	"Canada"	"Canada"
## [416]	"Canada"	"Iran"	"Iran"	"Iran"	"Iran"
## [421]	"Iran"	"Iran"	"Iran"	"Iran"	"Iran"
## [426]	"Iran"	"Iran"	"Iran"	"Iran"	"Iran"
## [431]	"Iran"	"Iran"	"Iran"	"Iran"	"Iran"
## [436]	"Iran"	"Iran"	"Iran"	"Iran"	"Iran"
## [441]	"Iran"	"Iran"	"Iran"	"Iran"	"Iran"

[446] "Iran" "UK" "USA" "UK" "UK"
 ## [451] "UK" "UK" "UK" "Iran" "UK"
 ## [456] "UK" "UK" "UK" "UK" "USA"
 ## [461] "USA" "USA" "USA" "USA" "Cyprus"
 ## [466] "USA" "USA" "USA" "USA" "USA"
 ## [471] "Malaysia" "Malaysia" "Malaysia" "Malaysia" "USA"
 ## [476] "China" "China" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [481] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [486] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [491] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [496] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [501] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [506] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [511] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [516] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [521] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [526] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [531] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [536] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [541] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [546] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [551] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [556] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [561] "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada" "UK or Canada"
 ## [566] "UK or Canada" "UK or Canada" "Saudi Arabia" "Turkey" "Turkey"
 ## [571] "Turkey" "Turkey" "Turkey" "Turkey" "Turkey"
 ## [576] "Turkey" "USA" "USA" "USA" "USA"
 ## [581] "USA" "USA" "USA" "USA" "USA"
 ## [586] "USA" "USA" "USA" "USA" "USA"
 ## [591] "USA" "USA" "USA" "USA" "USA"
 ## [596] "USA" "USA" "USA" "USA" "USA"

```

## [601] "USA"      "USA"      "USA"      "USA"      "USA"
## [606] "USA"      "USA"      "USA"      "USA"      "USA"
## [611] "USA"      "USA"      "USA"      "USA"      "USA"
## [616] "USA"      "USA"      "USA"      "USA"      "USA"
## [621] "USA"      "USA"      "USA"      "USA"      "USA"
## [626] "USA"      "USA"      "USA"      "USA"      "USA"
## [631] "USA"      "USA"      "USA"      "USA"      "USA"
## [636] "USA"      "USA"      "USA"      "USA"      "USA"
## [641] "USA"      "USA"      "USA"      "USA"      "USA"
## [646] "USA"      "USA"      "USA"      "USA"      "USA"
## [651] "USA"      "USA"      "USA"      "USA"      "USA"
## [656] "USA"      "USA"      "USA"      "USA"      "USA"
## [661] "USA"      "USA"      "USA"      "USA"      "USA"
## [666] "USA"      "USA"      "USA"      "Australia" "Australia"
## [671] "Australia" "Australia" "Australia" "Australia" "Australia"
## [676] "Australia" "Australia" "Australia" "Iran"      "Iran"
## [681] "Iran"      "Iran"      "Iran"      "Iran"      "Iran"
## [686] "Iran"      "Iran"      "Iran"      "Iran"      "Iran"
## [691] "Israel"    "Israel"    "USA"      "USA"      "USA"
## [696] "USA"      "USA"      "Canada"   "Australia" "Australia"
## [701] "Australia" "Australia" "Australia" "Australia" "Australia"
## [706] "Australia" "Australia" "Australia" "Australia" "Australia"
## [711] "Australia" "Australia" "Belgium"  "UK"       "USA"
## [716] "USA"      "USA"      "USA"      "USA"      "USA"
## [721] "USA"      "USA"      "USA"      "USA"      "USA"
## [726] "USA"      "USA"      "USA"      "USA"      "USA"
## [731] "USA"      "USA"      "USA"      "USA"      "USA"
## [736] "USA"      "USA"      "USA"      "USA"      "USA"
## [741] "USA"      "Australia" "Australia" "USA, Canada" "USA"
## [746] "USA"

# Calculate proportions and create a data frame

```

```

prop_table_f1 <- prop.table(table(country_data_f)) * 100
prop_table_f1 <- round(prop_table_f1, 2)
print(prop_table_f1)          # Proportions table
## country_data_f
##  Australia  Bahrain  Barbados  Belgium  Canada  China
##    5.63    0.13    0.80    0.13    6.84    0.27
##  Cyprus  Europe    India    Iran    Israel    Italy
##    0.13    0.80    0.13    7.91    0.27    0.40
##  Malaysia  Nigeria  Pakistan  Romania Saudi Arabia  Spain
##    2.41    0.54    0.94    0.13    0.27    2.01
##  Turkey    UAE      UK UK or Canada  USA  USA, Canada
##    1.07    0.94    3.22    12.06    52.82    0.13

# Create a data frame
prop_dff1 <- data.frame(Category = names(prop_table_f1), Percentage =
as.numeric(prop_table_f1))
print(prop_dff1)          # Final data frame
##   Category Percentage
## 1  Australia    5.63
## 2   Bahrain    0.13
## 3  Barbados    0.80
## 4   Belgium    0.13
## 5   Canada    6.84
## 6   China     0.27
## 7   Cyprus    0.13
## 8   Europe    0.80
## 9    India    0.13
## 10   Iran     7.91
## 11  Israel    0.27
## 12   Italy    0.40
## 13 Malaysia    2.41
## 14  Nigeria    0.54

```

```

## 15    Pakistan    0.94
## 16    Romania    0.13
## 17 Saudi Arabia    0.27
## 18     Spain    2.01
## 19    Turkey    1.07
## 20     UAE    0.94
## 21     UK    3.22
## 22 UK or Canada    12.06
## 23     USA    52.82
## 24 USA, Canada    0.13
# Country
length(filtered_mccann1$country)
## [1] 746
sum(is.na(filtered_mccann1$country))
## [1] 0
country_data_f <- filtered_mccann1$country[!is.na(filtered_mccann1$country)]
prop_table_f1 <- prop.table(table(country_data_f)) * 100
prop_table_f1 <- round(prop_table_f1, 2)
prop_dff1 <- data.frame(Category = names(prop_table_f1), Percentage = prop_table_f1)
print(prop_dff1)
##      Category Percentage.country_data_f Percentage.Freq
## 1   Australia           Australia           5.63
## 2    Bahrain           Bahrain           0.13
## 3  Barbados           Barbados           0.80
## 4   Belgium           Belgium           0.13
## 5    Canada           Canada           6.84
## 6    China           China           0.27
## 7   Cyprus           Cyprus           0.13
## 8   Europe           Europe           0.80
## 9    India           India           0.13
## 10   Iran           Iran           7.91

```

```
## 11    Israel          Israel      0.27
## 12     Italy          Italy       0.40
## 13  Malaysia          Malaysia    2.41
## 14   Nigeria          Nigeria     0.54
## 15   Pakistan        Pakistan     0.94
## 16   Romania          Romania     0.13
## 17 Saudi Arabia      Saudi Arabia 0.27
## 18    Spain          Spain       2.01
## 19   Turkey          Turkey      1.07
## 20    UAE            UAE         0.94
## 21     UK            UK          3.22
## 22 UK or Canada      UK or Canada 12.06
## 23     USA            USA        52.82
## 24 USA, Canada       USA, Canada  0.13
```

#Publication Type

```
length(filtered_mccann1$pub_type)
```

```
## [1] 746
```

```
sum(is.na(filtered_mccann1$pub_type))
```

```
## [1] 0
```

```
pub_type_data_f <- filtered_mccann1$pub_type[!is.na(filtered_mccann1$pub_type)]
```

```
prop_table_f2 <- prop.table(table(pub_type_data_f)) * 100
```

```
prop_table_f2 <- round(prop_table_f2, 2)
```

```
prop_dff2 <- data.frame(Category = names(prop_table_f2), Percentage = prop_table_f2)
```

```
print(prop_dff2)
```

```
##   Category Percentage.pub_type_data_f Percentage.Freq
```

```
## 1      0              0          53.22
```

```
## 2      1              1          35.52
```

```
## 3      3              3           4.96
```

```
## 4      4              4           6.30
```

Publication Year

```
length(filtered_mccann1$pub_year)
```



```

## [1] 746
sum(is.na(filtered_mccann1$pub_year))
## [1] 0
pub_year_data_f <- filtered_mccann1$pub_year[!is.na(filtered_mccann1$pub_year)]
prop_table_f3 <- prop.table(table(pub_year_data_f)) * 100
prop_table_f3 <- round(prop_table_f3, 2)
prop_dff3 <- data.frame(Category = names(prop_table_f3), Percentage = prop_table_f3)
print(prop_dff3)
##   Category Percentage.pub_year_data_f Percentage.Freq
## 1    1997           1997           0.13
## 2    1998           1998           0.13
## 3    2000           2000           0.80
## 4    2002           2002           0.67
## 5    2003           2003          14.08
## 6    2004           2004           5.63
## 7    2005           2005           3.62
## 8    2006           2006           4.16
## 9    2007           2007           7.91
## 10   2008           2008           5.09
## 11   2009           2009           5.36
## 12   2010           2010           3.49
## 13   2011           2011           6.97
## 14   2012           2012          14.61
## 15   2013           2013          10.32
## 16   2014           2014           5.50
## 17   2015           2015           2.82
## 18   2016           2016           2.41
## 19   2018           2018           5.90
## 20   2019           2019           0.40
#Education Level 1
length(filtered_mccann1$ed_level1)

```

```

## [1] 746
sum(is.na(filtered_mccann1$ed_level1))
## [1] 0
ed_level1_data_f <- filtered_mccann1$ed_level1[!is.na(filtered_mccann1$ed_level1)]
prop_table_f4 <- prop.table(table(ed_level1_data_f)) * 100
prop_table_f4 <- round(prop_table_f4, 2)
prop_dff4 <- data.frame(Category = names(prop_table_f4), Percentage = prop_table_f4)
print(prop_dff4)
##  Category Percentage.ed_level1_data_f Percentage.Freq
## 1      2                2          100
# EI Stream
length(filtered_mccann1$ei_stream)
## [1] 746
sum(is.na(filtered_mccann1$ei_stream))
## [1] 0
ei_stream_data_f <- filtered_mccann1$ei_stream[!is.na(filtered_mccann1$ei_stream)]
prop_table_f5 <- prop.table(table(ei_stream_data_f)) * 100
prop_table_f5 <- round(prop_table_f5, 2)
prop_dff5 <- data.frame(Category = names(prop_table_f5), Percentage = prop_table_f5)
print(prop_dff5)
##  Category Percentage.ei_stream_data_f Percentage.Freq
## 1      1                1          34.18
## 2      2                2           9.52
## 3      3                3          56.30
# EI Measure
length(filtered_mccann1$ei_measure)
## [1] 746
sum(is.na(filtered_mccann1$ei_measure))
## [1] 136
ei_measure_data_f <- filtered_mccann1$ei_measure[!is.na(filtered_mccann1$ei_measure)]
prop_table_f6 <- prop.table(table(ei_measure_data_f)) * 100

```

```

prop_table_f6 <- round(prop_table_f6, 2)
prop_dff6 <- data.frame(Category = names(prop_table_f6), Percentage = prop_table_f6)
print(prop_dff6)
##  Category Percentage.ei_measure_data_f Percentage.Freq
## 1      1.1              1.1      33.28
## 2      1.2              1.2       1.15
## 3      1.3              1.3       1.48
## 4      1.4              1.4       1.15
## 5      1.5              1.5       1.64
## 6      2.1              2.1       1.64
## 7      2.2              2.2      4.26
## 8      2.3              2.3       2.95
## 9      2.4              2.4       2.13
## 10     2.5              2.5       2.79
## 11     3.1              3.1     35.08
## 12     3.2              3.2       2.62
## 13     3.3              3.3      9.84
# Achievement Type
length(filtered_mccann1$achievement_type)
## [1] 746
sum(is.na(filtered_mccann1$achievement_type))
## [1] 0
achieve_type_data_f <-
filtered_mccann1$achievement_type[!is.na(filtered_mccann1$achievement_type)]
prop_table_f7 <- prop.table(table(achieve_type_data_f)) * 100
prop_table_f7 <- round(prop_table_f7, 2)
prop_dff7 <- data.frame(Category = names(prop_table_f7), Percentage = prop_table_f7)
print(prop_dff7)
##  Category Percentage.achieve_type_data_f Percentage.Freq
## 1      0              0      84.99
## 2      1              1     15.01

```

```
# Part 3: (Filter 2) By EI_Measure/ stream and Descriptive Statistics
```

```
# Filter by EI Measure 1.1 and 3.1
```

```
# Filter rows where ei_measure is either 1.1 or 3.1
```

```
filtered_mccann2 <- filtered_mccann1[filtered_mccann1$ei_measure == 1.1 |
```

```
filtered_mccann1$ei_measure == 3.1, ]
```

```
# View summary statistics of the filtered data frame
```

```
summary(filtered_mccann2)
```

```
##  study_id  sample_id  author      pub_year
```

```
## Min.   : 8.0  Min.   : 8.0  Length:553    Min.   :2000
```

```
## 1st Qu.: 53.0 1st Qu.: 55.0  Class :character 1st Qu.:2004
```

```
## Median : 98.0 Median :105.0  Mode  :character Median :2007
```

```
## Mean   :107.6 Mean   :116.2          Mean   :2008
```

```
## 3rd Qu.:163.0 3rd Qu.:177.0          3rd Qu.:2011
```

```
## Max.   :246.0 Max.   :276.0          Max.   :2019
```

```
## NA's   :136  NA's   :136          NA's   :136
```

```
##  yi        vi      pub_type      n
```

```
## Min.   :-0.58227 Min.   :0.00054 Min.   :0.0000  Min.   : 30.0
```

```
## 1st Qu.: 0.03025 1st Qu.:0.00269 1st Qu.:0.0000 1st Qu.: 85.0
```

```
## Median : 0.13734 Median :0.00563 Median :0.0000 Median :150.0
```

```
## Mean   : 0.15342 Mean   :0.00750 Mean   :0.6259 Mean   :287.9
```

```
## 3rd Qu.: 0.27553 3rd Qu.:0.01010 3rd Qu.:1.0000 3rd Qu.:330.0
```

```
## Max.   : 0.71828 Max.   :0.03421 Max.   :4.0000 Max.   :1771.0
```

```
## NA's   :136  NA's   :136  NA's   :136  NA's   :136
```

```
##  ed_level1  ed_level2  country      perc_white
```

```
## Min.   :2    Min.   :1.000  Length:553    Min.   : 0.00
```

```
## 1st Qu.:2    1st Qu.:2.000  Class :character 1st Qu.: 52.00
```

```
## Median :2    Median :2.000  Mode  :character Median : 67.80
```

```
## Mean   :2    Mean   :1.803          Mean   : 63.68
```

```
## 3rd Qu.:2    3rd Qu.:2.000          3rd Qu.: 89.25
```

```

## Max. :2    Max. :2.000          Max. :100.00
## NA's :136  NA's :136           NA's :333
##   age      perc_female   ei_construct   ei_stream
## Min. :17.74 Min. : 0.00 Min. :0.000 Min. :1.000
## 1st Qu.:19.74 1st Qu.: 62.02 1st Qu.:0.000 1st Qu.:1.000
## Median :21.30 Median : 68.70 Median :3.500 Median :3.000
## Mean :21.88 Mean : 67.95 Mean :3.702 Mean :2.026
## 3rd Qu.:23.00 3rd Qu.: 73.62 3rd Qu.:6.000 3rd Qu.:3.000
## Max. :40.00 Max. :100.00 Max. :9.000 Max. :3.000
## NA's :137  NA's :174  NA's :147  NA's :136
## ei_measure  subject    humanities  achievement_type
## Min. :1.100 Min. :0.0000 Min. :0.0000 Min. :0.0000
## 1st Qu.:1.100 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:0.0000
## Median :3.100 Median :0.0000 Median :0.0000 Median :0.0000
## Mean :2.126 Mean :0.9248 Mean :0.1679 Mean :0.1679
## 3rd Qu.:3.100 3rd Qu.:0.0000 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :3.100 Max. :7.0000 Max. :1.0000 Max. :1.0000
## NA's :136  NA's :141  NA's :136  NA's :136
# Get the number of rows and columns in the filtered data frame
nrow(filtered_mccann2) # Number of rows
## [1] 553
ncol(filtered_mccann2) # Number of columns
## [1] 20
# Proportion tables of important categorical variables to help inform next steps

# EI Measure
length(filtered_mccann2$ei_measure)
## [1] 553
sum(is.na(filtered_mccann2$ei_measure))
## [1] 136
ei_measure_data_f2 <- filtered_mccann2$ei_measure[!is.na(filtered_mccann2$ei_measure)]

```

```

prop_table_f62 <- prop.table(table(ei_measure_data_f2)) * 100
prop_table_f62 <- round(prop_table_f62, 2)
prop_dff62 <- data.frame(Category = names(prop_table_f62), Percentage = prop_table_f62)
print(prop_dff62)
##  Category Percentage.ei_measure_data_f2 Percentage.Freq
## 1      1.1              1.1          48.68
## 2      3.1              3.1          51.32
#Education Level 1
length(filtered_mccann2$ed_level1)
## [1] 553
sum(is.na(filtered_mccann2$ed_level1))
## [1] 136
ed_level1_data_f2 <- filtered_mccann2$ed_level1[!is.na(filtered_mccann2$ed_level1)]
prop_table_f42 <- prop.table(table(ed_level1_data_f2)) * 100
prop_table_f42 <- round(prop_table_f42, 2)
prop_dff42 <- data.frame(Category = names(prop_table_f42), Percentage = prop_table_f42)
print(prop_dff42)
##  Category Percentage.ed_level1_data_f2 Percentage.Freq
## 1      2              2          100
# EI Stream
length(filtered_mccann2$ei_stream)
## [1] 553
sum(is.na(filtered_mccann2$ei_stream))
## [1] 136
ei_stream_data_f2 <- filtered_mccann2$ei_stream[!is.na(filtered_mccann1$ei_stream)]
prop_table_f52 <- prop.table(table(ei_stream_data_f2)) * 100
prop_table_f52 <- round(prop_table_f52, 2)
prop_dff52 <- data.frame(Category = names(prop_table_f52), Percentage = prop_table_f52)
print(prop_dff52)
##  Category Percentage.ei_stream_data_f2 Percentage.Freq
## 1      1              1          48.68

```

```

## 2      3      3      51.32
# Achievement Type
length(filtered_mccann2$achievement_type)
## [1] 553
sum(is.na(filtered_mccann2$achievement_type))
## [1] 136
achieve_type_data_f2 <-
filtered_mccann2$achievement_type[!is.na(filtered_mccann2$achievement_type)]
prop_table_f72 <- prop.table(table(achieve_type_data_f2)) * 100
prop_table_f72 <- round(prop_table_f72, 2)
prop_dff72 <- data.frame(Category = names(prop_table_f72), Percentage = prop_table_f72)
print(prop_dff72)
##  Category Percentage.achieve_type_data_f2 Percentage.Freq
## 1      0      0      83.21
## 2      1      1      16.79
# Visualization of outliers for all variables in the `vars` list using boxplots
# Assuming you have a list of variable names in the `vars` list
vars <- c("pub_year", "yi", "vi", "n", "perc_white", "age")

# Check for non-finite values in each variable
summary(filtered_mccann2[, vars])
##  pub_year      yi      vi      n
## Min. :2000 Min. : -0.58227 Min. :0.00054 Min. : 30.0
## 1st Qu.:2004 1st Qu.: 0.03025 1st Qu.:0.00269 1st Qu.: 85.0
## Median :2007 Median : 0.13734 Median :0.00563 Median :150.0
## Mean :2008 Mean : 0.15342 Mean :0.00750 Mean :287.9
## 3rd Qu.:2011 3rd Qu.: 0.27553 3rd Qu.:0.01010 3rd Qu.:330.0
## Max. :2019 Max. : 0.71828 Max. :0.03421 Max. :1771.0
## NA's :136 NA's :136 NA's :136 NA's :136
##  perc_white      age
## Min. : 0.00 Min. :17.74

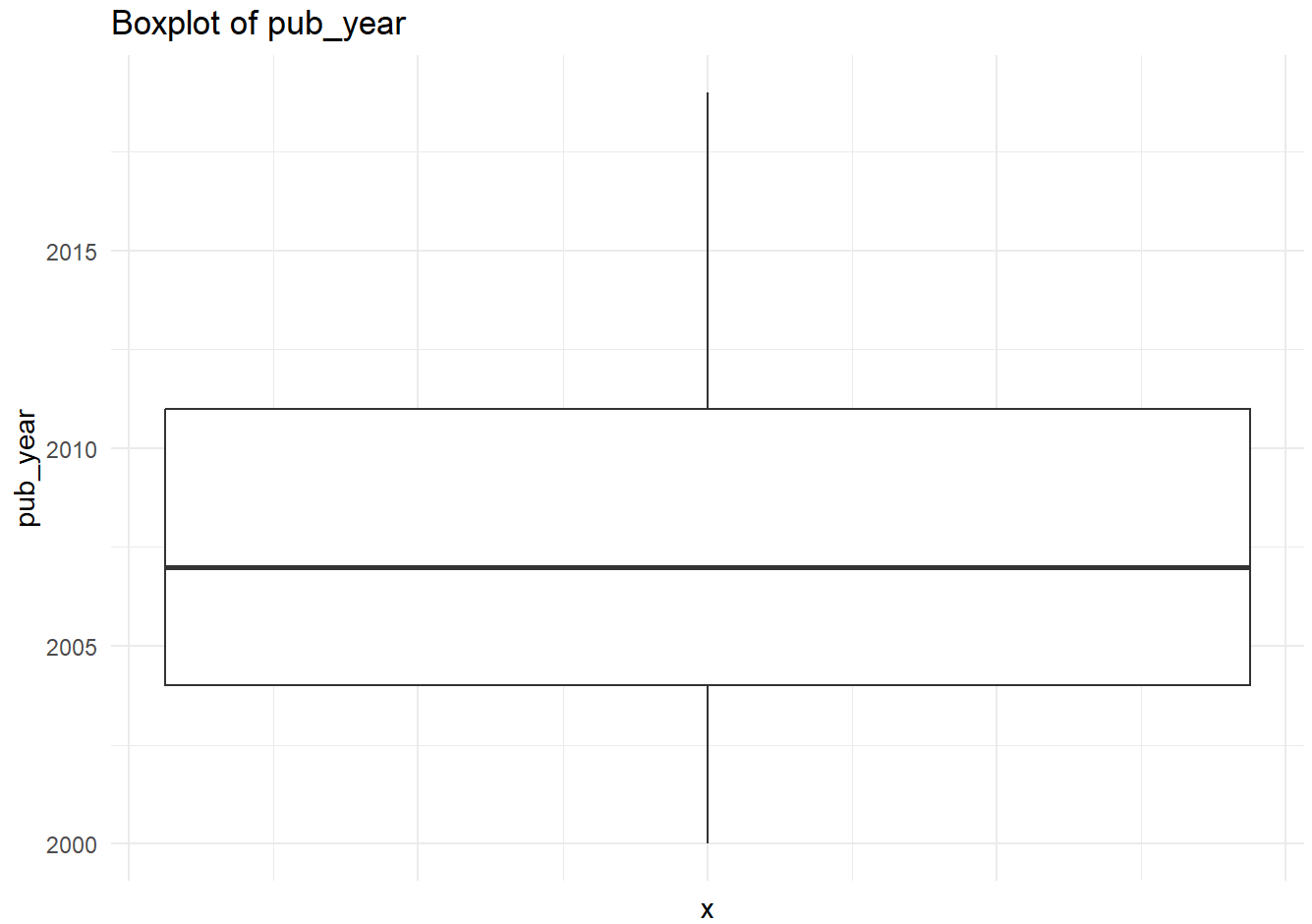
```

```

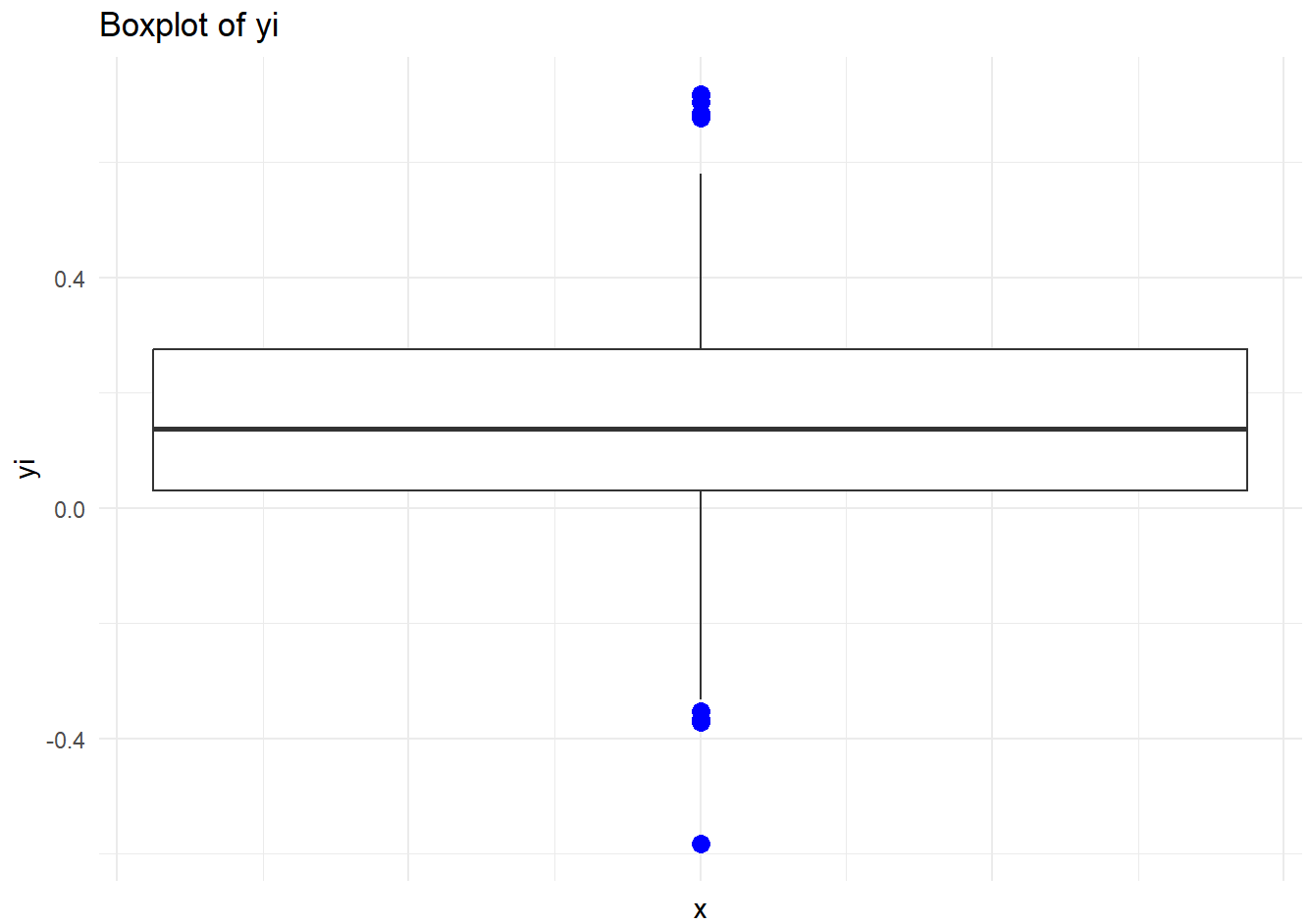
## 1st Qu.: 52.00 1st Qu.:19.74
## Median : 67.80 Median :21.30
## Mean : 63.68 Mean :21.88
## 3rd Qu.: 89.25 3rd Qu.:23.00
## Max. :100.00 Max. :40.00
## NA's :333 NA's :137
# Alternatively, check the specific variables for NAs
sapply(filtered_mccann2[, vars], function(x) sum(is.na(x)))
## pub_year yi vi n perc_white age
## 136 136 136 136 333 137
for (var in vars) {
  p <- ggplot(filtered_mccann2, aes(x = 1, y = !!sym(var))) + # Using 1 for x-axis as a dummy
variable
  geom_boxplot(outlier.colour = "blue", outlier.size = 3) +
  labs(title = paste("Boxplot of", var), y = var) +
  theme_minimal() +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())

  # Print the plot
  print(p)
}
## Warning: Removed 136 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

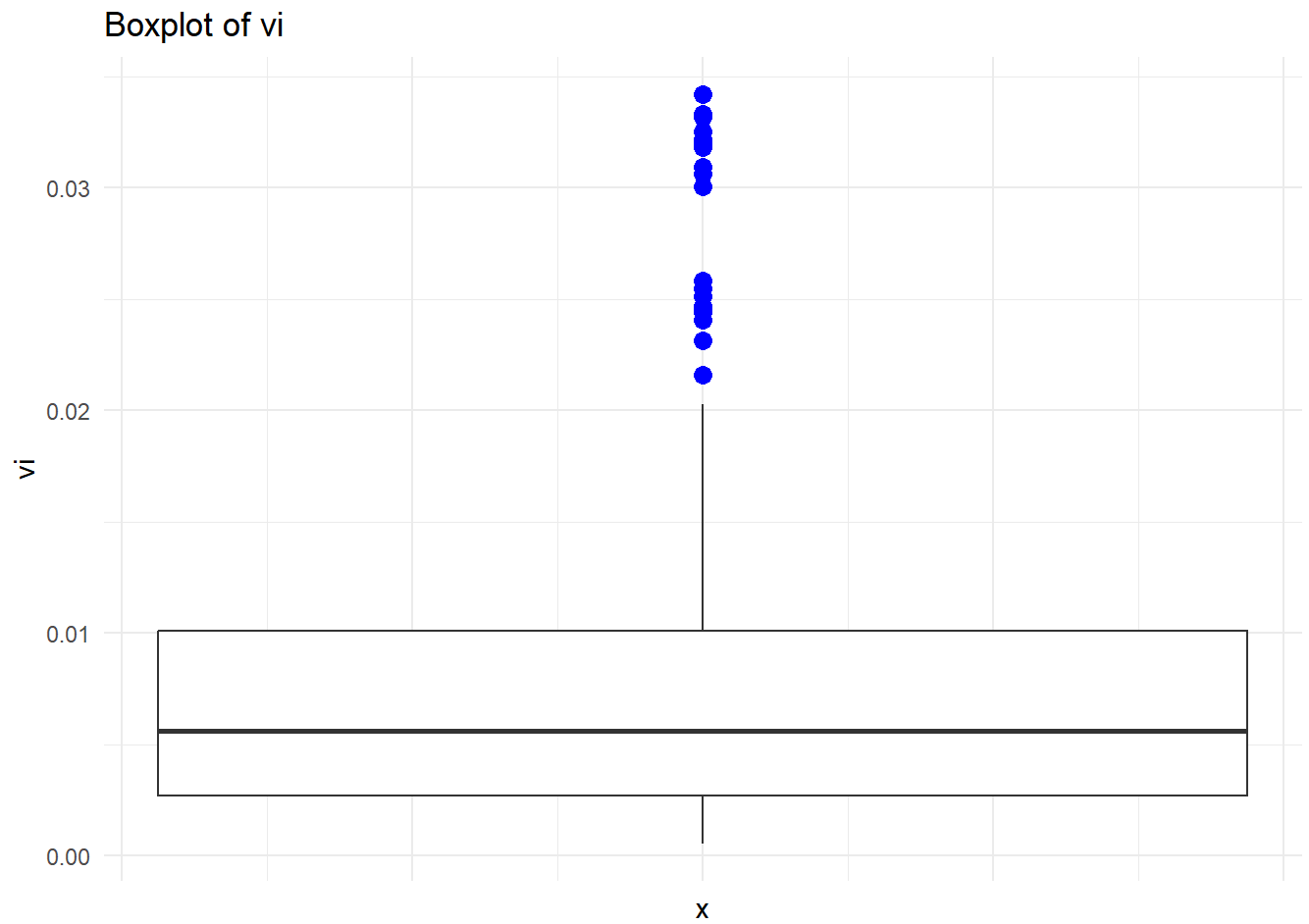
```

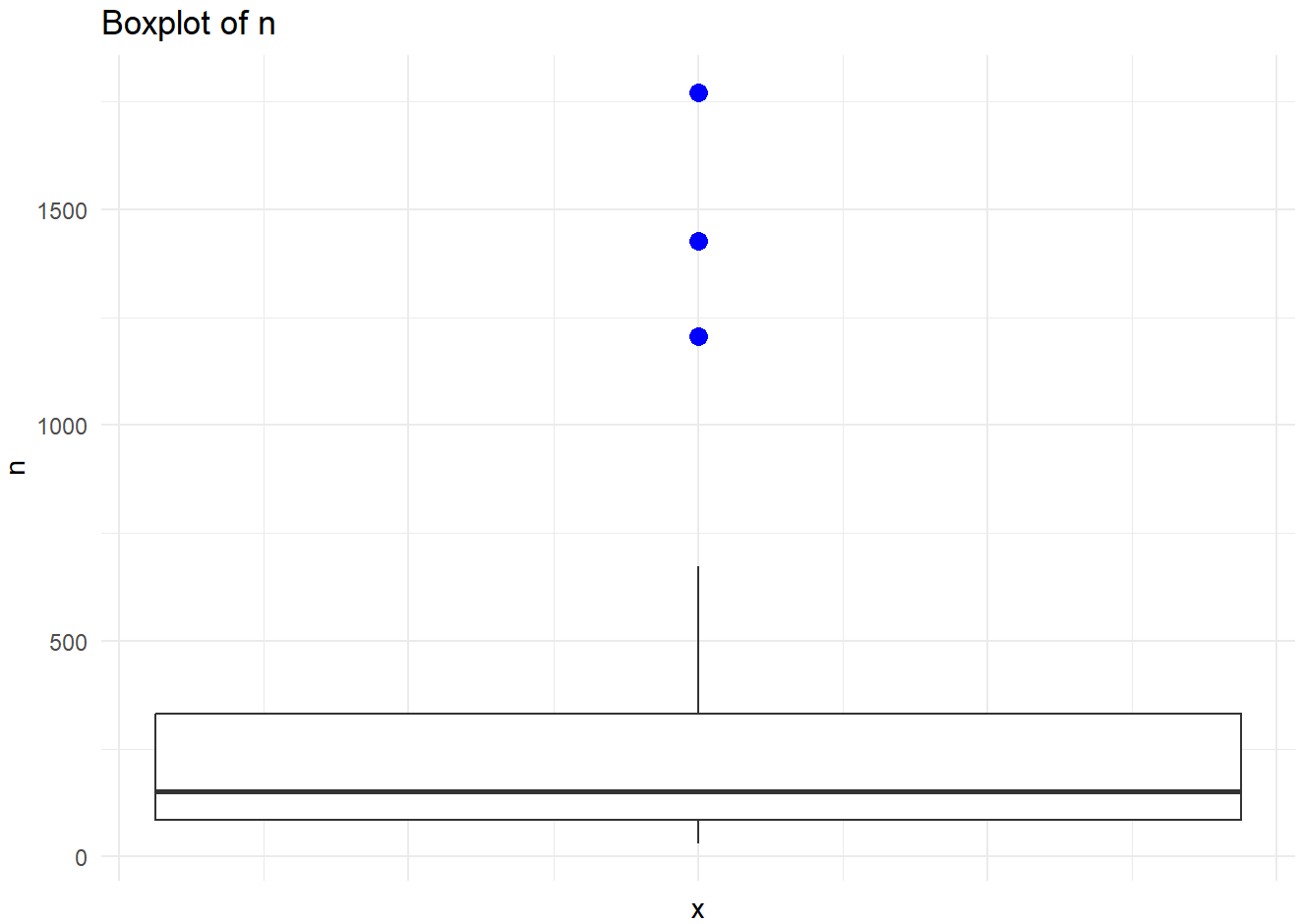
```
## Warning: Removed 136 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



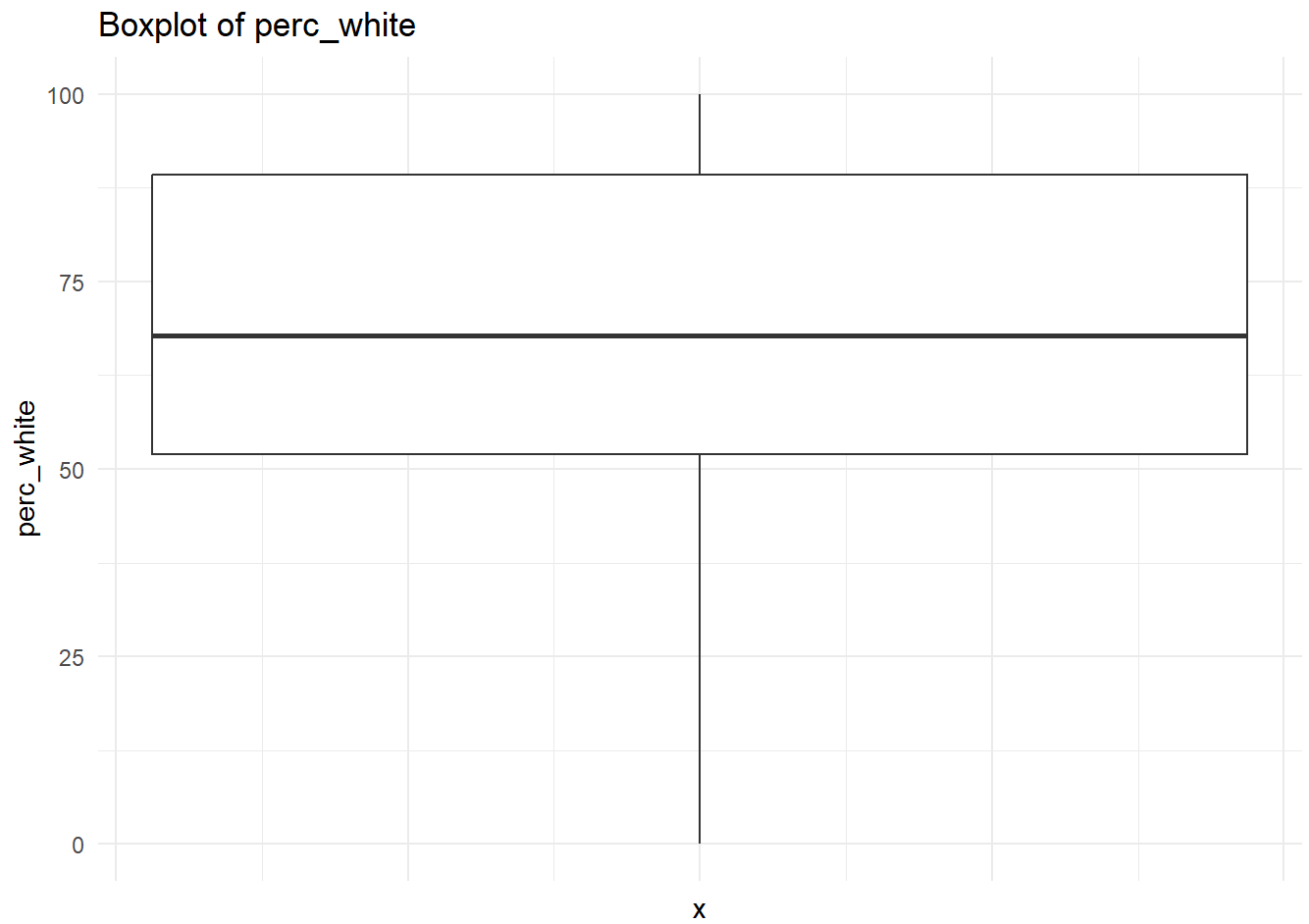
```
## Warning: Removed 136 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



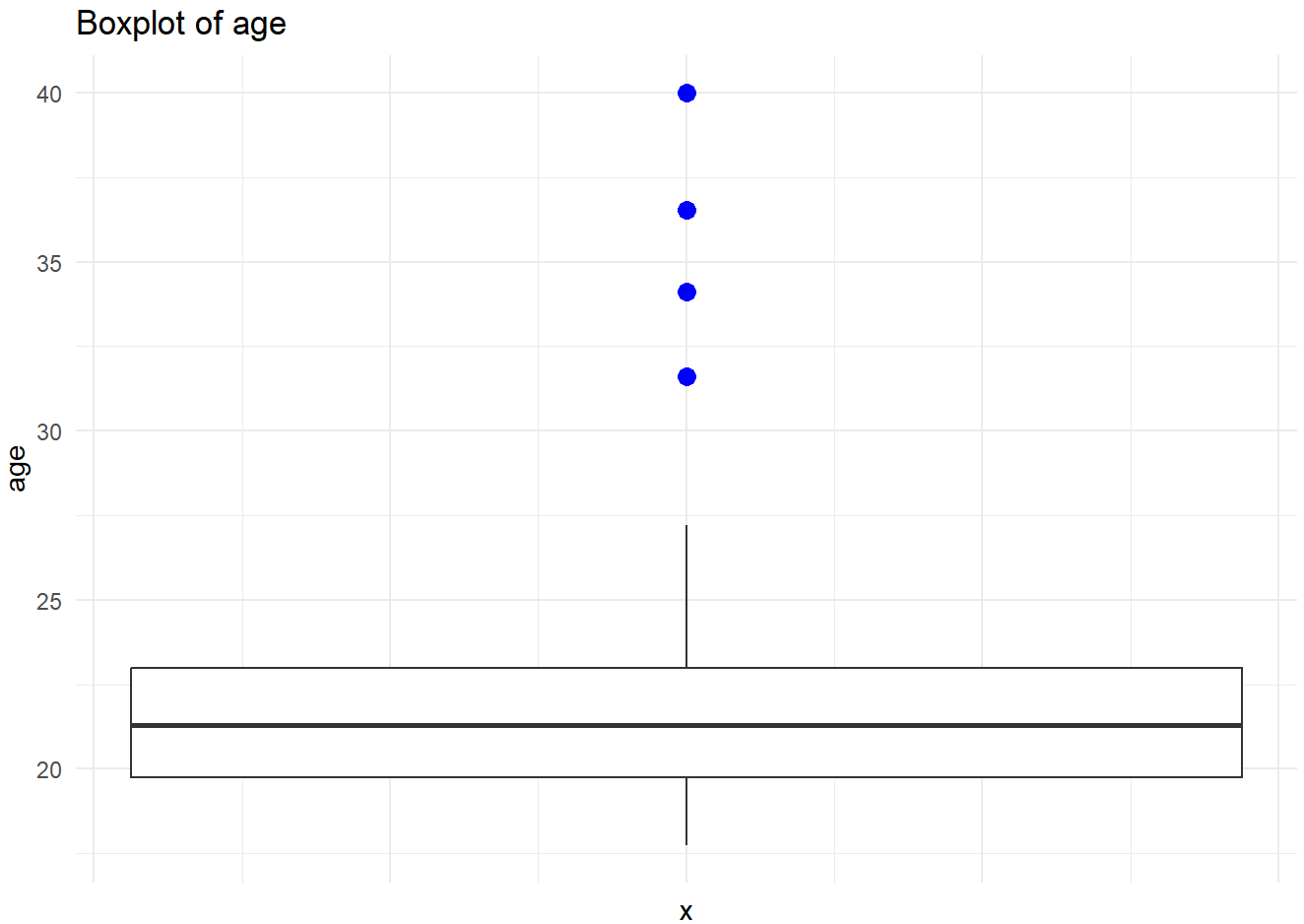
```
## Warning: Removed 136 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



```
## Warning: Removed 333 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



```
## Warning: Removed 137 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Part 4: Preliminary Data (filtered McCan 2)

Group by ei_measure and summarize r² statistics

```
summary_stats <- filtered_mccann2 %>%  
  group_by(ei_stream) %>%  
  summarize(  
    mean_r2 = mean(yi, na.rm = TRUE),  
    median_r2 = median(yi, na.rm = TRUE),  
    sd_r2 = sd(yi, na.rm = TRUE),  
    min_r2 = min(yi, na.rm = TRUE),  
    max_r2 = max(yi, na.rm = TRUE),  
    count = n())
```

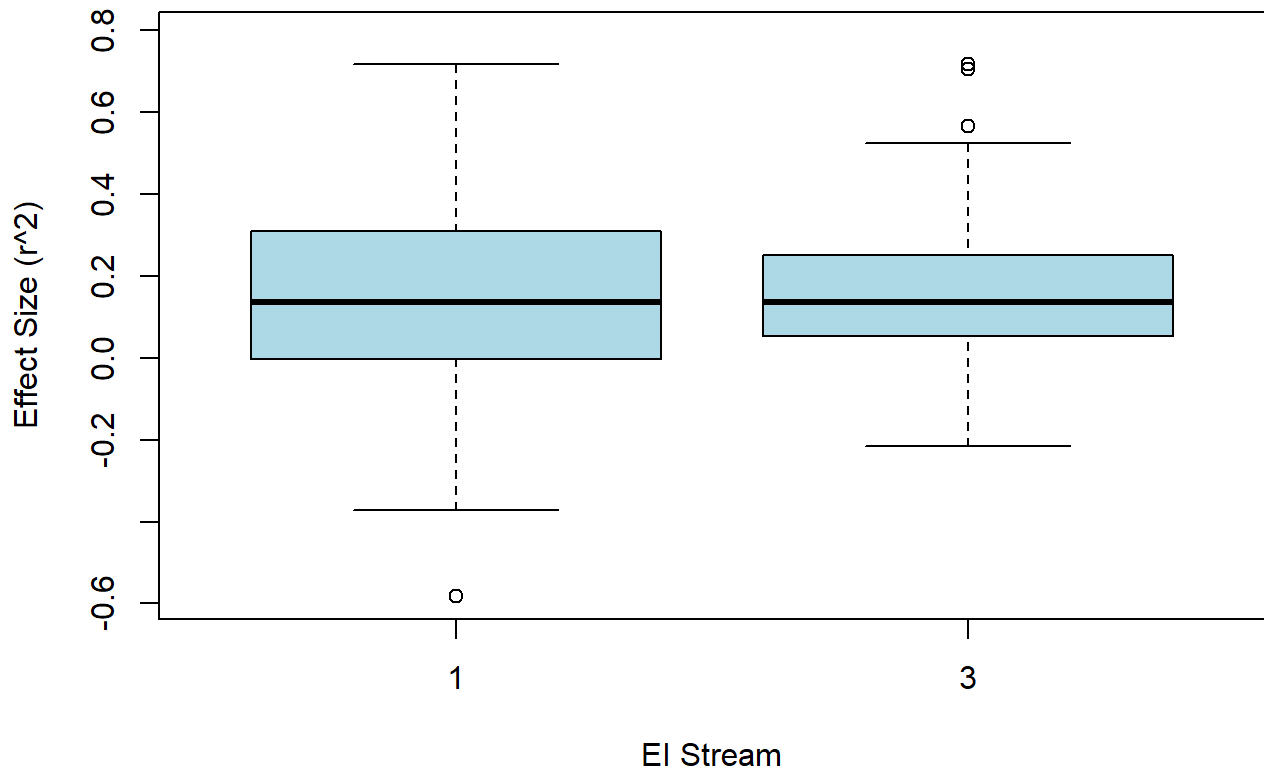
```

)
## Warning: There were 2 warnings in `summarize()`.
## The first warning was:
## i In argument: `min_r2 = min(yi, na.rm = TRUE)`.
## i In group 3: `ei_stream = NA`.
## Caused by warning in `min()`:
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
# View the summarized statistics
print(summary_stats)
## # A tibble: 3 × 7
##   ei_stream mean_r2 median_r2 sd_r2 min_r2 max_r2 count
##   <int>   <dbl>   <dbl> <dbl> <dbl>   <dbl> <int>
## 1     1  0.149   0.137 0.234 -0.582  0.718  203
## 2     3  0.158   0.137 0.151 -0.216  0.717  214
## 3    NA NaN    NA    NA    Inf   -Inf   136
# Boxplot of mean effect size grouped by ei_measure
y_range <- range(filtered_mccann2$yi, na.rm = TRUE)

boxplot(yi ~ ei_stream, data = filtered_mccann2,
  main = "Boxplot of Effect Size (r^2) by EI Stream",
  xlab = "EI Stream", ylab = "Effect Size (r^2)",
  col = "lightblue", ylim = c(y_range[1], y_range[2] * 1.1))

```

Boxplot of Effect Size (r^2) by EI Stream



```
# Barplot of mean  $r^2$  by ei_measure
# Ensures 'summary_stats' is created correctly:
summary_stats <- filtered_mccann2 %>%
  group_by(ei_stream) %>%
  summarize(mean_r2 = mean(yi^2, na.rm = TRUE) # Calculate mean of  $r^2$  (effect size
squared)
)
```

```
# Sort by mean_r2
summary_stats <- summary_stats %>% arrange(desc(mean_r2))
```

```
# Checking for NAN values and remove them
any(is.na(summary_stats$mean_r2)) # Checks if there are any NAs
```



```

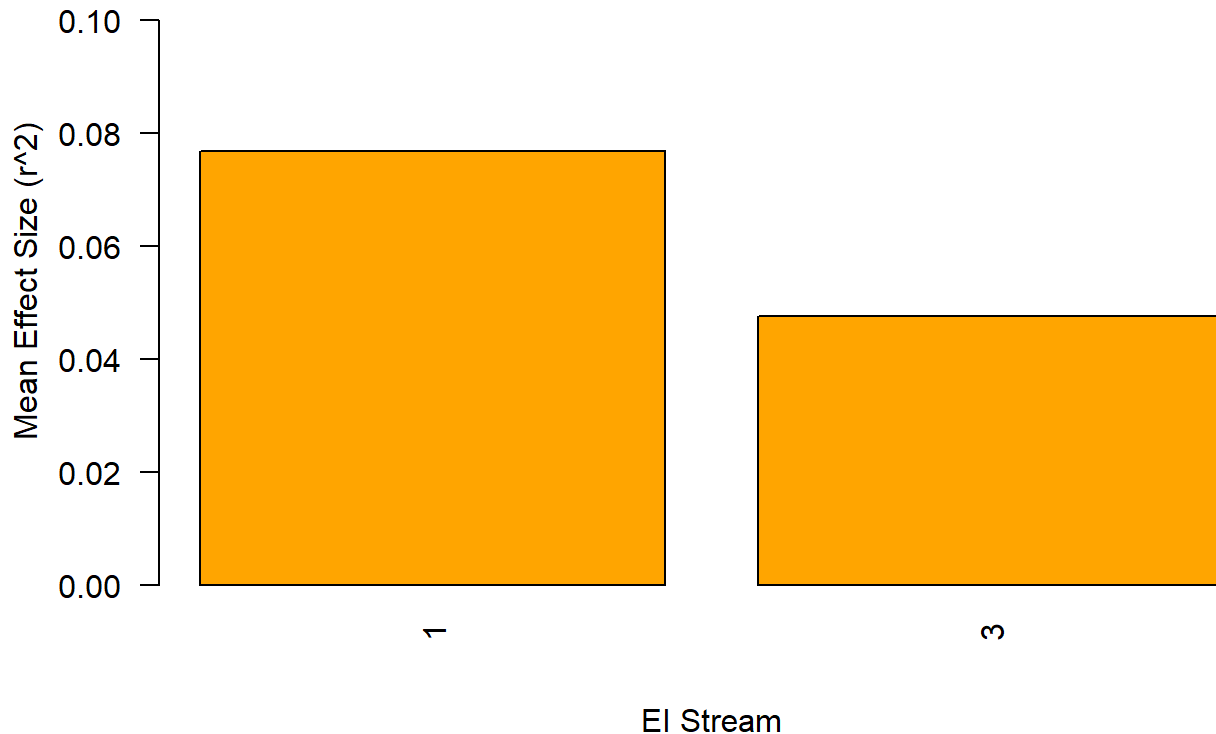
## [1] TRUE
any(is.nan(summary_stats$mean_r2)) # Checks if there are any NaNs
## [1] TRUE
any(is.infinite(summary_stats$mean_r2)) # Checks for infinite values
## [1] FALSE
# QC NaN values
summary_stats_clean <- summary_stats[!is.na(summary_stats$mean_r2) &
!is.nan(summary_stats$mean_r2) & !is.infinite(summary_stats$mean_r2), ]

# Check validity of the data
max_value <- max(summary_stats$mean_r2, na.rm = TRUE)
if (is.finite(max_value)) {
  ylim = c(0, max_value * 1.4)
} else {
  ylim = c(0, 1) # Default fallback
}

# Visualize with a barplot
barplot(
  summary_stats_clean$mean_r2,
  names.arg = summary_stats_clean$ei_stream,
  main = "Mean Effect Size ( $r^2$ ) by EI Stream",
  xlab = "EI Stream",
  ylab = "Mean Effect Size ( $r^2$ )",
  col = "orange",
  las = 2,
  ylim = c(0, max(summary_stats_clean$mean_r2) * 1.4)
)

```

Mean Effect Size (r^2) by EI Stream



```
# Save the summarized statistics to a CSV file
write.csv(summary_stats, "grouped_r2_summary.csv", row.names = FALSE)

# Part 5: Linear Regression Analysis (Ran 6 Models)

# Model 1: Mix of Important Variables
# Fitting the model
# yi = effect size dependent variable
# vi, n, ei_stream, ei_measure independent variables
model1 <- lm(yi ~ vi + n + ei_stream + ei_measure, data = filtered_mccann2)

# Summary
summary(model1)

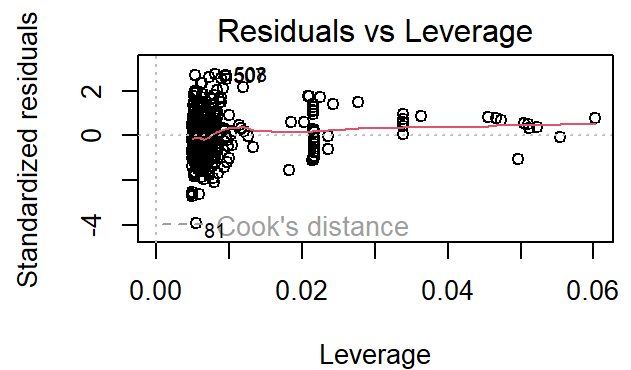
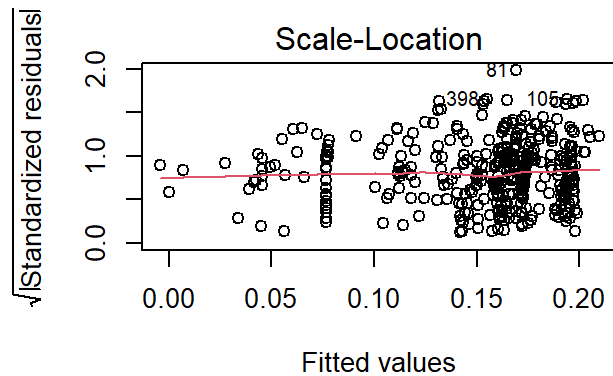
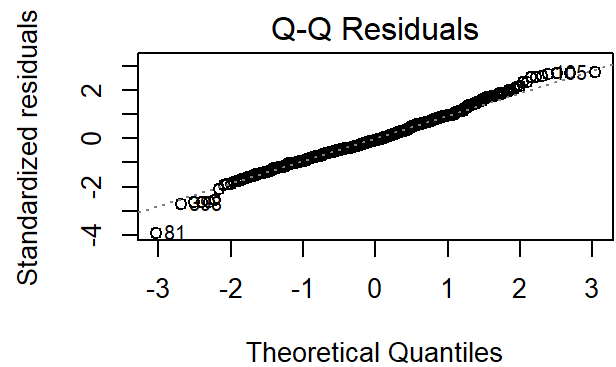
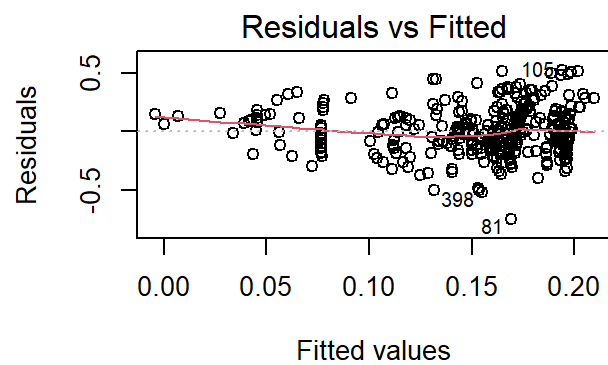
##

KB, JK, KS, and AZ
```

```

## Call:
## lm(formula = yi ~ vi + n + ei_stream + ei_measure, data = filtered_mccann2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.75165 -0.11863 -0.01518  0.12247  0.52410
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.170e-01  2.920e-02   7.434 6.15e-13 ***
## vi          -6.614e+00  1.749e+00  -3.781 0.000179 ***
## n           -1.465e-04  3.684e-05  -3.977 8.24e-05 ***
## ei_stream    1.390e-02  1.072e-02   1.297 0.195440
## ei_measure      NA         NA      NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1918 on 413 degrees of freedom
## (136 observations deleted due to missingness)
## Multiple R-squared:  0.04685,    Adjusted R-squared:  0.03992
## F-statistic: 6.766 on 3 and 413 DF,  p-value: 0.0001833
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model1)

```



```
# Save summary
```

```
capture.output(summary(model1), file = "model_summary.txt")
```

```
# Results
```

```
# Residuals
```

```
# The differences between the observed and predicted values of the effect size.
```

```
# Spread: range -0.75165 - 0.5241
```

```
# Coefficients Table
```

```
# Overview of the table
```

```
KB, JK, KS, and AZ
```

```

# Estimate: Effect size for each variable.
# Standard Error: SD of the effect size.
# t-value: Test statistic for the hypothesis -> H0 = 0
# P-value: measure of significance

# Summary of Coefficients
# Intercept: 0.2170, (p < 0.00) ***
# vi: -6.614, (p = 0.000179), negative relationship
# n: -0.0001465, (p < 0.001), *** weak negative relationship
# ei_stream: 0.01390, (p = 0.195), *** not significant
# ei_measure: not measured -> colinear with other variables

# Summary of Model Fit
# Residual standard error: 0.1918
# Multiple R-squared: 0.04685, 4.69% of the variability in the model
# is explained by this model -> weak explanatory power
# Adjusted R-squared: 0.03992
# F-statistic: 6.766, p = 0.0001833

#-----

# Linear Regression Analysis

# Model 2: All important variables
# Fitting the model
# yi = effect size dependent variable
# vi, n, ei_stream, ei_measure, ei_construct independent variables
model2 <- lm(yi ~ vi + n + ei_stream + ei_construct + ei_measure, data = filtered_mccann2)

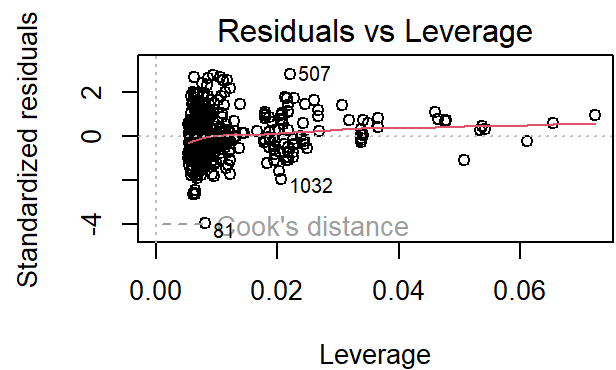
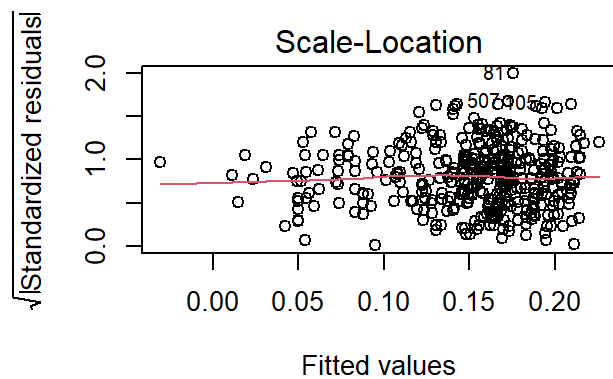
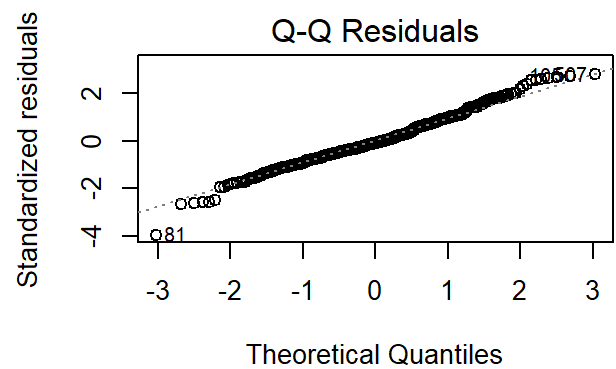
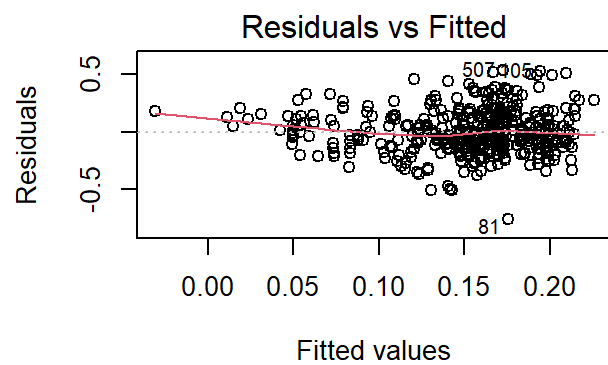
# Summary
summary(model2)

```

```

##
## Call:
## lm(formula = yi ~ vi + n + ei_stream + ei_construct + ei_measure,
##     data = filtered_mccann2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.75763 -0.11608 -0.01895  0.12192  0.53177
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.078e-01  3.009e-02  6.905 1.98e-11 ***
## vi          -6.296e+00  1.760e+00 -3.577 0.00039 ***
## n           -1.456e-04  3.673e-05 -3.965 8.68e-05 ***
## ei_stream    7.576e-03  1.321e-02  0.574 0.56652
## ei_construct  4.743e-03  3.869e-03  1.226 0.22096
## ei_measure      NA         NA     NA     NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1911 on 401 degrees of freedom
## (147 observations deleted due to missingness)
## Multiple R-squared:  0.05018,    Adjusted R-squared:  0.04071
## F-statistic: 5.297 on 4 and 401 DF,  p-value: 0.0003636
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model2)

```



```
# Save summary
```

```
capture.output(summary(model2), file = "model2_summary.txt")
```

```
# Results
```

```
# Residuals
```

```
# The differences between the observed and predicted values of the effect size.
```

```
# Spread: range -0.75763 - 0.53177
```

```
# Coefficients Table
```

```
# Overview of the table
```

```
KB, JK, KS, and AZ
```

```

# Estimate: Effect size for each variable.
# Standard Error: SD of the effect size.
# t-value: Test statistic for the hypothesis -> H0 = 0
# P-value: measure of significance

# Summary of Coefficients
# Intercept: 0.208, (p < 0.001) ***
# vi: -6.296, (p = 0.00039), *** strong negative relationship
# n: -0.0001465, (p < 0.001), *** weak negative relationship
# ei_stream: 0.007576, (p = 0.56652), not significant
# ei_construct: 0.004743, (p = 0.22096)
# ei_measure: not measured -> co linear with other variables

# Summary of Model Fit
# Residual standard error: 0.1911
# Multiple R-squared: 0.0518, 5.02% of the variability in the model
# is explained by this model -> weak explanatory power
# Adjusted R-squared: 0.04071
# F-statistic: 5.297, p = 0.00003636

#-----

# Linear Regression Analysis

# Model 3: vi
# Fitting the model
# yi = effect size dependent variable
# vi independent variable
model3 <- lm(yi ~ vi , data = filtered_mccann2)

# Summary

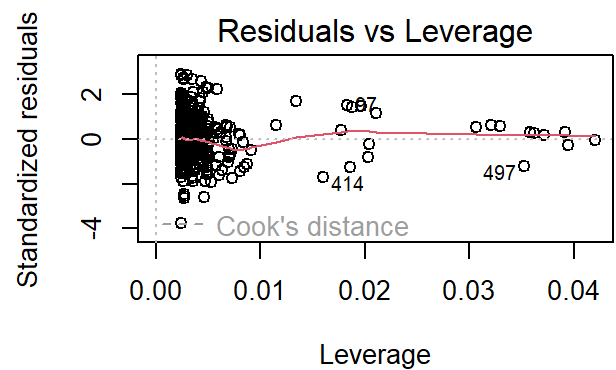
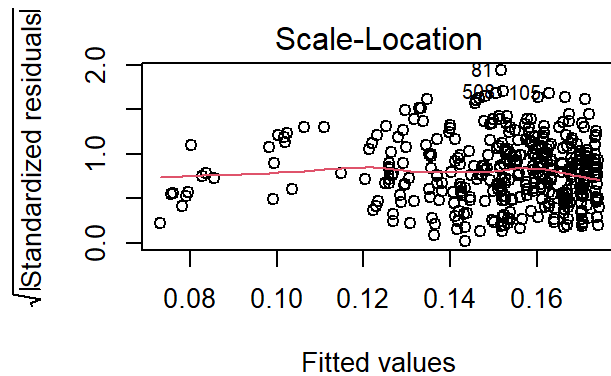
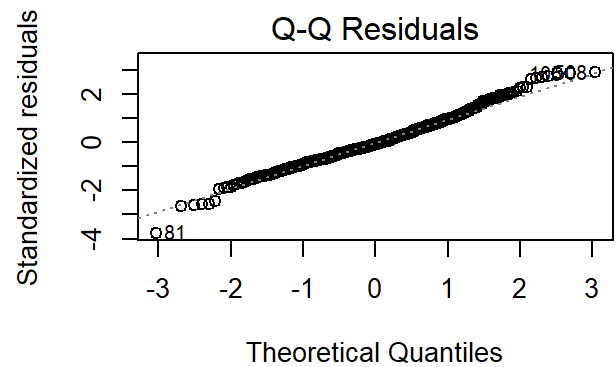
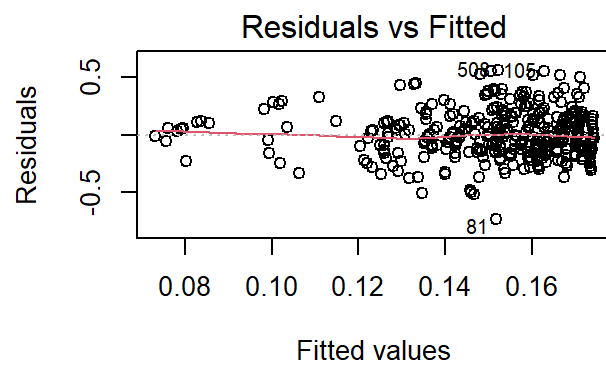
```



```

summary(model3)
##
## Call:
## lm(formula = yi ~ vi, data = filtered_mccann2)
##
## Residuals:
##   Min     1Q  Median     3Q      Max
## -0.7339 -0.1305 -0.0156  0.1195  0.5644
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.17602    0.01449   12.145  <2e-16 ***
## vi          -3.01211    1.45363   -2.072  0.0389 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.195 on 415 degrees of freedom
## (136 observations deleted due to missingness)
## Multiple R-squared:  0.01024,    Adjusted R-squared:  0.007855
## F-statistic: 4.294 on 1 and 415 DF,  p-value: 0.03887
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model3)

```



```
# Save summary
```

```
capture.output(summary(model3), file = "model3_summary.txt")
```

```
# Results
```

```
# Residuals
```

```
# The differences between the observed and predicted values of the effect size.
```

```
# Spread: range -0.7339 - 0.5644
```

```
# Coefficients Table
```

```
# Overview of the table
```

```
KB, JK, KS, and AZ
```

```

# Estimate: Effect size for each variable.
# Standard Error: SD of the effect size.
# t-value: Test statistic for the hypothesis ->  $H_0 = 0$ 
# P-value: measure of significance

# Summary of Coefficients
# Intercept: 0.17602, ( $p < 0.00$ ) ***
# vi: -3.01211, ( $p = 0.0389$ ), * strong negative relationship

# Summary of Model Fit
# Residual standard error: 0.195
# Multiple R-squared: 0.01024, 1.02% of the variability in the model
# is explained by this model -> weak explanatory power
# Adjusted R-squared: 0.007855
# F-statistic: 4.294,  $p = 0.03887$ 

```

```

#-----

```

```

# Linear Regression Analysis

```

```

# Model 4: n
# Fitting the model
# yi = effect size dependent variable
# n independent variable
model4 <- lm(yi ~ n , data = filtered_mccann2)

```

```

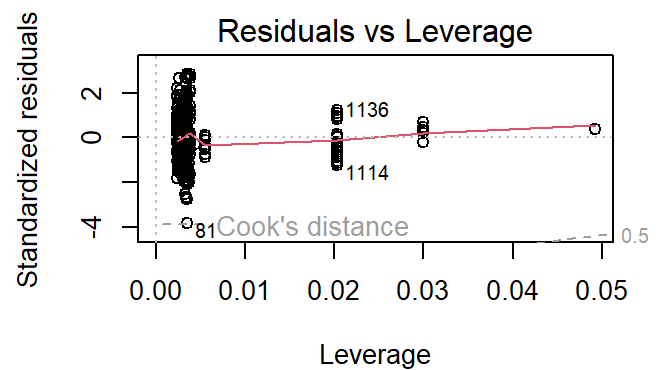
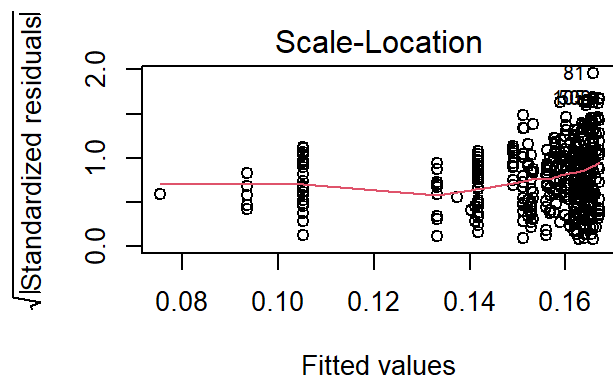
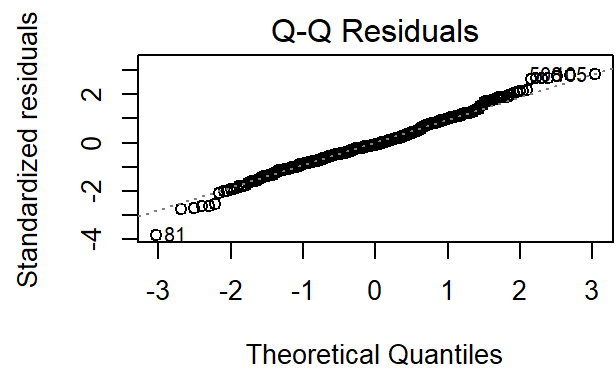
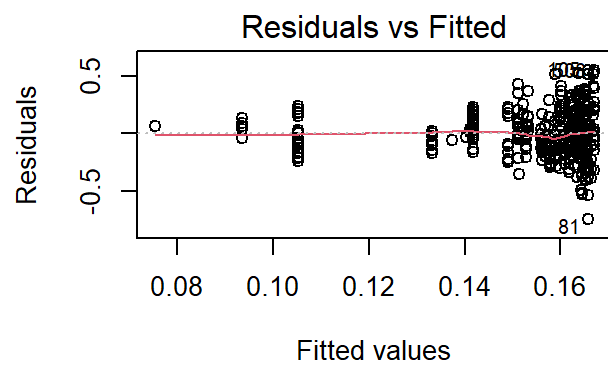
# Summary
summary(model4)
##
## Call:

```

```

## lm(formula = yi ~ n, data = filtered_mccann2)
##
## Residuals:
##   Min     1Q  Median     3Q    Max
## -0.74794 -0.11608 -0.01759  0.13014  0.55261
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.686e-01  1.259e-02  13.386  <2e-16 ***
## n           -5.259e-05  2.848e-05  -1.847   0.0655 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1952 on 415 degrees of freedom
## (136 observations deleted due to missingness)
## Multiple R-squared:  0.008152, Adjusted R-squared:  0.005762
## F-statistic: 3.411 on 1 and 415 DF, p-value: 0.06549
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model4)

```



```
# Save summary
```

```
capture.output(summary(model4), file = "model4_summary.txt")
```

```
# Results
```

```
# Residuals
```

```
# The differences between the observed and predicted values of the effect size.
```

```
# Spread: range -0.74794 - 0.55261
```

```
# Coefficients Table
```

```
# Overview of the table
```

```
KB, JK, KS, and AZ
```

```

# Estimate: Effect size for each variable.
# Standard Error: SD of the effect size.
# t-value: Test statistic for the hypothesis -> H0 = 0
# P-value: measure of significance

# Summary of Coefficients
# Intercept: 0.1686, (p < 0.0) ***
# n: -0.00005, (p = 0.0655), strong negative relationship

# Summary of Model Fit
# Residual standard error: 0.195
# Multiple R-squared: 0.008152, 0.08% of the variability in the model
# is explained by this model -> weak explanatory power
# Adjusted R-squared: 0.005762
# F-statistic: 3.4`, p = 0.06549

#-----

# Linear Regression Analysis

# Model 5: ei_measure
# Fitting the model
# yi = effect size dependent variable
# ei_stream independent variable
model5 <- lm(yi ~ ei_measure , data = filtered_mccann2)

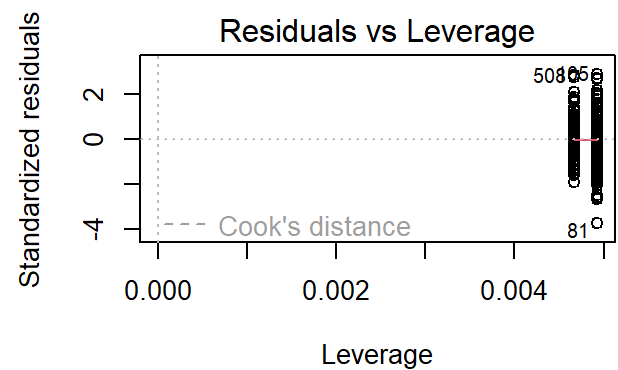
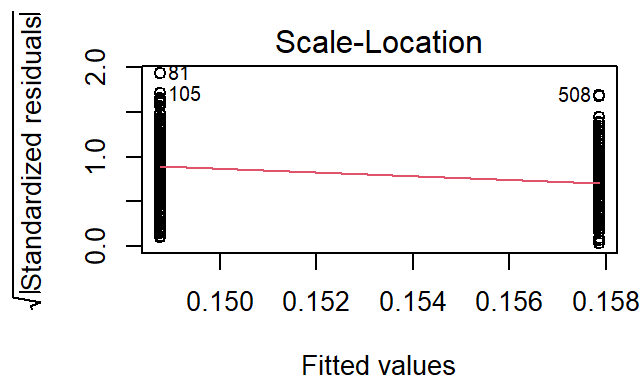
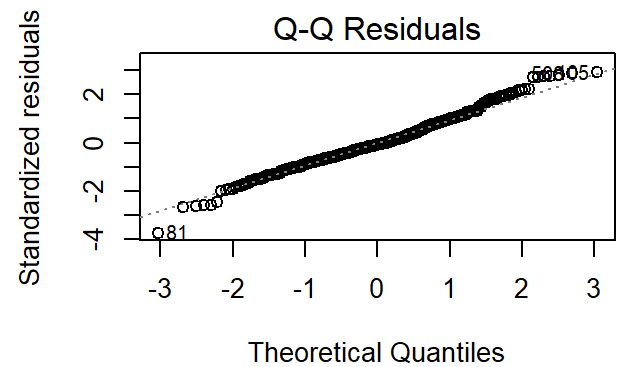
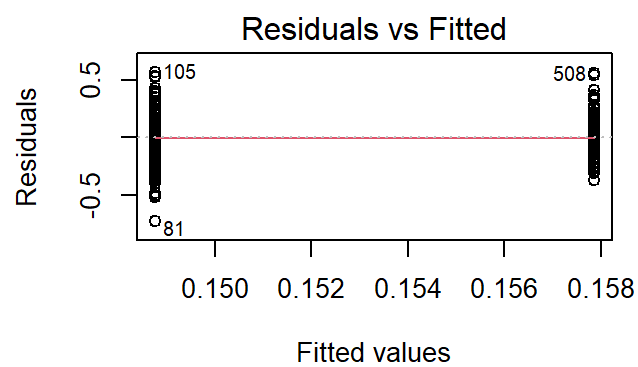
# Summary
summary(model5)
##
## Call:

```

```

## lm(formula = yi ~ ei_measure, data = filtered_mccann2)
##
## Residuals:
##   Min     1Q  Median     3Q    Max
## -0.73102 -0.12193 -0.01583  0.12596  0.56953
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.143746   0.022553   6.374 4.92e-10 ***
## ei_measure  0.004552   0.009599   0.474  0.636
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1959 on 415 degrees of freedom
## (136 observations deleted due to missingness)
## Multiple R-squared:  0.0005416, Adjusted R-squared: -0.001867
## F-statistic: 0.2249 on 1 and 415 DF, p-value: 0.6356
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model5)

```



```
# Save summary
```

```
capture.output(summary(model5), file = "model5_summary.txt")
```

```
# Results
```

```
# Residuals
```

```
# The differences between the observed and predicted values of the effect size.
```

```
# Spread: range -0.73102 - 0.56953
```

```
# Coefficients Table
```

```
# Overview of the table
```

```
KB, JK, KS, and AZ
```



```

# Estimate: Effect size for each variable.
# Standard Error: SD of the effect size.
# t-value: Test statistic for the hypothesis -> H0 = 0
# P-value: measure of significance

# Summary of Coefficients
# Intercept: 0.143746, (p < 0.0) ***
# ei_measure: 0.0045525, (p = 0.636), strong negative relationship

# Summary of Model Fit
# Residual standard error: 0.1959
# Multiple R-squared: 0.0005416, 0.05% of the variability in the model
# is explained by this model -> weak explanatory power
# Adjusted R-squared: -0.001867
# F-statistic: 0.2249, p = 0.6356

#-----

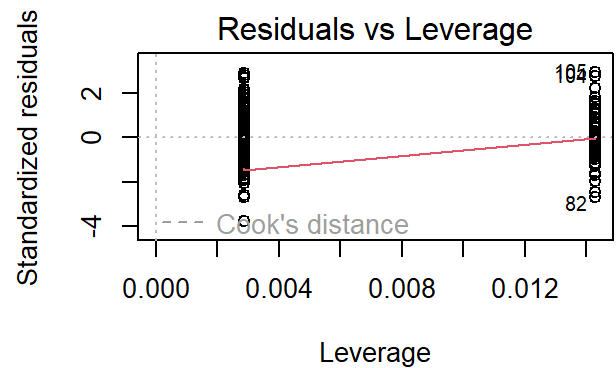
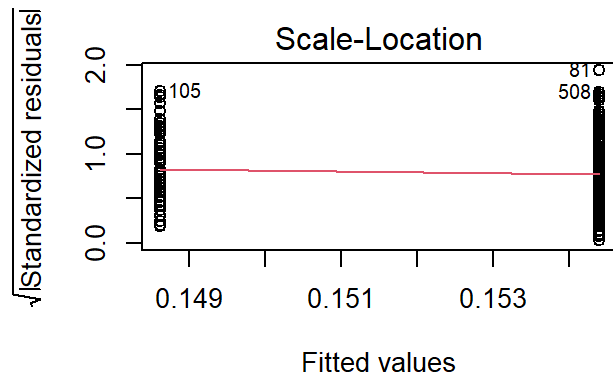
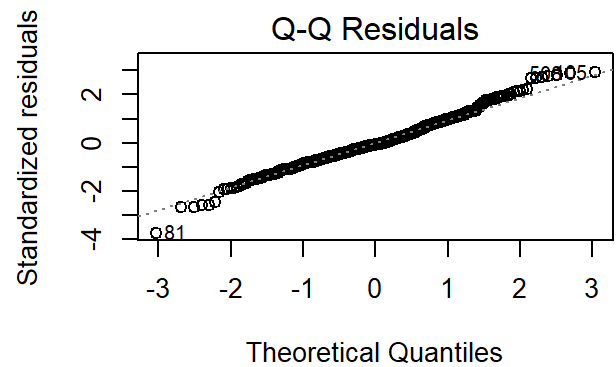
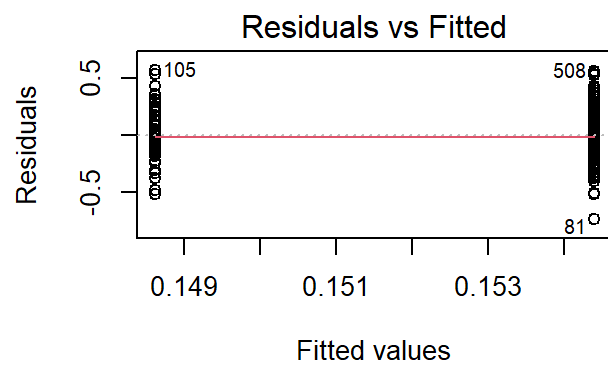
# Linear Regression Analysis

# Model 6: achievement type
# Fitting the model
# yi = effect size dependent variable
# ei_stream independent variable
model6 <- lm(yi ~ achievement_type , data = filtered_mccann2)

# Summary
summary(model6)
##
## Call:
## lm(formula = yi ~ achievement_type, data = filtered_mccann2)

```

```
##
## Residuals:
##   Min     1Q   Median     3Q      Max
## -0.73667 -0.12387 -0.01659  0.12114  0.56967
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.154395   0.010521  14.675  <2e-16 ***
## achievement_type -0.005785   0.025679  -0.225    0.822
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.196 on 415 degrees of freedom
## (136 observations deleted due to missingness)
## Multiple R-squared:  0.0001223, Adjusted R-squared: -0.002287
## F-statistic: 0.05075 on 1 and 415 DF, p-value: 0.8219
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model6)
```



```
# Save summary
```

```
capture.output(summary(model6), file = "model6_summary.txt")
```

```
# Results
```

```
# Residuals
```

```
# The differences between the observed and predicted values of the effect size.
```

```
# Spread: range -0.73667 - 0.56967
```

```
# Coefficients Table
```

```
# Overview of the table
```

```
KB, JK, KS, and AZ
```

```
# Estimate: Effect size for each variable.
# Standard Error: SD of the effect size.
# t-value: Test statistic for the hypothesis -> H0 = 0
# P-value: measure of significance

# Summary of Coefficients
# Intercept: 0.154395, (p < 0.00) ***
# achievement type: -0.005785, (p = 0.822), strong negative relationship
```

```
# Summary of Model Fit
# Residual standard error: 0.196
# Multiple R-squared: 0.0001223, 0.01% of the variability in the model
# is explained by this model -> weak explanatory power
# Adjusted R-squared: -0.002287
# F-statistic: 0.05075, p = 0.8219
```

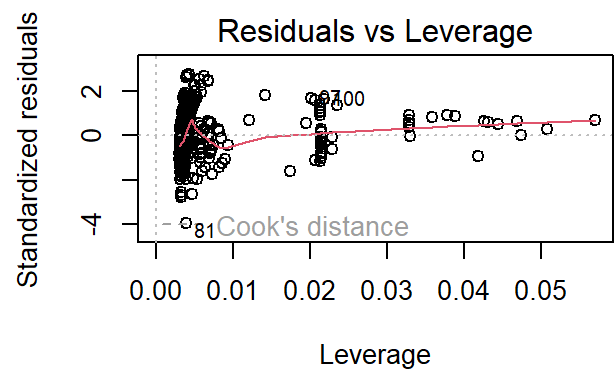
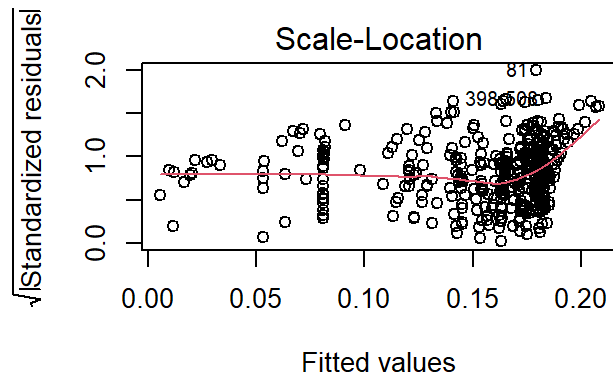
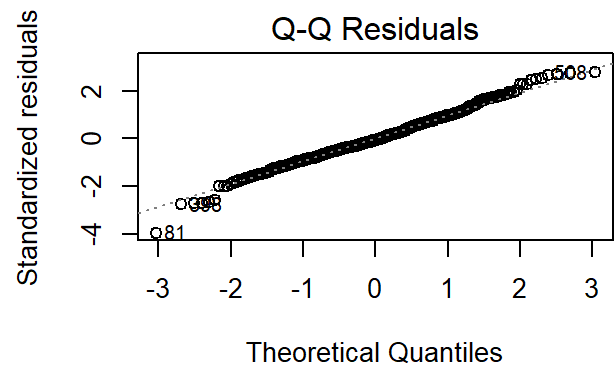
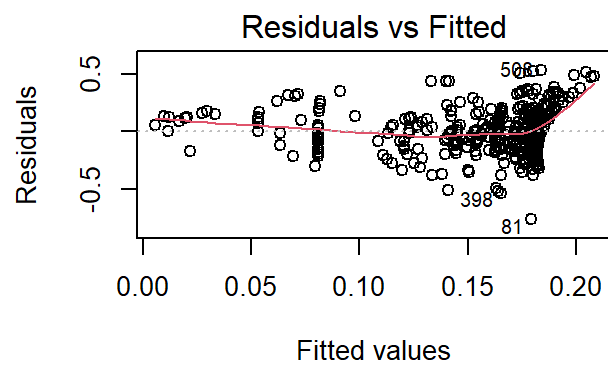
```
#-----
```

```
# Linear Regression Analysis
```

```
# Model 7: vi and n
# Fitting the model
# yi = effect size dependent variable
# vi and n independent variables
model7 <- lm(yi ~ vi + n , data = filtered_mccann2)
```

```
# Summary
summary(model7)
##
## Call:
## lm(formula = yi ~ vi + n, data = filtered_mccann2)
```

```
##
## Residuals:
##   Min     1Q   Median     3Q      Max
## -0.76164 -0.12101 -0.01275  0.12642  0.53280
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.413e-01  2.246e-02  10.743 < 2e-16 ***
## vi          -6.776e+00  1.746e+00  -3.881 0.000121 ***
## n           -1.286e-04  3.417e-05  -3.763 0.000192 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.192 on 414 degrees of freedom
## (136 observations deleted due to missingness)
## Multiple R-squared:  0.04297,    Adjusted R-squared:  0.03834
## F-statistic: 9.293 on 2 and 414 DF,  p-value: 0.0001127
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model7)
```



```
# Save summary
```

```
capture.output(summary(model7), file = "model7_summary.txt")
```

```
# Results
```

```
# Residuals
```

```
# The differences between the observed and predicted values of the effect size.
```

```
# Spread: range -0.76164 - 0.53280
```

```
# Coefficients Table
```

```
# Overview of the table
```

```
KB, JK, KS, and AZ
```

```

# Estimate: Effect size for each variable.
# Standard Error: SD of the effect size.
# t-value: Test statistic for the hypothesis ->  $H_0 = 0$ 
# P-value: measure of significance

# Summary of Coefficients
# Intercept: -0.2413, ( $p < 0.0$ ) ***
# vi: -6.77, ( $p = 0.000121$ ),*** strong negative relationship
# n: -0.0001, ( $p = 0.00192$ ), *** strong negative relationship

# Summary of Model Fit
# Residual standard error: 0.192
# Multiple R-squared: 0.04297, 4.3% of the variability in the model
# is explained by this model -> weak explanatory power
# Adjusted R-squared: 0.0384
# F-statistic: 9.293,  $p = 0.000127$ , ***
# Part 6: Decision Tree

# Install packages
install.packages("rpart")
## Installing package into 'C:/Users/angel/AppData/Local/R/win-library/4.4'
## (as 'lib' is unspecified)
## package 'rpart' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\angel\AppData\Local\Temp\RtmpEjErRu\downloaded_packages
install.packages("rpart.plot")
## Installing package into 'C:/Users/angel/AppData/Local/R/win-library/4.4'
## (as 'lib' is unspecified)
## package 'rpart.plot' successfully unpacked and MD5 sums checked
##

```

```

## The downloaded binary packages are in
## C:\Users\angel\AppData\Local\Temp\RtmpEjErRu\downloaded_packages
library(rpart)
library(rpart.plot)

# Fitted decision tree with restricted variables (author, sample_id, country, ei_measure,
achievement_type)
tree_model <- rpart(yi ~ vi + n + age + ei_construct + ei_measure + achievement_type,
                    data = filtered_mccann2,
                    method = "anova")

# Print tree model
print(tree_model)
## n=417 (136 observations deleted due to missingness)
##
## node), split, n, deviance, yval
##      * denotes terminal node
##
##  1) root 417 15.94194000  0.15342430
##    2) age< 17.87 10  0.30513820 -0.16451690 *
##    3) age>=17.87 407 14.60110000  0.16123610
##      6) n>=36 386 12.79360000  0.15244410
##        12) vi>=0.01347227 43  1.22025000  0.02961266 *
##        13) vi< 0.01347227 343 10.84325000  0.16784280
##          26) n>=73.5 325  8.47430000  0.15471480
##            52) age< 21.39133 227  5.82495700  0.13438130
##              104) age>=19.275 178  3.88334700  0.11439420
##                208) vi>=0.0007610781 170  3.61660000  0.10645660
##                  416) n>=632 33  0.18738960  0.03008624 *
##                  417) n< 632 137  3.19037900  0.12485230
##                    834) vi>=0.001835793 115  2.26219000  0.09437466

```



```

##          1668) n>=122.5 87  1.17301800  0.06610593
##          3336) vi>=0.004149624 43  0.37938000  0.01001535 *
##          3337) vi< 0.004149624 44  0.52614320  0.12092170 *
##          1669) n< 122.5 28  0.80362960  0.18220960
##          3338) vi>=0.008628262 21  0.29749120  0.10623900 *
##          3339) vi< 0.008628262 7  0.02132991  0.41012140 *
##          835) vi< 0.001835793 22  0.26297830  0.28416740 *
##          209) vi< 0.0007610781 8  0.02842561  0.28306900 *
##          105) age< 19.275 49  1.61219200  0.20698740
##          210) vi>=0.007264835 16  0.28747790  0.07843418 *
##          211) vi< 0.007264835 33  0.93209830  0.26931620
##          422) ei_measure>=2.1 15  0.24317270  0.18164640 *
##          423) ei_measure< 2.1 18  0.47756090  0.34237430 *
##          53) age>=21.39133 98  2.33809200  0.20181400
##          106) vi>=0.007777558 38  0.25572900  0.10220140 *
##          107) vi< 0.007777558 60  1.46649700  0.26490200
##          214) n>=193 19  0.38649420  0.14109940 *
##          215) n< 193 41  0.65383540  0.32227390
##          430) vi>=0.006098914 17  0.24210780  0.24062820 *
##          431) vi< 0.006098914 24  0.21813520  0.38010620 *
##          27) n< 73.5 18  1.30161700  0.40487560 *
##          7) n< 36 21  1.22922700  0.32284130
##          14) vi>=0.02794007 10  0.10175210  0.10328750 *
##          15) vi< 0.02794007 11  0.20721900  0.52243570 *
# Use cp table to prune the tree
printcp(tree_model)
##
## Regression tree:
## rpart(formula = yi ~ vi + n + age + ei_construct + ei_measure +
##   achievement_type, data = filtered_mccann2, method = "anova")
##

```

```

## Variables used in tree construction:
## [1] age      ei_measure n      vi
##
## Root node error: 15.942/417 = 0.03823
##
## n=417 (136 observations deleted due to missingness)
##
##      CP nsplit rel error  xerror   xstd
## 1 0.064967    0  1.00000 1.00339 0.079944
## 2 0.051687    1  0.93503 0.97227 0.093753
## 3 0.029078    5  0.72828 0.88526 0.094477
## 4 0.026732    7  0.67013 0.76008 0.089159
## 5 0.023390    8  0.64340 0.72340 0.088324
## 6 0.016779   15  0.47813 0.64275 0.086520
## 7 0.013258   16  0.46135 0.64296 0.088738
## 8 0.012144   17  0.44809 0.63840 0.089712
## 9 0.010000   18  0.43594 0.64118 0.089695
# Prune the tree based on the complexity parameter (cp) value that minimizes cross-validation
error
optimal_cp <- tree_model$scptable[which.min(tree_model$scptable[, "xerror"]), "CP"]
pruned_tree <- prune(tree_model, cp = optimal_cp)

# Print the pruned tree
print(pruned_tree)
## n=417 (136 observations deleted due to missingness)
##
## node), split, n, deviance, yval
##      * denotes terminal node
##
##  1) root 417 15.94194000  0.15342430
##    2) age< 17.87 10  0.30513820 -0.16451690 *
```

```

## 3) age>=17.87 407 14.60110000 0.16123610
## 6) n>=36 386 12.79360000 0.15244410
## 12) vi>=0.01347227 43 1.22025000 0.02961266 *
## 13) vi< 0.01347227 343 10.84325000 0.16784280
## 26) n>=73.5 325 8.47430000 0.15471480
## 52) age< 21.39133 227 5.82495700 0.13438130
## 104) age>=19.275 178 3.88334700 0.11439420
## 208) vi>=0.0007610781 170 3.61660000 0.10645660
## 416) n>=632 33 0.18738960 0.03008624 *
## 417) n< 632 137 3.19037900 0.12485230
## 834) vi>=0.001835793 115 2.26219000 0.09437466
## 1668) n>=122.5 87 1.17301800 0.06610593
## 3336) vi>=0.004149624 43 0.37938000 0.01001535 *
## 3337) vi< 0.004149624 44 0.52614320 0.12092170 *
## 1669) n< 122.5 28 0.80362960 0.18220960
## 3338) vi>=0.008628262 21 0.29749120 0.10623900 *
## 3339) vi< 0.008628262 7 0.02132991 0.41012140 *
## 835) vi< 0.001835793 22 0.26297830 0.28416740 *
## 209) vi< 0.0007610781 8 0.02842561 0.28306900 *
## 105) age< 19.275 49 1.61219200 0.20698740
## 210) vi>=0.007264835 16 0.28747790 0.07843418 *
## 211) vi< 0.007264835 33 0.93209830 0.26931620
## 422) ei_measure>=2.1 15 0.24317270 0.18164640 *
## 423) ei_measure< 2.1 18 0.47756090 0.34237430 *
## 53) age>=21.39133 98 2.33809200 0.20181400
## 106) vi>=0.007777558 38 0.25572900 0.10220140 *
## 107) vi< 0.007777558 60 1.46649700 0.26490200
## 214) n>=193 19 0.38649420 0.14109940 *
## 215) n< 193 41 0.65383540 0.32227390 *
## 27) n< 73.5 18 1.30161700 0.40487560 *
## 7) n< 36 21 1.22922700 0.32284130

```

```

##      14) vi>=0.02794007 10 0.10175210 0.10328750 *
##      15) vi< 0.02794007 11 0.20721900 0.52243570 *
# Plot cross-validation error
# Extract and print the cross-validation error for the unpruned tree
cv_error <- tree_model$cptable[, "xerror"]
print(cv_error)
##      1      2      3      4      5      6      7      8
## 1.0033899 0.9722696 0.8852556 0.7600785 0.7234023 0.6427503 0.6429581 0.6384002
##      9
## 0.6411798
# Find and print the minimum cross-validation error
min_cv_error <- min(cv_error)
cat("Minimum Cross-Validation Error:", min_cv_error, "\n")
## Minimum Cross-Validation Error: 0.6384002
# Find the complexity parameter (cp) that corresponds to the minimum cross-validation error
optimal_cp <- tree_model$cptable[which.min(cv_error), "CP"]
cat("Optimal Complexity Parameter (cp):", optimal_cp, "\n")
## Optimal Complexity Parameter (cp): 0.01214359
# Results:

# 1. Tree Structure
# Node = subset of data
# Splits based on conditions
# * -> terminal node (leaf)

# 2. Nodes
# node -> unique
# split -> condition that splits the data
# n -> # observations in the node
# deviance -> node variation
# yval -> predicted value for observations in the node

```

3. CP Values (Complexity Parameter)

CP: complexity parameter , low values = more branches, higher values = tree pruned earlier

nsplit: # branches at each split

Relative Error: relative error which grows as the tree grows

xerror: cross-validation error

xstd: sd of the cross-validation error

root node: initial split at the root of the tree

Optimal CP -> minimizes xerror

Summary of Findings

1. Effect size (y_i) decreases with increased variance (v_i) or larger population sizes (n)

2. Age affected earlier splits, possible significant effect on effect size

3. Min CV error: 0.646

Optimal CP: 0.012

#-----

Measurement of Tree Performance

Cross-validation Model

printcp(pruned_tree) # (xerror) for each split

##

Regression tree:

rpart(formula = $y_i \sim v_i + n + \text{age} + \text{ei_construct} + \text{ei_measure} +$

achievement_type, data = filtered_mccann2, method = "anova")

##

Variables actually used in tree construction:

[1] age ei_measure n vi

##

```

## Root node error: 15.942/417 = 0.03823
##
## n=417 (136 observations deleted due to missingness)
##
##      CP nsplit rel error  xerror   xstd
## 1 0.064967    0  1.00000 1.00339 0.079944
## 2 0.051687    1  0.93503 0.97227 0.093753
## 3 0.029078    5  0.72828 0.88526 0.094477
## 4 0.026732    7  0.67013 0.76008 0.089159
## 5 0.023390    8  0.64340 0.72340 0.088324
## 6 0.016779   15  0.47813 0.64275 0.086520
## 7 0.013258   16  0.46135 0.64296 0.088738
## 8 0.012144   17  0.44809 0.63840 0.089712
# Root node error = 0.03823
# n = 417

# Add to limitations: #NaN values/ low population after NaN removal
#-----

# Measurement of Variable Effects

# Which variables are important?
importance <- pruned_tree$variable.importance
importance <- sort(importance, decreasing = TRUE)
print(importance)
##      vi      n      age  ei_measure
## 5.7290335 4.3578561 2.6134019 0.7242815
## ei_construct achievement_type
## 0.4275492 0.2232582
# Results from most to least important
# vi = 5.923

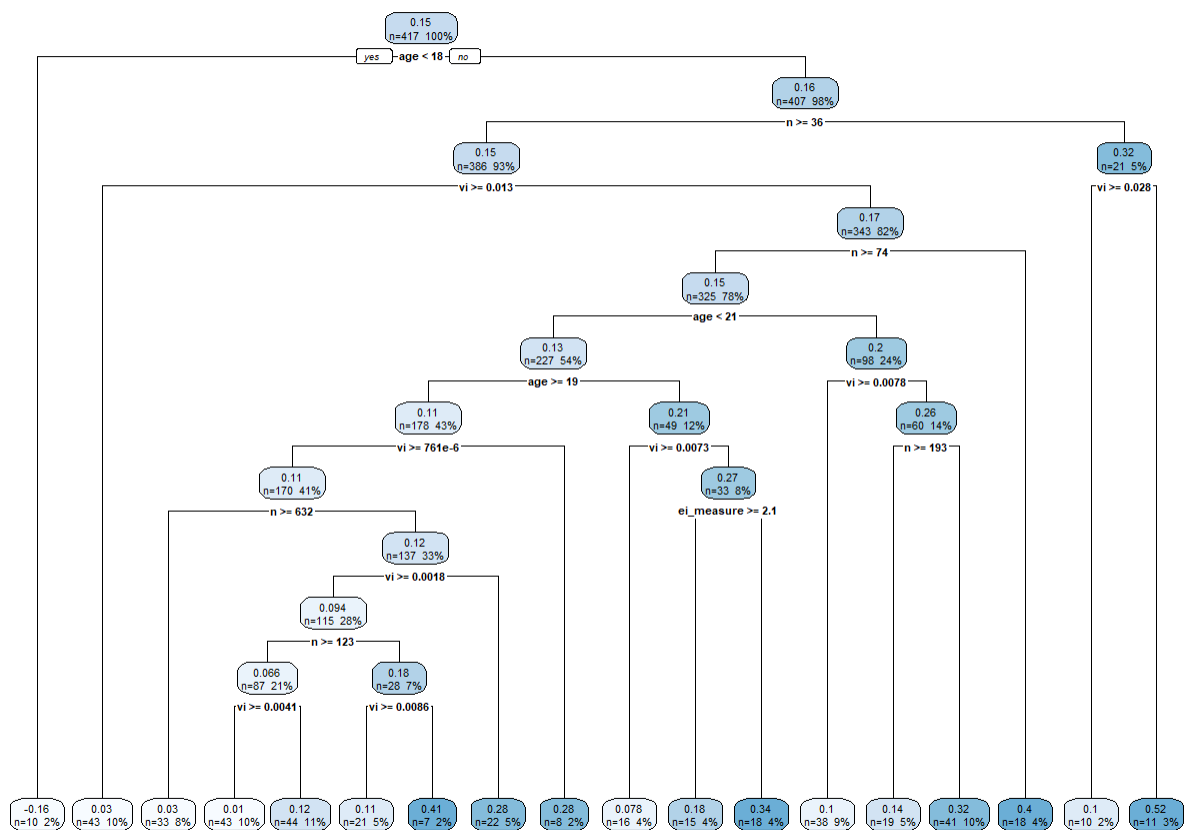
```

```
# n = 4.438
# age = 2.67
# ei_measure = 0.7698
# ei_construct = 0.4275492
# achievement type = 0.223
```

```
# View variable interactions
```

```
library(rpart.plot)
```

```
rpart.plot(pruned_tree, type = 2, extra = 101, fallen.leaves = TRUE)
```



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